

SS2 TECHNICAL DRAWING LESSON NOTES

FIRST TERM

WEEK 1

TOPIC: REVISION OF SS1 WORK

CONTENT

1. Introduction

At the beginning of a new session, it is essential to review and consolidate the knowledge and skills learned in SS1. This ensures that students can recall foundational concepts, apply them in more advanced work, and correct any misconceptions. The revision focuses on **basic drawing techniques, drawing instruments, lettering, geometrical constructions, orthographic projection, and isometric drawing**—all of which are the bedrock of Technical Drawing.

2. Review of Technical Drawing Instruments and Materials

Technical drawing requires accuracy and neatness, which can only be achieved through the proper use of instruments and materials.

Common Drawing Instruments include:

- **Drawing board:** A smooth flat board used for securing drawing paper.
- **T-square:** Used for drawing horizontal lines and guiding set squares.
- **Set squares (45° and 60°):** Used for drawing vertical and inclined lines.
- **Protractor:** Used to measure and construct angles.
- **Compass:** Used to draw circles and arcs.
- **Dividers:** Used to transfer measurements from one part of a drawing to another.
- **Drawing pencils:** Different grades (e.g., 2H, H, HB, B, 2B) are used for different line types and shades.
- **Eraser:** For correcting mistakes neatly.
- **Masking tape or clips:** Used to hold paper securely on the board.
- **French curves and flexible curves:** For drawing irregular curves.
- **Drawing paper or cartridge sheet:** The surface on which drawings are made.

Maintenance of Instruments:

- Keep instruments clean and dry.

- Store in instrument boxes after use.
- Sharpen pencils properly and avoid breaking compass points.
- Prevent T-square and set squares from warping or bending.

3. Review of Lettering and Line Types

a. Lettering Lettering is the process of writing titles, dimensions, and notes on technical drawings. It ensures legibility and uniformity.

Types of Lettering:

- **Uppercase (capital) lettering**
- **Lowercase lettering**
- **Italic lettering (slanting form)**

Qualities of Good Lettering:

- Uniformity of size and spacing
- Neatness and legibility
- Proper alignment
- Consistent line thickness

b. Line Types and Their Uses:

Line Type	Description	Use
Continuous thick line	Heavy and unbroken	Visible outlines
Continuous thin line	Light and unbroken	Dimension lines, projection lines
Dashed line	Short, evenly spaced dashes	Hidden details
Chain thin line	Alternating long and short dashes	Centre lines
Zig-zag line	Thin, irregular	Breaks or interruptions

4. Review of Geometrical Constructions

Geometrical constructions form the foundation for complex engineering and architectural designs. Students should be able to:

- Bisect lines and angles
- Construct perpendicular and parallel lines
- Divide a line into equal parts
- Construct regular polygons (triangle, square, pentagon, hexagon, etc.)
- Construct tangents to circles
- Construct arcs and circles through given points

Accuracy and precision are vital in all constructions.

5. Review of Orthographic Projection

Orthographic projection is the method of representing a 3D object on 2D planes using several views.

Key Concepts:

- **Projection planes:** Horizontal (HP) and Vertical (VP)
- **Types of orthographic projection:**
 - *First Angle Projection*
 - *Third Angle Projection*
- **Principal views:** Front, plan (top), and side views.

Applications: Used in engineering and architectural drawing to represent objects accurately with full dimensions.

6. Review of Isometric Drawing

Isometric drawing is a type of pictorial drawing where all dimensions are drawn at **30° to the horizontal**. It provides a 3D representation of an object to show how it looks in reality.

Key Principles:

- Three isometric axes are used: vertical and two inclined at 30° to the horizontal.
- Equal scaling along all axes.
- Circles appear as ellipses in isometric view.

Importance: Isometric drawings help visualize and interpret orthographic projections.

7. Importance of Revision in Technical Drawing

- Reinforces foundational knowledge for advanced topics.
 - Helps identify areas needing improvement.
 - Refreshes accuracy, neatness, and presentation skills.
 - Builds confidence for more complex tasks in SS2 and SS3.
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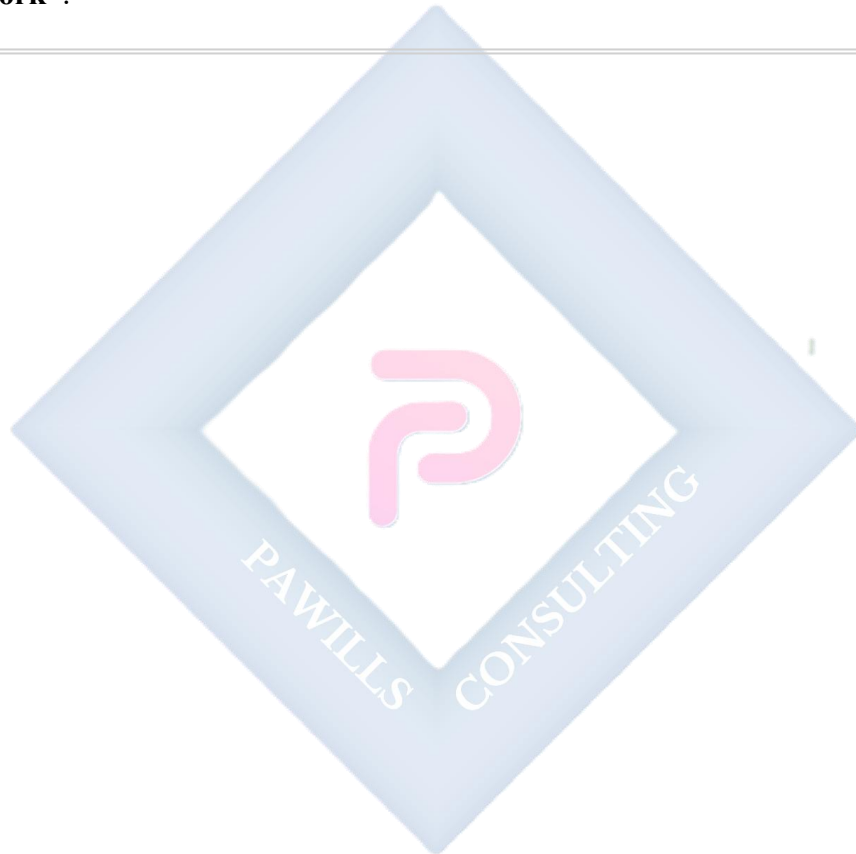
EVALUATION

1. List five common drawing instruments and state their uses.
2. Mention three qualities of good lettering.
3. Differentiate between orthographic and isometric drawings.

4. What are the two main types of projection used in technical drawing?
 5. Draw and label five different line types and state their uses.
 6. Explain how a perpendicular bisector of a line can be constructed.
 7. State two advantages of proper instrument maintenance.
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ASSIGNMENT

1. Draw five different line types, label them, and state one use for each.
 2. Construct a regular hexagon with each side measuring 40 mm.
 3. Prepare a neat, properly lettered title block for your drawing sheet titled **“Revision of SS1 Work”**.
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WEEK 2

TOPIC: GEOMETRICAL CONSTRUCTION

CONTENT

1. Meaning of Geometrical Construction

Geometrical construction is the **method of drawing geometrical figures accurately using only drawing instruments such as a compass, straightedge (ruler), and set squares**—without measuring directly with a protractor or scale (except when necessary).

It involves **constructing shapes, angles, tangents, bisectors, and other figures** by applying geometric principles.

Geometrical construction is a **fundamental skill in technical drawing**, architecture, and engineering design. It ensures **accuracy, precision, and clarity** in technical communication.

2. Importance of Geometrical Construction

- It forms the **foundation of technical and engineering drawing**.
 - It teaches accuracy, neatness, and discipline in drawing.
 - It helps in **designing mechanical and architectural components**.
 - It aids in the understanding of **geometric properties** and relationships.
 - It helps in developing **spatial reasoning and visualization** skills.
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3. Drawing Instruments Required

To perform geometrical constructions, the following instruments are essential:

- **Drawing board** – for securing paper.
 - **T-square** – for drawing horizontal lines.
 - **Set squares (45° and 60°)** – for vertical and inclined lines.
 - **Compass** – for drawing circles and arcs.
 - **Dividers** – for transferring measurements.
 - **Ruler (straightedge)** – for drawing straight lines.
 - **Pencil (2H, H)** – for light and dark lines.
 - **Eraser** – for correcting mistakes.
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4. Types of Geometrical Constructions

Geometrical constructions can be grouped into the following categories:

1. Basic Constructions

- Bisecting lines and angles
- Drawing perpendicular and parallel lines
- Dividing lines into equal parts

2. Polygon Constructions

- Constructing regular polygons (triangle, square, pentagon, hexagon, etc.)

3. Circle-Related Constructions

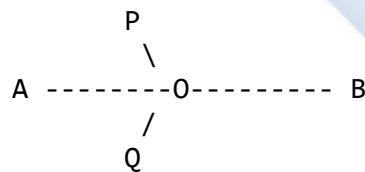
- Tangents to circles
- Arcs and circles passing through given points
- Inscribing and circumscribing circles about polygons

5. Basic Constructions and Illustrations

(a) Bisecting a Line Segment

To bisect line AB:

1. Draw a line AB.
2. With **A** as centre, draw an arc above and below AB using a radius more than half of AB.
3. With **B** as centre and the same radius, draw arcs cutting the previous arcs at points **P** and **Q**.
4. Join **PQ**. PQ bisects AB at **O**, the midpoint.



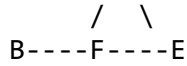
Result: Line AB is divided into two equal parts at O.

(b) Bisecting an Angle

To bisect $\angle ABC$:

1. Draw angle ABC.
2. With B as centre, draw an arc to cut both arms at **D** and **E**.
3. With **D** and **E** as centres and a convenient radius, draw two arcs to intersect at **F**.
4. Draw line **BF** — it bisects the angle $\angle ABC$.

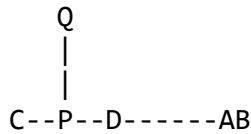




(c) Drawing a Perpendicular from a Point on a Line

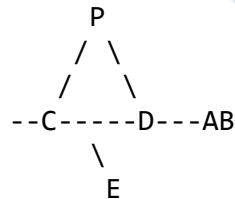
To draw a perpendicular from a point P on line AB:

1. Mark a point **P** on line **AB**.
2. With **P** as centre, draw an arc to cut **AB** at **C** and **D**.
3. With **C** and **D** as centres, and radius $>$ half **CD**, draw arcs to intersect at **Q** above **AB**.
4. Draw line **PQ** — it is perpendicular to **AB** at **P**.



(d) Drawing a Perpendicular from a Point Outside a Line

1. Draw line **AB** and mark point **P** above it.
2. With **P** as centre, draw an arc to cut **AB** at **C** and **D**.
3. With **C** and **D** as centres and equal radii, draw arcs intersecting at **E**.
4. Draw line **PE** — it is perpendicular to **AB**.



(e) Drawing a Line Parallel to Another Line

1. Draw a line **AB**.
2. At any point **C** above **AB**, draw a transversal line cutting **AB** at **D**.
3. With compass, measure the angle between the transversal and **AB**.
4. Reproduce the same angle at **C** using the compass arc method.
5. Draw a line through **C** — this line is parallel to **AB**.

6. Construction of Regular Polygons

Regular polygons are plane figures with **equal sides and equal angles**.

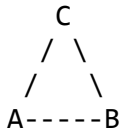
Examples:

- Triangle (3 sides)

- Square (4 sides)
- Pentagon (5 sides)
- Hexagon (6 sides)
- Octagon (8 sides)

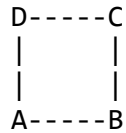
(a) Construction of an Equilateral Triangle (given one side)

1. Draw a straight line AB (say 50 mm).
2. With A as centre, draw an arc with radius AB.
3. With B as centre, draw another arc cutting the first at C.
4. Join AC and BC. $\therefore \triangle ABC$ is equilateral.



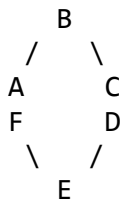
(b) Construction of a Square (given one side)

1. Draw line AB (say 50 mm).
2. At A and B, erect perpendiculars using set squares.
3. From A and B, mark equal lengths AD and BC equal to AB.
4. Join CD to complete square ABCD.



(c) Construction of a Regular Hexagon (given one side)

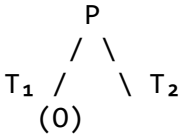
1. Draw a circle with radius equal to the side of the hexagon.
2. Mark a point A on the circumference.
3. With the same radius, step off the radius around the circle six times to get points A, B, C, D, E, F.
4. Join all points consecutively to form hexagon ABCDEF.



7. Constructions Involving Circles

(a) Tangent to a Circle from an External Point

1. Draw a circle with centre **O** and radius r .
2. Mark an external point **P**.
3. Join **OP**.
4. Bisect **OP** at **M**.
5. With **M** as centre and radius OM , draw a circle cutting the first at **T₁** and **T₂**.
6. Join **PT₁** and **PT₂**. These are tangents from **P** to the circle.



(b) Inscribing and Circumscribing a Circle

- **To inscribe a circle:** Draw a circle inside a given polygon such that it touches all sides internally.
- **To circumscribe a circle:** Draw a circle around a given polygon such that it passes through all vertices.

8. Practical Applications

- Engineering design (machine components, gears, etc.)
- Architectural plans and layouts
- Structural drawings (frames, beams, supports)
- Geometric design and pattern creation

EVALUATION

1. Define geometrical construction.
2. State four instruments used for geometrical construction.
3. Describe how to bisect a given angle.
4. Explain the steps involved in constructing an equilateral triangle.
5. What is the difference between inscribed and circumscribed circles?
6. Draw and label a perpendicular line from a given point on a line.
7. Construct a regular hexagon with each side measuring 40 mm.

ASSIGNMENT

1. Using your drawing instruments, construct: a. A line $AB = 60$ mm and bisect it. b. An angle of 90° using only compass and straightedge.

2. Construct a regular pentagon given one side = 40 mm.
 3. Draw tangents to a circle of radius 25 mm from an external point 50 mm from its centre.
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WEEK 3

TOPIC: GEOMETRICAL CONSTRUCTION (CONTINUED)

CONTENT

1. Introduction
 2. Construction of Triangles
 3. Construction of Quadrilaterals
 4. Construction of Regular Polygons
 5. Bisection of Lines and Angles
 6. Application of Geometrical Construction in Engineering and Design
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1. INTRODUCTION

Geometrical construction continues from the previous lesson but focuses on **practical shapes** used extensively in **engineering, building design, and architecture** — namely **triangles, quadrilaterals, and polygons**. In all constructions, **accuracy, neatness, and use of proper instruments** (compass, straightedge, set square, and protractor where necessary) are essential.

2. CONSTRUCTION OF TRIANGLES

Triangles are fundamental geometric shapes used in structural and mechanical design because of their **stability and rigidity**.

Types of Triangles (by sides)

- **Equilateral Triangle:** All sides and angles are equal (each angle = 60°).
 - **Isosceles Triangle:** Two sides and two angles are equal.
 - **Scalene Triangle:** All sides and angles are unequal.
 - **Right-angled Triangle:** One angle is 90° .
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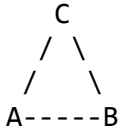
(a) Construction of an Equilateral Triangle (given one side)

Steps:

1. Draw a line AB (say 60 mm).
2. With A as centre and radius AB, draw an arc.
3. With B as centre and radius BA, draw another arc cutting the first at C.

4. Join AC and BC.

Result: Triangle ABC is equilateral.



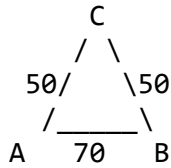
(b) Construction of an Isosceles Triangle (given base and sides)

Example: Base AB = 70 mm; equal sides = 50 mm.

Steps:

1. Draw base AB = 70 mm.
2. With A as centre, radius 50 mm, draw an arc.
3. With B as centre, radius 50 mm, draw another arc to cut the first at C.
4. Join AC and BC.

Result: $\triangle ABC$ is isosceles.



(c) Construction of a Scalene Triangle (given three sides)

Example: AB = 60 mm, BC = 50 mm, AC = 40 mm.

Steps:

1. Draw base AB = 60 mm.
2. With A as centre, radius 40 mm, draw an arc.
3. With B as centre, radius 50 mm, draw another arc cutting the first at C.
4. Join AC and BC.

Result: $\triangle ABC$ is scalene.

(d) Construction of a Right-Angled Triangle (given base and height)

Example: Base AB = 60 mm, height = 40 mm.

Steps:

1. Draw line $AB = 60$ mm.
2. At A, draw a perpendicular using set square.
3. Mark $AD = 40$ mm on the perpendicular.
4. Join BD. **Result:** $\triangle ABD$ is right-angled at A.



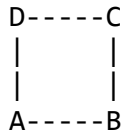
3. CONSTRUCTION OF QUADRILATERALS

Quadrilaterals are four-sided plane figures widely used in civil and mechanical design. They include **square, rectangle, rhombus, parallelogram, trapezium, and kite**.

(a) Square (given one side)

Steps:

1. Draw base $AB = 50$ mm.
2. At A and B, erect perpendiculars using a set square.
3. Mark AD and $BC = 50$ mm.
4. Join CD.



(b) Rectangle (given length and breadth)

Example: $AB = 80$ mm, $AD = 50$ mm.

Steps:

1. Draw base $AB = 80$ mm.
2. Erect perpendiculars at A and B.
3. Mark $AD = 50$ mm, $BC = 50$ mm.
4. Join CD to complete rectangle.

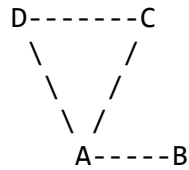


(c) Parallelogram (given sides and included angle)

Example: $AB = 80 \text{ mm}$, $AD = 50 \text{ mm}$, $\angle BAD = 60^\circ$.

Steps:

1. Draw $AB = 80 \text{ mm}$.
2. At A, draw $\angle BAD = 60^\circ$.
3. On AD, mark 50 mm .
4. From D, draw a line parallel to AB.
5. From B, draw a line parallel to AD to meet the first line at C.



(d) Rhombus (given one side and one angle)

Example: Side = 60 mm , $\angle A = 70^\circ$.

Steps:

1. Draw $AB = 60 \text{ mm}$.
2. At A, draw $\angle BAD = 70^\circ$.
3. On AD, mark 60 mm .
4. Complete rhombus by parallel lines.

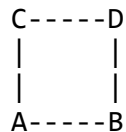
All sides are equal but angles differ.

(e) Trapezium (one pair of opposite sides parallel)

Example: $AB = 80 \text{ mm}$, $CD = 50 \text{ mm}$, height = 40 mm .

Steps:

1. Draw $AB = 80 \text{ mm}$ (base).
2. Draw a perpendicular from A and B.
3. Mark height = 40 mm .
4. Join C and D to form top side.



(f) Kite

Properties: Two pairs of adjacent sides are equal.

Steps:

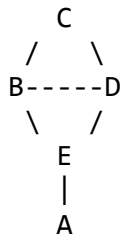
1. Draw a diagonal (say $AC = 70$ mm).
 2. From A and C, construct arcs to locate B and D such that $AB = AD$, and $CB = CD$.
 3. Join A–B–C–D–A.
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4. CONSTRUCTION OF REGULAR POLYGONS

Regular polygons are plane figures with **equal sides and angles**. They are fundamental in tiling, gear design, and architectural forms.

(a) Regular Pentagon (given one side)

1. Draw side $AB = 40$ mm.
2. With A as centre, draw an arc of radius AB.
3. With B as centre, draw another arc cutting at O.
4. Draw a circle with centre O and radius OA.
5. With compass, step off the radius around the circle five times to locate points C, D, E.
6. Join A–B–C–D–E–A.



(b) Regular Hexagon (using circle method)

1. Draw a circle with given radius (say 30 mm).
 2. Choose a point A on the circle.
 3. Step off the radius six times around the circumference to get six points.
 4. Join points consecutively to form hexagon.
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(c) Regular Octagon (using square method)

1. Draw a square of given side.
2. Draw diagonals to find the centre.
3. At each corner, draw 45° lines cutting diagonals.

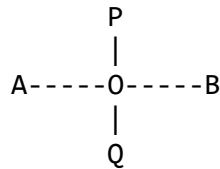
4. Connect the new intersection points to form an octagon.

5. BISECTION OF LINES AND ANGLES

(a) Bisecting a Line

Steps:

1. Draw a line AB.
2. With A as centre, radius $> \frac{1}{2} AB$, draw arcs above and below.
3. With B as centre, same radius, draw arcs to intersect at P and Q.
4. Join PQ. PQ bisects AB at O.



(b) Bisecting an Angle

Steps:

1. Draw $\angle ABC$.
2. With B as centre, draw an arc cutting BA and BC at D and E.
3. With D and E as centres, draw arcs to meet at F.
4. Draw BF to bisect the angle.

6. APPLICATION OF GEOMETRICAL CONSTRUCTION IN ENGINEERING AND DESIGN

- **Civil Engineering:** Design of beams, columns, roofs, and arches.
- **Mechanical Engineering:** Construction of gears, cams, and linkages.
- **Architecture:** Planning rooms, windows, and decorative patterns.
- **Technical Illustration:** Pictorial and orthographic drawings depend on accurate geometry.
- **Graphic Design and Pattern Making:** In textiles, logos, and ornamental designs.

EVALUATION

1. Define geometrical construction.
2. Draw and label three types of triangles and state their properties.
3. Construct a rhombus of side 60 mm and one angle of 70° .

4. Distinguish between a parallelogram and a rectangle.
 5. Construct a regular hexagon given one side = 30 mm.
 6. State the steps to bisect an angle.
 7. Explain two practical applications of geometrical construction in engineering.
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ASSIGNMENT

1. Using your drawing instruments, construct: a. An isosceles triangle (base = 70 mm, equal sides = 50 mm). b. A trapezium with parallel sides 80 mm and 50 mm, height 40 mm.
 2. Construct a regular pentagon with each side = 35 mm.
 3. Bisect a line 60 mm long and then bisect one of the angles formed by the bisector with the line.
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WEEK 4

TOPIC: GEOMETRICAL CONSTRUCTION (CONTINUED)

SUB-TOPICS

1. Division of Lines into Equal Parts
 2. Construction of Tangents to Circles
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CONTENT

1. INTRODUCTION

Geometrical construction forms the foundation of **engineering**, **architectural**, and **technical drawing**. Every accurate shape or figure—whether polygonal, circular, or irregular—relies on geometric principles for correctness. In this lesson, we will study two important construction techniques:

- **Division of lines into equal parts, and**
- **Construction of tangents to circles.**

Both are indispensable in **design**, **pattern development**, and **machine drawing**, ensuring accuracy, proportionality, and visual balance.

2. DIVISION OF A STRAIGHT LINE INTO EQUAL PARTS

Meaning

Dividing a straight line into equal parts means splitting it into smaller, congruent segments using geometrical methods (not direct measurement with a ruler). This method ensures **mathematical precision**, especially in scaling or drawing parts proportionally.

Instruments Required

- Drawing board and T-square
 - Set squares (30°, 45°, 60°)
 - Compass and divider
 - Scale rule
 - Drawing pencil (2H, H, HB)
 - Eraser and masking tape
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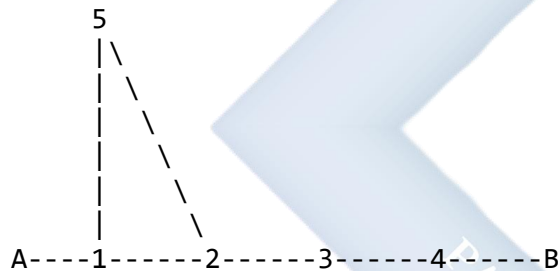
Step-by-Step Procedure

To divide a straight line **AB** into, for example, **5 equal parts**:

1. **Draw the given line AB.** Let $AB = 100$ mm (or any given length).
2. **At point A, draw an inclined line AC** at about 30° – 45° to AB using a set square.
3. **On line AC, mark off five equal segments** using a **divider** or **compass**. Label them as **1, 2, 3, 4, and 5** starting from A.
4. **Join point 5 to B.** This line (5B) acts as the reference for parallel divisions.
5. **Draw lines through points 1, 2, 3, and 4 on AC, each parallel to line 5B,** using a set square or compass.
6. **Label the points of intersection on AB as P_1 , P_2 , P_3 , and P_4 .**

Now, **AB** is divided into **five equal parts**.

Illustration (Text Diagram)



The lines drawn from points 1–4 to meet AB divide it into **five equal parts**.

Geometrical Principle Behind the Method

All triangles drawn between parallel lines having the same base and height are **equal in area**. Hence, parallel lines drawn through equal divisions on the inclined line create equal segments on the main line.

Applications

The division of lines into equal parts is useful in:

- Creating **equal spacing** for rivets, bolts, or beams.
- Designing **staircases, columns, and panels**.
- Dividing scales and arcs into equal divisions.
- Constructing **gears, threads, and machine parts**.

- Dividing **maps, patterns, or diagrams** into equal proportions.

Alternative Method (Using Compass Only)

To divide a line AB into **n equal parts**:

1. With A as center, draw an arc intersecting AB.
2. With the same radius, continue marking arcs along AB.
3. If slight errors occur, bisect the last small segment geometrically. This is best used for small-scale quick constructions.

3. CONSTRUCTION OF TANGENTS TO CIRCLES

Meaning

A **tangent** is a straight line that touches a circle at **exactly one point**, called the **point of tangency**. It does not pass through the circle. The **radius** at the point of tangency is always **perpendicular** to the tangent.

Types of Tangents

1. Tangent at a point on the circle
2. Tangents from a point outside the circle
3. Common tangents to two circles
 - *External tangents*
 - *Internal tangents*

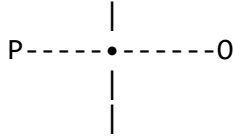
A. Tangent at a Point on a Circle

Steps:

1. Draw a circle with center **O** and radius **r**.
2. Choose a point **P** on the circumference.
3. Draw the radius **OP**.
4. Construct a **perpendicular** to **OP** at point **P** using a set square or compass.
5. The line through **P** is the **tangent** to the circle.

Illustration:

Tangent
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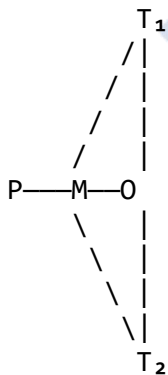
The tangent touches the circle at **P** and is **perpendicular to the radius OP**.

B. Tangents from a Point Outside the Circle

Steps:

1. Draw a circle with center **O** and radius **r**.
2. Mark an external point **P** outside the circle.
3. Join **OP**.
4. Bisect **OP** to find the midpoint **M**.
5. With **M** as center and **OM** as radius, draw a new circle.
6. The new circle cuts the original one at points **T₁** and **T₂**.
7. Join **P–T₁** and **P–T₂**.
8. These are the **two tangents** to the circle from point **P**.

Illustration:



C. Common Tangents to Two Circles

Types:

1. **External Tangents:** Do not intersect the line joining the centers.
2. **Internal Tangents:** Cross the line joining the centers.

Construction (External Tangent):

1. Draw two circles with centers **O₁** and **O₂**.
2. Draw the line joining the centers.
3. Reduce one circle's radius by the difference between the two radii.

4. Draw a tangent from the external point (as in the previous construction).
5. Restore the true radius and transfer tangent points.
6. Draw the two external tangents touching both circles.

Geometrical Principle

If **OP** is the radius and **PT** is the tangent, then ($OP \perp PT$).

This principle is the basis of tangent constructions and is used in determining **touching points** in mechanical motion, such as gears, pulleys, and cams.

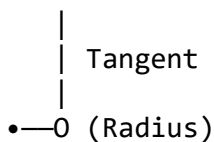
Practical Applications

Tangents are used in:

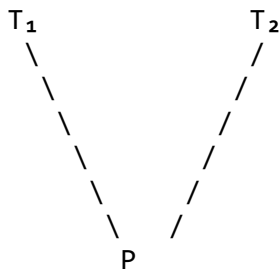
- **Engineering Design:** Contact between gears, belts, and pulleys.
- **Architectural Drafting:** Smooth curve-to-wall transitions.
- **Civil Engineering:** Designing curved road paths and roundabouts.
- **Mechanical Drawing:** Designing cams, flywheels, and rollers.
- **Art and Graphics:** Smooth blending between straight and curved surfaces.

Illustrations (Text-Based Summary)

Tangent at a point:



Tangents from external point:



4. RELATIONSHIP BETWEEN RADIUS AND TANGENT

- The **tangent** to a circle is always **perpendicular to the radius** at the point of contact. Mathematically: [$OP \perp PT$]
- Tangents drawn from the same external point are **equal in length**.

$$[PT_1 = PT_2]$$

EVALUATION

1. Define the term *tangent to a circle*.
 2. State the relationship between a tangent and the radius at the point of tangency.
 3. Explain step-by-step how to divide a straight line into six equal parts.
 4. What are the two main types of tangents that can be drawn to two circles?
 5. List three applications of tangent constructions in engineering or architecture.
 6. Describe how to construct tangents from an external point to a given circle.
 7. Why is geometric division preferred to direct measurement for accuracy?
-

ASSIGNMENT

1. Construct a straight line **AB = 150 mm**, and divide it into **seven equal parts**.
 2. Draw a **circle of radius 25 mm**, and construct **two tangents** from a point 50 mm outside the circle.
 3. Construct **two circles** of radii **30 mm** and **20 mm**, their centers 90 mm apart. Draw **two common external tangents** to touch both circles.
-

WEEK 6

TOPIC:

SIMPLE SCALE DRAWING – PLAIN AND DIAGONAL SCALES

CONTENT

1. INTRODUCTION

In Technical Drawing, it is often impossible to draw an object in its actual size because it may be **too large** (like a building) or **too small** (like a screw). A **scale drawing** is therefore a **representation of an object in proportionate but reduced or enlarged size**.

A **scale** is the **ratio between the dimension of the drawing and the actual dimension** of the object.

$$[\text{Scale} = \frac{\text{Drawing size}}{\text{Actual size}}]$$

2. TYPES OF SCALES

1. **Full Scale (1:1)** – Drawing size equals the real size. [example: Drawing a 50 mm rod as 50 mm on paper.]
 2. **Reduced Scale (1:n)** – The drawing is smaller than the actual object. [example: 1:100 for a building plan (1 cm = 100 cm).]
 3. **Enlarged Scale (n:1)** – The drawing is larger than the real object. [example: 10:1 for small tools or machine parts.]
-

3. REPRESENTATIVE FRACTION (R.F.)

The **Representative Fraction** expresses the ratio between the length on the drawing and the actual length using the same units.

$$[\text{R.F.} = \frac{\text{Length on Drawing}}{\text{Actual Length}}]$$

Example: If 1 cm on a drawing represents 2 m (200 cm) in reality, $[\text{R.F.} = \frac{1}{200}]$

4. PURPOSE OF SCALE DRAWING

- To represent **large objects** on a manageable drawing sheet.
- To **enlarge** small components for better detail.
- To ensure **uniform proportion** and **accuracy** in all measurements.

- To allow for **quick reading** of measurements during construction or fabrication.
- To promote **standardization** in engineering and architectural documentation.

5. PLAIN SCALE

Definition

A **Plain Scale** is a line divided into equal parts to measure or read **two units** (usually a main unit and a subunit), such as **metres and decimetres** or **kilometres and hectometres**.

Parts of a Plain Scale

1. The **main divisions** – represent major units.
2. The **subdivisions** – represent smaller units.
3. The **zero point** – often lies between the main and sub-divisions.
4. **Labeling** – each major and minor division is marked with its corresponding value.

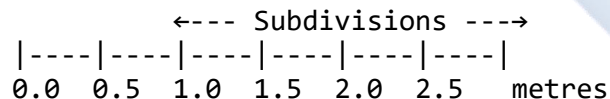
Formula for the Length of a Scale

$$[L = R.F. \times \text{Actual Length to be Represented}]$$

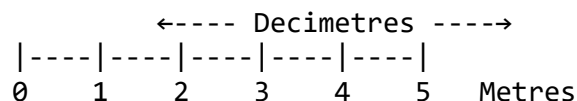
Steps for Constructing a Plain Scale

1. Draw a straight **base line (AB)** of calculated length.
2. Divide it into **equal main parts** representing the major units.
3. Subdivide the **first division on the left** into ten smaller parts.
4. Number the main divisions (1, 2, 3, ...) to the right and sub-divisions (1, 2, 3, ...) to the left.
5. Label with appropriate **units** (e.g., metres, centimetres).

ILLUSTRATION 1: PLAIN SCALE



Or, more detailed:



(Each small division on the left = 0.1 m or 10 cm)

Example:

Construct a plain scale of R.F. = 1/50 to read up to 5 metres.

Solution: $L = \frac{1}{50} \times 500 \text{ cm} = 10 \text{ cm}$. So, draw a 10 cm long line and divide it as above.

6. DIAGONAL SCALE

Definition

A **Diagonal Scale** is used for **more precise readings**, enabling measurement up to **three units** (e.g., metres, decimetres, and centimetres).

It works on the **principle of similar triangles**—using diagonals to subdivide divisions into smaller, measurable parts.

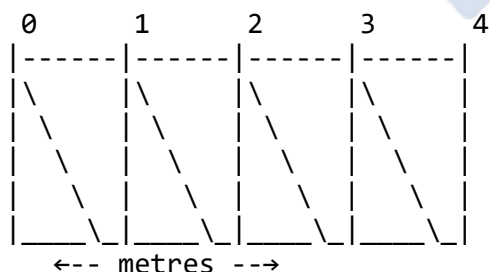
Principle

In a diagonal scale, the diagonal line divides small squares into smaller parts that are **proportional**, allowing for finer readings (e.g., 0.1 cm or 0.01 m).

Steps for Constructing a Diagonal Scale

1. Draw a **plain scale** as the base.
2. Erect a **perpendicular line (AC)** at the zero point.
3. Mark equal **vertical divisions** on AC (e.g., 10 equal parts).
4. From the top of the first vertical, draw a line **parallel to AB**.
5. Draw **diagonals** from each top division to the next lower intersection on the next vertical.
6. The intersection points of diagonals with vertical lines show **subdivided measurements**.

ILLUSTRATION 2: DIAGONAL SCALE



Each diagonal intersection allows reading **one-tenth** of the smallest division, e.g., 0.01 m.

Worked Example

Construct a **Diagonal Scale** to read metres, decimetres, and centimetres with R.F. = $1/50$, to measure up to 5 metres.

Solution:

1. ($L = \frac{1}{50} \times 500, \text{cm} = 10, \text{cm}$)
2. Draw a 10 cm line (AB).
3. Divide into 5 equal parts (each = 1 m).
4. Subdivide the first division on the left into 10 (each = 0.1 m).
5. Erect perpendiculars of 2 cm height.
6. Subdivide vertically into 10 equal parts and draw diagonals.

This allows readings up to **centimetres** (0.01 m).

7. DIFFERENCES BETWEEN PLAIN AND DIAGONAL SCALES

Feature	Plain Scale	Diagonal Scale
Units measured	Two (e.g., m and cm)	Three (e.g., m, dm, cm)
Accuracy	Moderate	High
Construction	Simple	Complex
Principle	Linear division	Similar triangles
Application	General drawing	Precision engineering, mapping

8. PRACTICAL APPLICATIONS OF SCALES

- **Architecture:** Building plans and site layouts.
- **Mechanical engineering:** Machine parts and fittings.
- **Civil engineering:** Bridges, roads, and drainage plans.
- **Surveying and mapping:** Land area representation.
- **Town planning:** Layouts of cities and estates.

EVALUATION

1. Define a scale and give three types of scales.
 2. What is the Representative Fraction (R.F.) and how is it used?
 3. State three differences between plain and diagonal scales.
 4. Explain the principle of the diagonal scale.
 5. List three uses of scale drawing in technical work.
 6. Using R.F. = $1/100$, construct a plain scale to represent 5 metres.
-

ASSIGNMENT

1. Construct a **Plain Scale** to measure up to **6 m** using R.F. = $1/50$. Show 4.8 m on the scale.
 2. Construct a **Diagonal Scale** to read up to **5 m**, showing metres, decimetres, and centimetres (R.F. = $1/100$).
 3. Explain how similar triangles help in obtaining smaller subdivisions on a diagonal scale.
-



WEEK 8

TOPIC:

PICTORIAL DRAWINGS – OBLIQUE PROJECTION

CONTENT

1. INTRODUCTION

In Technical Drawing, pictorial drawings are used to **represent three-dimensional (3D) objects on a two-dimensional (2D) surface** such as paper. They help the viewer **visualize** the object as it would appear in reality.

There are three major types of pictorial drawings:

1. **Isometric Projection**
2. **Oblique Projection**
3. **Perspective Projection**

In this lesson, we focus on **Oblique Projection** — its meaning, principles, types, and method of construction.

2. DEFINITION OF OBLIQUE PROJECTION

Oblique Projection is a type of pictorial drawing in which the **front face of the object is drawn in true shape**, while the other faces are drawn at an **angle (usually 45°)** to the horizontal line and are **reduced in scale** to give a sense of depth.

In oblique projection:

- The **front face** is **parallel** to the picture plane and drawn in **true dimensions**.
 - The **receding (side) faces** are **inclined** to the picture plane and drawn at an **angle (typically 30°, 45°, or 60°)** to represent depth.
 - The **depth dimension** is sometimes **foreshortened** to make the object look more realistic.
-

3. CHARACTERISTICS OF OBLIQUE PROJECTION

1. The **front face** of the object is shown in **true size and shape**.
2. The **receding edges** (depth lines) are drawn **obliquely** (usually at **45°**, but can also be **30° or 60°**).
3. The **depth (receding dimension)** may be drawn at **full scale (Cavalier)** or **half scale (Cabinet)**.

4. The **projectors** (lines of projection) are drawn **oblique** to the picture plane.
5. It shows the object in a **more realistic 3D form** than orthographic projection, though slightly distorted.

4. TYPES OF OBLIQUE PROJECTION

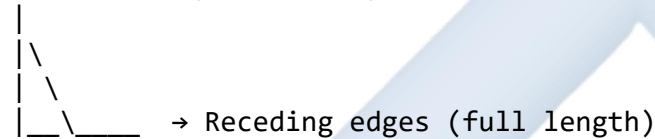
There are **two main types** of oblique projection:

A. Cavalier Oblique Projection

- The **receding edges** are drawn **at full length** (true length).
- The angle of projection is **usually 45°**.
- The result appears **distorted** because the depth is exaggerated.

Example: A cube drawn with all edges at full length, receding at 45°.

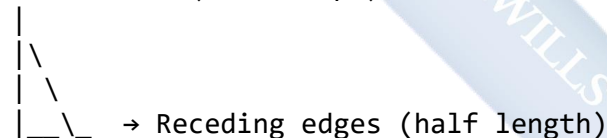
Front Face (True Shape)



B. Cabinet Oblique Projection

- The **receding edges** are drawn **half the true length**.
- The angle of projection is also **usually 45°**.
- The object appears **more natural and realistic** than in the cavalier method.

Front Face (True Shape)



Note: Cabinet Oblique is the **preferred method** in technical practice because it gives a better visual appearance.

5. COMPARISON BETWEEN CAVALIER AND CABINET OBLIQUE PROJECTION

Feature	Cavalier Oblique	Cabinet Oblique
Scale on receding axis	Full scale (1:1)	Half scale (1:2)
Appearance	Exaggerated depth	Realistic depth
Common usage	Mechanical drawings	Furniture and architectural drawings
Accuracy	Less realistic	More realistic

Feature	Cavalier Oblique	Cabinet Oblique
Receding angle	45°, sometimes 30° or 60°	45°, sometimes 30°

6. CONSTRUCTION OF OBLIQUE PROJECTION

Instruments Required

- Drawing board
- T-square
- 45° and 30°/60° set squares
- Ruler or scale
- Pencil and eraser
- Compass (if arcs/circles are needed)

STEP-BY-STEP PROCEDURE (Example: Oblique Drawing of a Rectangular Block)

Given Dimensions: Length = 100 mm Height = 60 mm Width = 50 mm

STEP 1: Draw the Front View (True Shape)

1. Draw a **horizontal baseline (XY)**.
2. On this line, mark **length = 100 mm (AB)**.
3. At A and B, erect **perpendiculars** 60 mm high to represent height.
4. Complete the **rectangle ABCD** — this is the **front face**, drawn in **true size**.

Front Face:

A(0,0), B(100,0), C(100,60), D(0,60)

STEP 2: Draw the Receding Axis

From points A and D, draw lines at **45°** to the horizontal (for the depth).

STEP 3: Mark the Depth

On each 45° line, measure **50 mm** (width). If using **Cabinet Oblique**, measure **25 mm** instead (half of 50 mm).

Label these points as E (from A) and H (from D).

STEP 4: Complete the Top and Side Faces

1. From points E and H, draw verticals (same height as front face).

2. Join the upper ends to complete the **top face** (E–F–G–H).
3. Draw the **visible edges** with thick lines and **hidden ones** (if needed) with dashed lines.

ILLUSTRATION 1: OBLIQUE PROJECTION OF A RECTANGULAR BLOCK

Front View (True Shape)

A(0,0)-----B(100,0)

|

|

D(0,60)-----C(100,60)

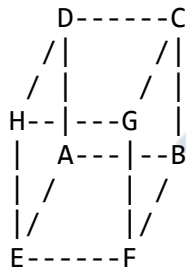
Receding edges at 45°:

A–E and D–H = 50 mm (or 25 mm for Cabinet)

Top face:

E–F–G–H connects at same height as C–D

Visual (Text Diagram):



7. OBLIQUE PROJECTION OF CIRCULAR OBJECTS

When a **circle** appears on an **oblique surface**, it appears as an **ellipse**.

Steps:

1. Draw the true circle on the **front face** or **top view**.
2. On the oblique plane, construct a **square** enclosing the circle.
3. Transfer all points of tangency (top, bottom, left, right) to the oblique surface.
4. Connect them smoothly to form an **ellipse** (using the oblong method).

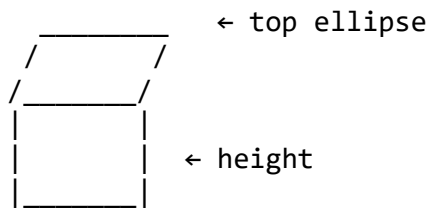
ILLUSTRATION 2: OBLIQUE PROJECTION OF A CYLINDER

Given: Diameter = 50 mm, Height = 70 mm.

Steps:

1. Draw the front circle (true size).

2. From its diameter ends, draw receding lines at 45° to represent depth.
3. Draw the **top ellipse** using transferred points.
4. Join the visible sides with **vertical lines**.



8. ADVANTAGES OF OBLIQUE PROJECTION

1. Easy to construct — only one face is drawn in true shape.
2. The front face retains true dimensions.
3. Simple to understand visually.
4. Easier to measure directly from the front face.
5. Useful for quick sketches and presentations.

9. DISADVANTAGES

1. Receding faces appear distorted (especially in Cavalier type).
2. Less realistic than isometric or perspective projections.
3. Circles on receding planes must be drawn as ellipses — more complex.
4. Not suitable for engineering working drawings.

10. PRACTICAL APPLICATIONS OF OBLIQUE DRAWINGS

- Used in **furniture design** (Cabinet Oblique).
- Used in **woodwork and carpentry** for visual presentation.
- Used in **architectural sketches** for early visualization.
- Used in **technical illustrations** and **assembly manuals**.
- Used in **engineering sketches** for quick communication.

EVALUATION

1. Define Oblique Projection.
 2. List two types of oblique projection and state their differences.
 3. State three characteristics of oblique projection.
 4. Draw a rectangular block using the Cavalier Oblique method (given dimensions).
 5. Explain why the Cabinet Oblique projection is preferred for realistic drawings.
 6. Describe how a circle is represented in an oblique projection.
-

ASSIGNMENT

1. Draw a **Cavalier Oblique projection** of a cube with **40 mm sides**.
 2. Draw a **Cabinet Oblique projection** of a rectangular block $100\text{ mm} \times 60\text{ mm} \times 50\text{ mm}$.
 3. Construct the **oblique projection of a cylinder** (diameter 40 mm, height 60 mm).
-



WEEK 9

TOPIC:

PICTORIAL DRAWINGS – ISOMETRIC PROJECTION

CONTENT

1. INTRODUCTION

In Technical Drawing, **pictorial drawings** are used to **represent three-dimensional (3D) objects on a two-dimensional (2D) surface** such as paper. They allow a viewer to visualize the object as it appears in reality.

There are three major forms of pictorial drawing:

1. **Isometric Projection**
2. **Oblique Projection**
3. **Perspective Projection**

This lesson focuses on **Isometric Projection**, which is one of the most widely used in engineering, architectural, and design drawings.

2. DEFINITION OF ISOMETRIC PROJECTION

Isometric Projection is a form of pictorial drawing in which the **object is tilted** so that **all three dimensions (length, breadth, and height)** are shown in **one view**.

The projection is based on the **isometric axes**, which are **three lines inclined at 120°** to one another. In this type of projection:

- The three principal axes (X, Y, Z) represent the length, width, and height of the object.
 - The angles between these axes are equal (120° each).
 - The dimensions along these axes are drawn **at the same scale** but **foreshortened** by a constant factor (0.816).
-

3. MEANING OF "ISOMETRIC"

The word "**Isometric**" comes from Greek words:

- *Iso* = Equal
- *Metron* = Measure

Hence, "isometric" means **equal measure** — because all axes are scaled equally and at equal angles.

4. CHARACTERISTICS OF ISOMETRIC PROJECTION

1. The three isometric axes are **120° apart** from each other.
2. Each axis represents one of the three dimensions of the object (length, breadth, height).
3. All edges parallel to these axes are drawn **to the same scale**.
4. Lines parallel to any isometric axis are shown in **true length**.
5. Circles appear as **ellipses** in isometric projection.
6. Angles and shapes on inclined planes appear **distorted** because of the projection.

5. TYPES OF ISOMETRIC DRAWINGS

Type	Description	Remarks
Isometric Projection	Drawn using isometric scale (foreshortened dimensions).	More technically accurate.
Isometric Drawing	Drawn using full (true) dimensions.	More common in practice for simplicity.

6. DIFFERENCE BETWEEN ISOMETRIC PROJECTION AND ISOMETRIC DRAWING

Feature	Isometric Projection	Isometric Drawing
Scale	Reduced to 0.816 of true length (Isometric scale)	Full (true) length
Appearance	Smaller, more accurate	Slightly larger, less accurate
Use	Technical projections	General pictorial representation
Lines	Foreshortened	True size

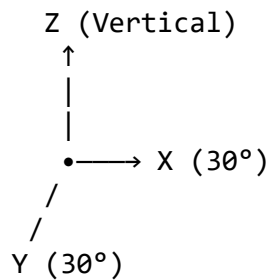
7. ISOMETRIC AXES AND PLANES

Isometric Axes

Three lines meeting at one point (the isometric centre) and equally inclined at 120° to one another:

- One vertical axis (height)
- Two at 30° to the horizontal (length and width)

Illustration 1: Isometric Axes Diagram



Isometric Planes

The isometric planes are imaginary planes formed by any two of the isometric axes:

1. **Horizontal Plane (X–Y)**
2. **Left Vertical Plane (Y–Z)**
3. **Right Vertical Plane (X–Z)**

These planes help locate features like holes, edges, and inclined surfaces.

8. ISOMETRIC SCALE

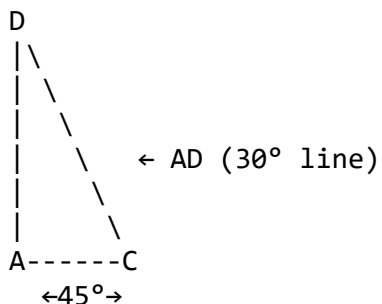
Because the isometric projection foreshortens the actual dimensions, an **isometric scale** is used to transfer the true lengths to the reduced (foreshortened) lengths.

Construction of Isometric Scale

Step-by-step:

1. Draw a **horizontal line AB**.
2. At A, draw a line **AC at 45°** and **AD at 30°**.
3. Mark true lengths along AC (10 mm, 20 mm, 30 mm, etc.).
4. From each division, draw **vertical lines** to meet line AD.
5. The distances measured on AD represent **isometric lengths** ($0.816 \times \text{true length}$).

Illustration 2: Isometric Scale Construction



Note: The isometric scale is mainly used in **isometric projection**, not in **isometric drawing**.

9. STEPS FOR CONSTRUCTING AN ISOMETRIC DRAWING

Example: Rectangular Block (100 mm × 60 mm × 40 mm)

Step 1: Draw the **isometric axes** — one vertical and two lines at 30° to the horizontal.

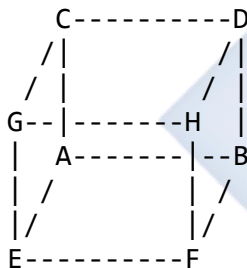
Step 2: From the isometric centre, mark:

- 100 mm along one 30° axis (length)
- 60 mm along the other 30° axis (width)
- 40 mm along the vertical axis (height)

Step 3: Complete the **isometric box** using parallel lines to the three axes.

Step 4: Use darker lines for visible edges and lighter dashed lines for hidden edges.

Illustration 3: Isometric Drawing of a Rectangular Block



(Axes at 120°; all sides parallel to isometric axes)

10. ISOMETRIC PROJECTION OF CIRCLES

A **circle** in isometric projection appears as an **ellipse**.

Steps to Draw a Circle in Isometric View (Oblong Method)

1. Draw the square that encloses the true circle.
 2. Draw the same square in isometric projection.
 3. Locate the midpoints of each side of the square.
 4. Join these midpoints diagonally to locate tangential points.
 5. Use a compass to draw four arcs that smoothly form an ellipse.
-

Illustration 4: Isometric Circle (Ellipse) Construction



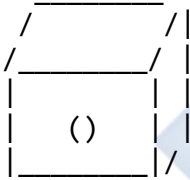
(Appears as an ellipse in the isometric view)

11. ISOMETRIC PROJECTION OF COMBINED SOLIDS

When two or more solids (e.g., cylinder on a block, cone on a cube) are combined:

1. Draw the **isometric box** for the base solid first.
2. Add the **top solid** in the correct position.
3. For circular solids (cylinders, cones), use the **ellipse method** for top and bottom.

Illustration 5: Cylinder on a Block



12. ADVANTAGES OF ISOMETRIC PROJECTION

1. Provides a **clear 3D view** of the object.
2. Shows all three dimensions (length, width, height) in one view.
3. **Measurements can be taken directly** along the isometric axes.
4. Easy to construct with simple geometrical tools.
5. Useful for **visual communication** between designers and manufacturers.

13. DISADVANTAGES

1. Cannot represent **true shapes** of inclined surfaces.
2. Circles appear as **ellipses** — more difficult to construct.
3. **Angles** other than 90° are **distorted**.
4. Not used for precise engineering production drawings.

14. APPLICATIONS OF ISOMETRIC DRAWINGS

- **Engineering and manufacturing:** machine parts and assemblies.

- **Architecture:** presentation of rooms, buildings, and components.
- **Furniture design:** visual representation of products.
- **Graphics and animation:** 3D modeling and design visualization.
- **Education:** teaching and learning visualization of solid geometry.

EVALUATION

1. Define **isometric projection**.
 2. Explain the meaning of “isometric axes.”
 3. Distinguish between **isometric drawing** and **isometric projection**.
 4. State the **angle between isometric axes**.
 5. Describe how circles appear in isometric drawings.
 6. Draw the **isometric projection of a cube** of 40 mm sides.
-

ASSIGNMENT

1. Construct an **isometric projection** of a rectangular block 80 mm × 60 mm × 40 mm.
 2. Using the **oblong method**, draw the **isometric view of a circle** of diameter 50 mm.
 3. Draw the **isometric projection of a cylinder** (base diameter 40 mm, height 70 mm).
-

DIAGRAM SUMMARY

Diagram	Description
Fig. 1: Isometric Axes	Three 120° axes showing X, Y, and Z directions.
Fig. 2: Isometric Scale	45° and 30° line showing true and isometric lengths.
Fig. 3: Isometric Block	3D rectangular solid with all edges parallel to axes.
Fig. 4: Isometric Circle	Ellipse representing a circular hole or cylinder top.
Fig. 5: Combination of Solids	Cylinder placed on a rectangular block (complex solid).

WEEK 10

TOPIC:

FREEHAND SKETCHING

CONTENT

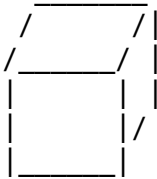
1. Meaning and importance of freehand sketching
 2. Materials and instruments for freehand sketching
 3. Basic principles and techniques of freehand sketching
 4. Steps in freehand sketching
 5. Proportion in sketching
 6. Types of freehand sketches
 7. Freehand sketching of simple objects
 8. Care and presentation of sketches
-

1. MEANING OF FREEHAND SKETCHING

Freehand sketching is the process of drawing objects **without the use of drawing instruments** such as rulers, compasses, or protractors. It is done **manually**, using only the hand and eye to represent an object's shape, proportion, and appearance.

In technical drawing, freehand sketching serves as a **quick visual tool** to communicate ideas before producing a detailed instrument drawing.

Illustration 1: Example of a Freehand Sketch of a Simple Object



(A freehand sketch of a rectangular block – drawn without instruments)

2. IMPORTANCE OF FREEHAND SKETCHING

1. **Communication Tool:** Helps engineers, designers, and students express ideas quickly and clearly.

2. **Speed:** Saves time — no need for instruments or long setups.
3. **Flexibility:** Useful in the field or during brainstorming sessions.
4. **Foundation for Design:** Serves as a preliminary step before detailed mechanical or architectural drawings.
5. **Visual Thinking:** Develops the ability to visualize and proportion objects accurately.
6. **Problem-Solving Aid:** Enables visualization of modifications, mechanisms, or design alternatives.

3. MATERIALS AND INSTRUMENTS USED IN FREEHAND SKETCHING

Although no measuring tools are used, **basic materials** are needed:

MATERIAL	USE
HB or 2B Pencil	For light sketching lines.
Eraser	For correcting mistakes.
Plain or Graph Paper	Surface for sketching.
Sharpener	Keeps pencil point fine.
Drawing Board (Optional)	Provides stable drawing support.

4. BASIC PRINCIPLES AND TECHNIQUES OF FREEHAND SKETCHING

To produce neat, proportionate, and legible sketches, the following techniques are applied:

(a) Good Posture and Grip

- Sit upright with the forearm supported.
- Hold the pencil **lightly but firmly** about 25–30 mm from the point.
- Move the pencil **freely from the wrist and elbow**, not the fingers alone.

(b) Line Quality

- Use **light lines** first for construction, then **darken** the outlines.
- Lines should be **uniform in thickness** and **smooth**, not shaky.

(c) Proportion

- Ensure the **parts of the object relate correctly** to one another in size.
- Avoid exaggeration of any dimension.

(d) Observation

- Observe the object carefully before sketching.
- Identify main shapes, axes, and relative dimensions.

(e) Simplicity

- Avoid unnecessary details.

- Focus on the essential shape and structure.

5. STEPS IN FREEHAND SKETCHING

Step 1: Observation

Study the object and note its general shape and main dimensions.

Step 2: Enclosing Box

Visualize or lightly sketch an **imaginary box** around the object (also called the “proportion box”).

Step 3: Light Construction Lines

Draw the main outlines **lightly**, indicating height, length, and width.

Step 4: Refine the Shape

Add minor features — edges, corners, holes, etc.

Step 5: Final Outlining

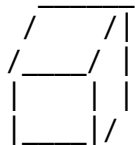
Darken the final lines neatly and erase unnecessary construction marks.

Illustration 2: Stages in Freehand Sketching of a Block

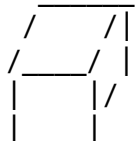
STAGE 1: Outline Box



STAGE 2: Add Details



STAGE 3: Final Outline



6. PROPORTION IN SKETCHING

Proportion is the correct relationship between the sizes of different parts of an object. It ensures that no part of the sketch looks too big or too small compared to others.

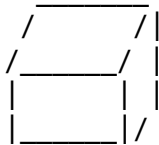
Example:

If a box is twice as long as it is high, the same ratio must appear in the sketch.

Actual Object	Proportion in Sketch
Length = 100 mm	Draw length twice the height
Height = 50 mm	Maintain ratio 2 : 1

Illustration 3: Proportion Comparison

Correct Proportion: Wrong Proportion:

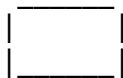


7. TYPES OF FREEHAND SKETCHES

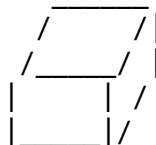
Type	Description	Example
Orthographic Sketch	Sketch showing different views (front, top, side) on flat planes.	Front and top view of a block.
Pictorial Sketch	Shows object in 3D form (Isometric or Oblique).	Cube or cylinder drawn in 3D.
Exploded Sketch	Parts of an assembly are separated to show how they fit together.	Exploded view of a nut and bolt.
Diagrammatic Sketch	Simplified sketch showing relationships or principles.	Circuit layout, linkage, etc.

Illustration 4: Types of Freehand Sketches

(a) Orthographic Views:

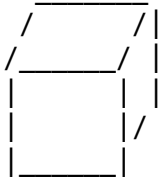


(b) Pictorial (3D) View:



8. FREEHAND SKETCHING OF SIMPLE OBJECTS

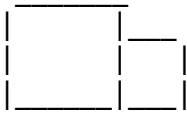
Example 1: Freehand Sketch of a Cube



Example 2: Freehand Sketch of a Cylinder



Example 3: Freehand Sketch of a Simple Machine Part



(Each sketch is approximate but proportionate and recognizable.)

9. TIPS FOR IMPROVING FREEHAND SKETCHING SKILL

1. Practice drawing **straight lines** without rulers.
2. Draw **boxes and geometric shapes** repeatedly to develop control.
3. Sketch **real-life objects** (e.g., books, cups, tools) in 3D.
4. Focus on **accuracy and proportion**, not perfection.
5. Keep a **sketchbook** for daily practice.

10. CARE AND PRESENTATION OF FREEHAND SKETCHES

- Keep sketches **clean and smudge-free**.
- Use **sharp pencils** for clarity.
- Label all parts neatly and legibly.
- Present sketches in **logical arrangement** — front, side, top views.
- Use **light guidelines** to align titles and annotations.

EVALUATION

1. Define **freehand sketching**.
2. List **five materials** required for freehand sketching.
3. Mention **four techniques** for producing good freehand sketches.

4. State **three advantages** of freehand sketching.
5. Explain the meaning of **proportion** in freehand sketching.
6. Draw a **freehand pictorial sketch** of a cube or cylinder.

ASSIGNMENT

1. Sketch in freehand the **orthographic views** (front, top, side) of a rectangular block of 80 mm × 50 mm × 30 mm.
2. Draw in freehand a **pictorial sketch** of a mug or cup.
3. Produce a **freehand sketch of a simple mechanical part**, showing proper proportion and line quality.

DIAGRAM SUMMARY

Figure No.	Description
Fig. 1:	Example of a freehand sketch (block)
Fig. 2:	Stages in freehand sketching
Fig. 3:	Proportion comparison (correct vs wrong)
Fig. 4:	Orthographic vs pictorial sketches
Fig. 5:	Freehand sketches of simple objects (cube, cylinder, part)

WEEK 11

TOPIC:

FREEHAND SKETCHING (CONTINUED)

CONTENT OUTLINE

1. Meaning and importance of freehand sketching
 2. Revision of basic sketching principles
 3. Sketching of complex geometric and mechanical objects
 4. Sketching of objects with curves, holes, and irregular details
 5. Representation of fillets, chamfers, and rounds
 6. Sectional freehand sketches
 7. Shading, proportion, and line control
 8. Labeling, dimensioning, and presentation of sketches
 9. Common mistakes and how to correct them
 10. Practical exercises
-

1. MEANING AND IMPORTANCE OF FREEHAND SKETCHING

Freehand sketching is the **rapid drawing of objects without using technical instruments**. It is the first step in transforming an idea into a visible concept. It is mainly used in engineering, architecture, and design as a means of **visual communication** before precise drawings are made with instruments or CAD.

It allows a designer to:

- Visualize and communicate ideas quickly.
 - Develop creativity and spatial reasoning.
 - Record design observations conveniently.
 - Produce preliminary drawings for discussion and modification.
-

Illustration 1: Example of a Freehand Technical Object

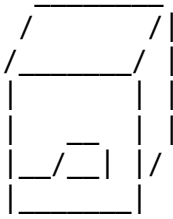


Fig. 1 – A freehand sketch of an L-block (complex shape made of prisms).

2. REVISION OF BASIC PRINCIPLES OF FREEHAND SKETCHING

Before advancing to complex sketches, recall that a good freehand sketch must demonstrate:

Feature	Description
Accuracy	Correct shape, proportion, and relative dimensions
Clarity	Neat, readable lines without clutter
Proportion	Proper ratio of parts
Balance	Symmetrical and stable appearance
Line Quality	Uniform, confident lines with variation for edges and outlines

Essential Techniques:

- Draw from **shoulder and wrist**, not fingers only.
- Start with **light guiding lines**; darken only final outlines.
- Observe and visualize before drawing.
- Use **consistent light source direction** for shading.
- Practice frequently — skill improves with repetition.

3. SKETCHING COMPLEX GEOMETRIC OBJECTS

Complex forms are built by **combining simple shapes** such as cubes, prisms, cylinders, and cones. For example, a *block with a cylindrical hole*, or a *base joined to a vertical plate*, are complex objects.

Step-by-Step Method

1. **Study the Object:** Identify geometric elements (e.g., cuboid + cylinder).
2. **Outline the Enclosing Box:** Draw a light rectangular or cubic boundary.

3. **Mark Major Dimensions:** Estimate length, width, and height proportionally.
4. **Draw Major Edges:** Sketch the main outlines lightly.
5. **Add Secondary Features:** Holes, curves, chamfers, fillets, recesses, or notches.
6. **Darken Visible Edges:** After confirming shapes, darken outlines neatly.
7. **Erase Unwanted Lines:** Remove all construction marks.
8. **Label Key Parts:** Add neat, horizontal annotations.

Illustration 2: Complex Object – L-Shaped Block with Slot

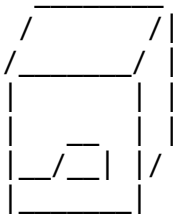


Fig. 2 – Freehand pictorial sketch of an L-block showing internal slot detail.

4. SKETCHING OBJECTS WITH CURVES, HOLES, AND IRREGULAR FEATURES

Objects often contain **rounded corners**, **curved surfaces**, or **circular holes**. These require careful observation to keep them proportional and realistic.

(a) Representing Holes

- Use **ellipses** (not circles) to represent circular holes in 3D view.
- Align both ends of the hole correctly.
- The inner ellipse is drawn lighter to represent hidden curvature.

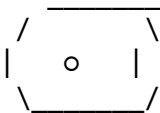


Fig. 3 – Hole shown as ellipse in freehand sketch.

(b) Representing Fillets and Rounds

Fillet: a small, rounded interior corner. **Round:** a small, rounded exterior corner.

Technique:

- Observe where edges meet.
- Replace the corner with a smooth curve drawn using free wrist motion.

- Ensure the curve is tangent to both adjoining edges.

Sharp corner: γ Correct fillet: $\frown$

Fig. 4 – Comparison between sharp and rounded corner.

(c) Representing Chamfers

Chamfers are **beveled edges** connecting two surfaces at an angle (often 45°).

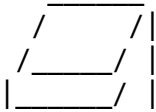


Fig. 5 – Chamfered top edge in freehand sketch.

5. FREEHAND SECTIONAL SKETCHES

When an object has **internal details** (holes, slots, cavities) that cannot be clearly shown from the outside, we use **sectional sketches**.

Types of Sections

Type	Description	Purpose
Full Section	The object is cut entirely in half	Shows complete interior
Half Section	Only one-half is sectioned	Shows both interior and exterior
Offset Section	Cutting plane changes direction	Reveals multiple features in one view

Principles:

- Section lines (hatching) are drawn at **45°**, evenly spaced.
- Hidden lines inside the sectioned part are **omitted**.
- External uncut features remain visible.

Illustration 6: Half-Section of Cylindrical Part (Freehand)

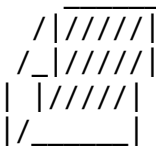


Fig. 6 – Simplified freehand sectional sketch showing hatching on cut surface.

6. SHADING, PROPORTION, AND LINE CONTROL

To make freehand sketches realistic, students must master **line weight and tonal variation**.

(a) Shading

Used to indicate **light, form, and depth**.

- Assume light falls from **top-left** direction.
- Surfaces facing light are **lightly shaded**; opposite surfaces are **darker**.
- Use parallel lines or light strokes to apply tone.

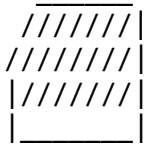


Fig. 7 – Example of light and dark shading to indicate depth.

(b) Proportion

Proportion ensures each part maintains correct relative size. For instance, if an object's length is twice its height, the sketch must reflect a **2:1 ratio**.

Tips:

- Visualize or mark equal divisions.
- Compare parts by eye before darkening lines.

(c) Line Quality and Control

Line Type	Appearance	Use
Construction line	Light, thin	Initial outline
Visible edge	Thick, dark	Object boundaries
Hidden edge	Dashed	Hidden features (if not sectioned)
Centre line	Long-short-long	Axis or symmetry indicator

7. LABELING, DIMENSIONING, AND PRESENTATION

Every freehand technical sketch should be **self-explanatory** through neat labeling and basic dimensions.

Labeling Rules

- Use **uppercase lettering**.
- Keep all text **horizontal**.
- Avoid crossing lines over labels.

Dimensioning Rules

- Dimension lines must be **thin** and **parallel** to the feature measured.
- Arrowheads touch the outline neatly.
- Figures are placed **above the dimension line**, clearly legible.

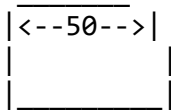


Fig. 8 – Example of dimensioning in freehand sketch.

8. COMMON ERRORS AND CORRECTIONS

Common Error	Cause	Correction
Shaky or broken lines	Poor hand control	Draw from shoulder; relax hand
Poor proportion	Wrong visual judgment	Use reference lines or grids
Over-shading	Excessive strokes	Keep tone even and light
Crooked labels	Careless placement	Use imaginary horizontal guide
Cluttered drawing	Lack of organization	Arrange views clearly

9. PRACTICAL CLASS EXERCISES

Exercise 1: Sketch in freehand an **L-shaped bracket** with a **circular hole** and **filleted corners**. Label all visible parts.

Exercise 2: Draw a **sectional freehand sketch** of a **flanged base** with a **central cylindrical hole**. Show hatching for the cut area.

Exercise 3: Produce a **freehand pictorial sketch** of a **combined object** — cube and half-cylinder joined at one face. Indicate shading to show depth.

10. APPLICATIONS OF FREEHAND SKETCHING IN INDUSTRY

- **Design concept sketches** before production drawings.
- **Field sketches** of components for repair or modification.

- **Workshop instructions** for fabricators and machinists.
- **Architectural site sketches** for space layout.
- **Communication between engineers, builders, and clients.**

EVALUATION

1. Define *complex freehand sketching*.
 2. Describe the steps involved in sketching a **mechanical part** with holes.
 3. Explain how **fillets** and **chamfers** are represented in freehand sketches.
 4. Differentiate between **full**, **half**, and **offset** sections.
 5. Draw a **sectional freehand sketch** of a hollow block showing internal detail.
 6. Explain the use of **shading** in freehand sketches.
 7. State **five rules for labeling and dimensioning**.
-

ASSIGNMENT

1. Sketch in freehand a **complex object** made of a rectangular prism with a semicircular hole through it.
 2. Produce a **sectional freehand sketch** of a **piston** showing bore, crown, and skirt.
 3. Draw a **flanged bracket** in freehand, showing top and side views, proportionally shaded and labeled.
-

ILLUSTRATION SUMMARY

Figure	Description
Fig. 1	Example of a freehand L-block
Fig. 2	Complex L-block with slot
Fig. 3	Freehand representation of hole
Fig. 4	Fillet and round representation
Fig. 5	Chamfered edge
Fig. 6	Sectional sketch of cylindrical part
Fig. 7	Shading example
Fig. 8	Dimensioning and labeling in freehand sketch

SS2 TECHNICAL DRAWING LESSON NOTES

SECOND TERM

WEEK 1: REVISION OF FIRST TERM'S WORK

CONTENT OUTLINE

1. Overview of First Term topics
 2. Summary of key concepts
 3. Revision of geometrical constructions
 4. Review of scale drawing (plain and diagonal scales)
 5. Revision of pictorial drawings (oblique and isometric projections)
 6. Revision of freehand sketching techniques
 7. Application exercises and problem-solving
 8. Common errors and corrections
 9. Evaluation and assignment
-

1. OVERVIEW OF FIRST TERM TOPICS

In the first term, SS2 students were introduced to **advanced geometric and pictorial construction methods** which are foundational for technical drawing at higher levels. The major topics covered include:

1. **Geometrical Constructions**
 - Construction of triangles, quadrilaterals, and polygons.
 - Division of lines into equal parts.
 - Tangents to circles.
2. **Scale Drawing**
 - Plain and diagonal scales.
 - Construction of scales to represent real-life dimensions accurately.
3. **Pictorial Drawings**
 - Oblique and isometric projections.
 - Visualization and construction of 3D objects from orthographic views.
4. **Freehand Sketching**
 - Basic and complex sketching.
 - Proportion, shading, labeling, and presentation techniques.

These topics together form the **foundation of technical representation**, bridging **theory, visualization, and practical drawing skills**.

2. SUMMARY OF KEY CONCEPTS

(a) Geometrical Construction

Geometrical constructions involve using **compass, ruler, and set squares** to create exact shapes without measurement. These skills are vital in producing **accurate, standardized, and reproducible designs**.

Examples of constructions:

- Bisecting lines and angles
 - Drawing perpendicular and parallel lines
 - Constructing regular polygons (hexagon, pentagon, octagon)
 - Constructing tangents to circles
-

Illustration 1: Bisection of a Line

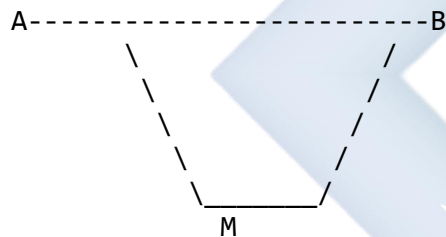


Fig. 1 – The line AB bisected at point M.

Procedure:

1. With A and B as centers, draw arcs above and below AB with equal radius.
 2. Join the intersection points of arcs — line passes through M, the midpoint of AB.
-

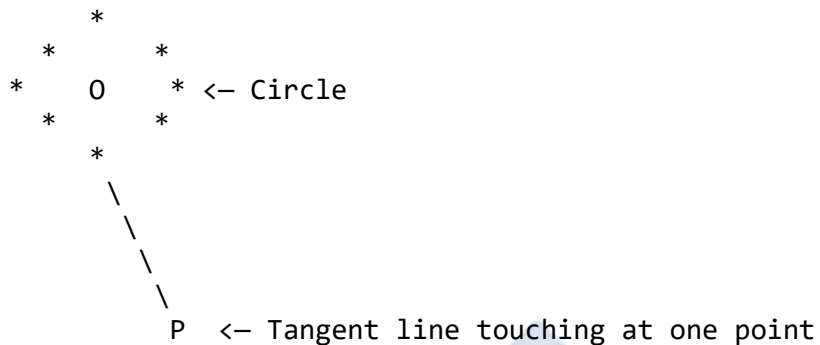
(b) Division of a Line into Equal Parts

Used when a line must be split proportionally into specific segments — important for scale construction and object dimensioning.

(c) Tangents to Circles

A tangent is a line that touches a circle at **one point only**. Applications include mechanical design (e.g., pulleys, wheels).

Illustration 2: Tangent to a Circle



3. REVIEW OF SCALE DRAWING

Scale drawing allows large or small objects to be represented accurately on paper.

(a) Plain Scales

A **plain scale** shows **two units or a unit and its subunit** (e.g., metres and decimetres).

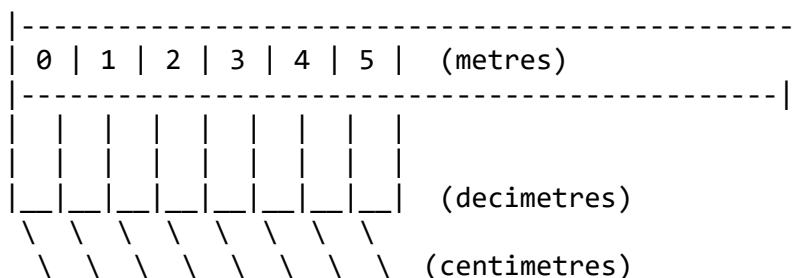
Steps:

1. Choose a suitable Representative Fraction (R.F.).
2. Convert actual length to drawing length.
3. Construct a straight line and divide it according to R.F.
4. Label major and minor divisions.

(b) Diagonal Scales

Used to show **three units or fractional parts of the smallest division** (e.g., metres, decimetres, and centimetres).

Illustration 3: Diagonal Scale Layout



4. REVIEW OF PICTORIAL DRAWINGS

(a) Oblique Projection

- Drawn with one face of the object parallel to the picture plane.
 - The receding axes are drawn at **45°**, and dimensions along these axes are **half their true lengths** to avoid distortion.
-

(b) Isometric Projection

- All three axes (X, Y, Z) are inclined at **120° to each other**.
 - Used for realistic 3D representation.
 - All sides are drawn at their **true lengths**.
-

Illustration 4: Isometric Projection Axes

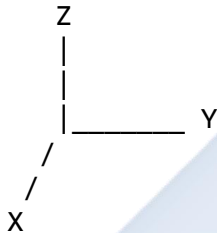
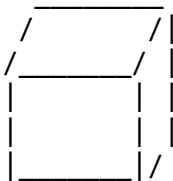


Illustration 5: Isometric View of a Cube



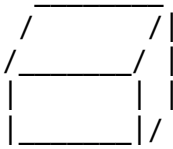
5. REVIEW OF FREEHAND SKETCHING

Freehand sketching helps in **quickly visualizing ideas** without the use of drawing tools.

Key Features:

- Correct proportion and line control.
 - Clear labeling and dimensioning.
 - Ability to convey shape and function.
-

Illustration 6: Freehand Pictorial Sketch of a Block



6. APPLICATION EXERCISES

Exercise 1:

Construct a triangle ABC, given that $AB = 70$ mm, $BC = 60$ mm, and $\angle ABC = 45^\circ$.

Exercise 2:

Draw a plain scale to represent 1 metre, showing decimetres and centimetres. R.F. = $1/20$.

Exercise 3:

Sketch freehand an isometric block with a circular hole through it.

Exercise 4:

Construct tangents to a circle of 30 mm radius from an external point 60 mm from the centre.

7. COMMON ERRORS IDENTIFIED IN FIRST TERM

Error	Cause	Correction
Unequal sides in polygons	Inaccurate compass setting	Use firm, precise arcs
Poor proportion in sketches	Lack of visualization	Practice estimation and balance
Misaligned isometric axes	Wrong angles	Maintain 30° from horizontal on each side
Distorted scales	Wrong R.F. calculation	Review formula $R.F. = \text{Drawing Length} / \text{Actual Length}$

8. CLASS DISCUSSION QUESTIONS

1. Why is geometrical construction important in technical drawing?
2. Differentiate between oblique and isometric projections.
3. What are the main differences between plain and diagonal scales?
4. Explain why freehand sketching is necessary for a technical designer.

5. How can errors in proportion be avoided during freehand sketching?
-

9. EVALUATION

1. Define geometrical construction.
 2. What is a tangent, and how is it constructed on a circle?
 3. Explain the term "Representative Fraction" (R.F.).
 4. Draw and label the isometric axes.
 5. List three advantages of freehand sketching.
 6. Differentiate between full and half sections.
 7. Construct a line 80 mm long and divide it into 5 equal parts.
-

10. ASSIGNMENT

1. Draw a **diagonal scale** to represent 4 metres, showing metres, decimetres, and centimetres. (R.F. = $1/25$)
 2. Sketch an **isometric view** of a cube with a hole through its centre.
 3. Construct tangents from a point outside a circle of radius 25 mm and state all steps clearly.
-

11. TEACHER'S INSTRUCTIONAL GUIDE

- Begin the lesson by reviewing students' notebooks and exercise books from the previous term.
 - Use chalkboard and drawing instruments to re-demonstrate geometric constructions.
 - Ask students to reproduce diagrams from memory before correction.
 - Encourage group work on pictorial drawings and scales.
 - Conduct a short test at the end of the class to assess readiness for the new term.
-

12. TEACHING AIDS

- Drawing board and T-square
 - Set squares and compass
 - Ruler and divider
 - Model objects (cubes, prisms, cylinders)
 - Chalkboard diagrams
 - Prepared freehand and isometric sketches for display
-

13. SUMMARY OF LESSON OBJECTIVES

By the end of this revision class, students should be able to:

1. Recall and explain all first term topics.
 2. Demonstrate geometric constructions using appropriate instruments.
 3. Apply scale drawing methods accurately.
 4. Distinguish between oblique and isometric projections.
 5. Produce neat, proportionate freehand sketches.
 6. Identify and correct common drawing errors.
-



WEEK 2

TOPIC:

ORTHOGRAPHIC PROJECTION

CONTENT OUTLINE

1. Definition of projection and orthographic projection
 2. Importance and uses of orthographic projection
 3. Types of orthographic projection
 4. Explanation of first-angle and third-angle projection
 5. Differences between the two methods
 6. Conventional symbols used in orthographic projection
 7. Step-by-step procedures for drawing objects in both methods
 8. Practical examples and exercises
 9. Evaluation
 10. Assignment
-

1. DEFINITION OF PROJECTION

Projection is the method of representing a **three-dimensional object** on a **two-dimensional surface (paper)** by imagining lines, called **projectors**, drawn from various points on the object to the plane of projection.

In simple terms, it is the **process of transferring the shape of an object** from space onto paper.

Types of Projection

Projection can be classified into:

- **Pictorial projections** (e.g., isometric, oblique)
 - **Orthographic projections**
 - **Perspective projections**
-

2. DEFINITION OF ORTHOGRAPHIC PROJECTION

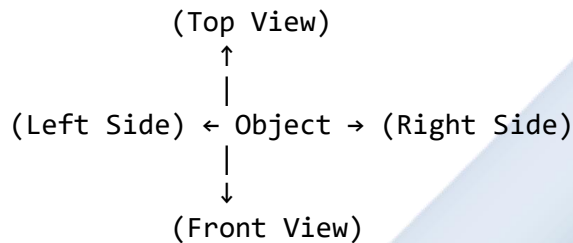
Orthographic projection is a method of drawing an object by projecting its features onto different planes **perpendicular (at right angles)** to one another.

It shows **different views** of the same object as seen from:

- The **front**
- The **top (plan)**
- The **side (end view)**

Each view is a **true shape** of the object's surface, not distorted by perspective.

Illustration 1: Basic Concept of Orthographic Projection



This shows how an object is imagined to be placed inside a **transparent box**, where each face of the box represents a **plane of projection**.

3. IMPORTANCE AND USES OF ORTHOGRAPHIC PROJECTION

Orthographic projection is vital in:

1. **Engineering and Architecture:** For creating working drawings for manufacturing or construction.
 2. **Design and Drafting:** To express design details precisely.
 3. **Technical Communication:** It eliminates misunderstanding between designers, engineers, and builders.
 4. **Dimensioning and Detailing:** Provides accurate views for measurement and assembly.
-

4. TYPES OF ORTHOGRAPHIC PROJECTION

There are two main types based on the position of the object relative to the projection planes:

1. **First Angle Projection**
2. **Third Angle Projection**

Both are standardized internationally by **ISO (International Organization for Standardization)**.

5. FIRST-ANGLE PROJECTION

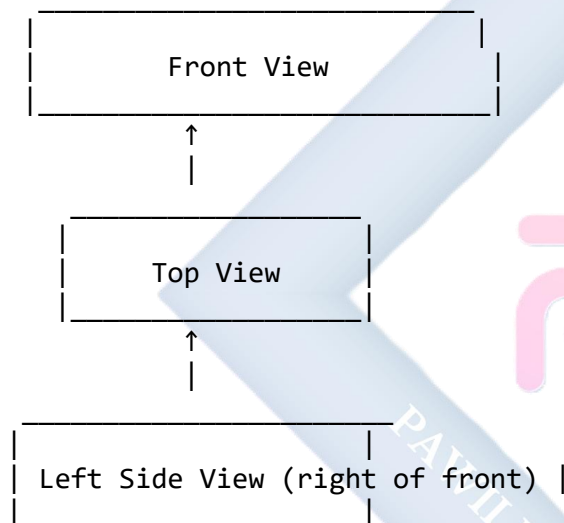
Definition:

In **first-angle projection**, the object is placed **in the first quadrant**, that is, **between the observer and the plane of projection**.

The views are arranged as follows:

- **Front view** – on the vertical plane (V.P.)
- **Top view** – below the front view (on the horizontal plane, H.P.)
- **Side view** – on the opposite side of the object

Illustration 2: First-Angle Projection Layout

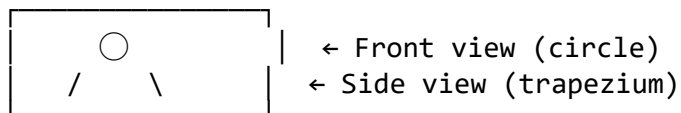


Note:

- The **plan (top view)** is drawn **below** the front view.
- The **left side view** is drawn **on the right side** of the elevation.

First-Angle Projection Symbol (as used in title blocks):

[Truncated Cone Symbol]



1st ANGLE

This symbol shows the viewing relationship of the object.

6. THIRD-ANGLE PROJECTION

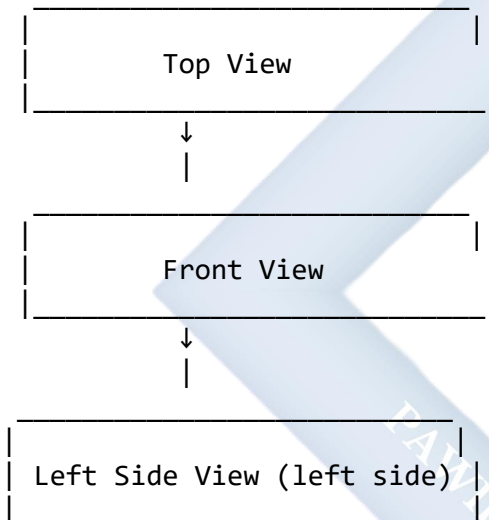
Definition:

In **third-angle projection**, the object is placed **in the third quadrant**, i.e., the plane of projection is **between the observer and the object**.

The views are arranged as follows:

- **Front view** – on the vertical plane
- **Top view** – above the front view
- **Side view** – on the same side as it appears to the observer

Illustration 3: Third-Angle Projection Layout

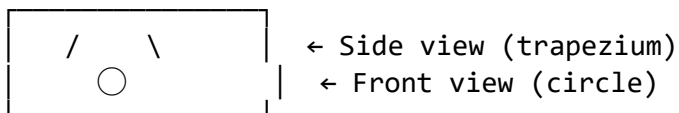


Note:

- The **plan (top view)** is drawn **above** the front view.
- The **left side view** is drawn **on the left side** of the elevation.

Third-Angle Projection Symbol:

[Truncated Cone Symbol]



3rd ANGLE

7. DIFFERENCES BETWEEN FIRST AND THIRD ANGLE PROJECTION

S/N	Feature	First Angle Projection	Third Angle Projection
1	Position of object	Between observer and plane	Between plane and observer
2	Position of top view	Below the front view	Above the front view
3	Position of left side view	Drawn on the right side	Drawn on the left side
4	Symbol representation	Circle on left, cone on right	Circle on right, cone on left
5	Common usage	Widely used in Europe, Asia, Africa	Widely used in the USA and Canada

8. CONVENTIONAL SYMBOLS AND LABELS

In every technical drawing that uses orthographic projection, one of the following symbols must be used in the **title block**:

Symbol for First Angle Projection:



Symbol for Third Angle Projection:



These symbols help the reader know how the drawing views are arranged.

9. STEPS IN ORTHOGRAPHIC PROJECTION DRAWING

Step 1: Choose a simple object (e.g., a rectangular block or L-shaped object).

Step 2: Decide which face should be the front view.

Step 3: Draw the front view (elevation) on the vertical plane.

Step 4: Project construction lines at 90° to locate other views.

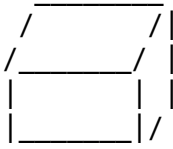
Step 5: Draw the top view (plan) and side view accordingly.

Step 6: Label each view neatly — *Front View*, *Top View*, *Side View*.

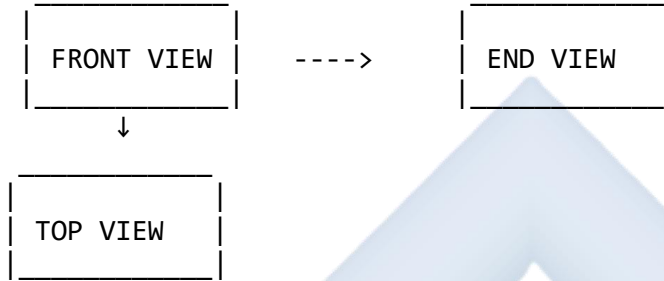
Step 7: Apply line conventions (thick for outlines, thin for construction).

Illustration 4: Orthographic Projection of a Simple Block

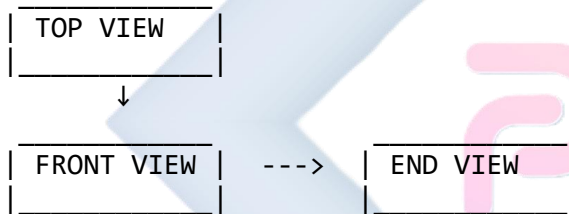
3D Object (Pictorial View):



First Angle Projection Layout:



Third Angle Projection Layout:



10. APPLICATIONS OF ORTHOGRAPHIC PROJECTION

Orthographic drawings are used for:

- **Manufacturing:** To produce exact shapes and dimensions of components.
 - **Architecture:** To show elevations and plans of buildings.
 - **Engineering Design:** To prepare working drawings and assembly details.
 - **Communication:** Between designer, manufacturer, and client.
-

11. PRACTICAL CLASS EXERCISES

Exercise 1: Draw a cube (40 mm sides) in **first-angle projection**, showing *front, top, and right-side* views.

Exercise 2: Repeat the same in **third-angle projection**.

Exercise 3: Sketch the **symbol** for both projections on the same sheet.

Exercise 4: Identify the arrangement of views in a given orthographic drawing and state which projection it follows.

12. COMMON ERRORS TO AVOID

Error	Cause	Correction
Placing the plan above in first-angle projection	Confusing projection types	Review layout rules carefully
Inconsistent projection lines	Poor line discipline	Use faint construction lines
Omitted side view	Poor visualization	Always use three main views
Wrong labeling	Carelessness	Double-check placement before labeling

13. CLASS DISCUSSION QUESTIONS

1. What is projection in technical drawing?
2. Differentiate between orthographic and pictorial projection.
3. List and explain the four principal views used in orthographic projection.
4. State the main difference between first-angle and third-angle projection.
5. Explain the importance of projection symbols in drawings.

14. EVALUATION

1. Define orthographic projection.
2. State three uses of orthographic projection.
3. Draw and label the layout of views in **first-angle projection**.
4. Draw and label the layout of views in **third-angle projection**.
5. List five differences between the two projection methods.
6. Sketch the **symbol** used for first-angle projection.

15. ASSIGNMENT

1. Using an **L-shaped block**, draw its **front, top, and right-side views** using both **first-angle** and **third-angle projections**.
 2. Explain the importance of the position of the object in each projection method.
 3. Draw neatly the **projection symbols** for first and third-angle projection and label them correctly.
-

16. TEACHER'S NOTE / INSTRUCTIONAL GUIDE

- Begin with a recap of pictorial views (isometric and oblique) before introducing orthographic projection.
 - Demonstrate by placing a **wooden or plastic model inside a transparent box** to show projection planes.
 - Encourage students to visualize each face before drawing.
 - Use chalkboard diagrams and instrument drawings for step-by-step demonstration.
 - Reinforce the importance of line quality and view arrangement.
-

17. TEACHING MATERIALS

- Drawing board, T-square, set squares, compass
 - Models of cubes, cylinders, brackets
 - Transparent “glass box” model
 - Sample drawings of both projection types
 - Chalkboard sketches and rulers
-

18. LEARNING OBJECTIVES

By the end of this lesson, students should be able to:

1. Define projection and orthographic projection.
 2. Distinguish between first-angle and third-angle projection.
 3. Arrange views correctly in both methods.
 4. Interpret projection symbols in title blocks.
 5. Produce neat and correctly arranged orthographic views of simple objects.
-

WEEK 3

TOPIC: ORTHOGRAPHIC PROJECTION (CONTINUED)

CONTENT

1. INTRODUCTION

Orthographic projection is one of the most important methods of representing three-dimensional objects accurately on two-dimensional planes such as paper or drawing sheets. In the previous lesson, we defined orthographic projection, studied the differences between **First Angle** and **Third Angle** projections, and introduced their symbols.

In this lesson, we will **construct orthographic views of given objects**, such as the **front elevation**, **plan (top view)**, and **end elevation (side view)**, using the correct projection method, scales, and conventions.

2. DEFINITION AND PURPOSE

Orthographic projection is a method of representing a 3D object by projecting its features onto a plane using perpendicular projectors.

Purpose:

- To communicate the exact shape and size of an object.
- To show different sides (views) accurately.
- To provide manufacturing and construction details.

Typical views obtained are:

- **Front Elevation (Front View)** — view seen from the front.
- **Plan (Top View)** — view seen from above.
- **End Elevation (Side View)** — view seen from the right or left side.

3. PRINCIPLE OF ORTHOGRAPHIC PROJECTION

Orthographic projection works on the principle of **parallel projection**, where **projectors** are drawn perpendicular to the **projection plane**.

There are two main projection systems used worldwide:

Feature	First Angle Projection	Third Angle Projection
Object position	Between observer and plane	Plane between observer and object
Layout on paper	Plan below, right side view on the	Plan above, right side view on the

Feature	First Angle Projection	Third Angle Projection
	left	right
Standard symbol		

4. STEPS IN CONSTRUCTING ORTHOGRAPHIC VIEWS

Let us construct the orthographic views of a simple object — for example, a rectangular block with a slot.

Step 1: Study the Object

Examine the shape and decide which side gives the **best description** of the object. This will be taken as the **front view**.

Step 2: Draw Projection Lines

Lightly draw **construction lines** (horizontal and vertical) to help position the views properly.

Step 3: Draw the Front Elevation

Project the outline of the object as seen from the front. Use a **continuous thick line** for visible edges and **dashed lines** for hidden parts.

Step 4: Project the Plan (Top View)

From the front elevation, draw **vertical projectors** downward (in first angle) or upward (in third angle) to locate the plan view.

Step 5: Project the Side View

Extend **horizontal projectors** to the side (usually the right). Use a **45° miter line** to align height and depth dimensions between views.

Step 6: Add Dimensions and Title Block

Label each view — *Front Elevation*, *Plan*, and *End Elevation* — and dimension all necessary features.

5. EXAMPLES OF ORTHOGRAPHIC PROJECTION

Example 1: Simple Block

Object: A rectangular block measuring 60 mm × 40 mm × 30 mm.

Required Views:

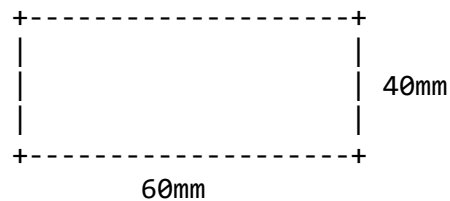
1. Front Elevation
2. Plan
3. End Elevation

Procedure:

1. Draw the front rectangle (60×40 mm).
2. Project downwards for the plan (60×30 mm).
3. Project sideways for the end elevation (40×30 mm).
4. Label each view appropriately.

Illustration:

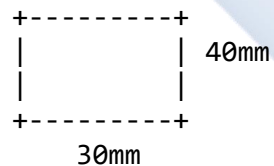
Front View (Elevation)



Top View (Plan)



Side View

**Example 2: Object with Hole**

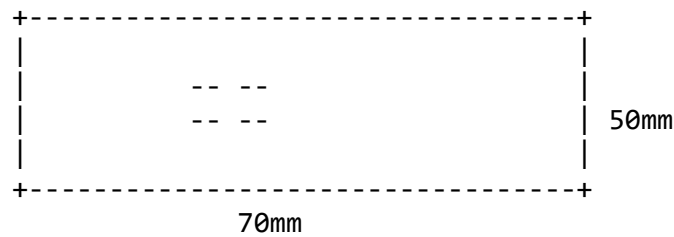
Object: A block $70 \text{ mm} \times 50 \text{ mm} \times 40 \text{ mm}$ with a 20 mm diameter hole at the center.

Procedure:

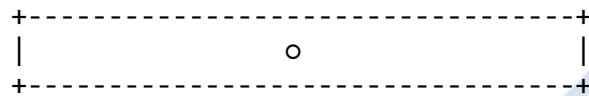
1. Draw the front elevation as a rectangle (70×50 mm).
2. Mark the center point and represent the hole using **center lines** and a **hidden circle** in the plan view.
3. Draw the plan below the elevation, showing the hole as a visible circle.
4. Project the right-side view.

Illustration (Simplified):

Front Elevation (with hidden hole)



Top View (Plan)



6. HINTS FOR GOOD ORTHOGRAPHIC DRAWING

- Use **light construction lines** before finalizing outlines.
- Maintain **alignment** between views.
- Ensure **dimensions** are consistent across all views.
- Use appropriate **line types**:
 - Thick lines for visible outlines.
 - Dashed lines for hidden edges.
 - Thin lines for projection.
 - Chain lines for center marks.

7. APPLICATIONS OF ORTHOGRAPHIC PROJECTION

Orthographic drawings are used in:

- **Engineering design and manufacturing.**
- **Architectural building plans.**
- **Machine component detailing.**
- **Furniture and product design.**

They provide **clear, measurable, and interpretable** details that guide production and assembly.

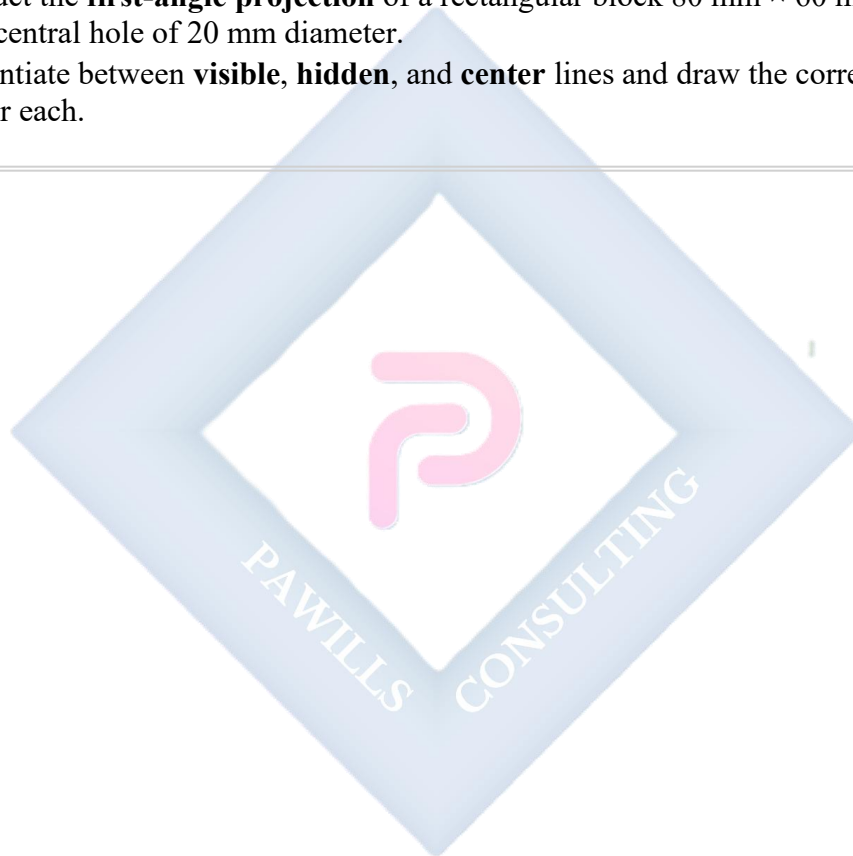
EVALUATION

1. Define orthographic projection.
2. Mention three types of views obtained in orthographic projection.

3. Differentiate between first-angle and third-angle projection.
4. Explain the purpose of using construction lines in orthographic drawing.
5. Sketch the layout arrangement of views in first-angle projection.
6. Sketch the layout arrangement of views in third-angle projection.
7. List four applications of orthographic projection in engineering practice.

ASSIGNMENT

1. Draw the **orthographic projections** (front, plan, and side views) of a cube of 40 mm sides.
 2. Construct the **first-angle projection** of a rectangular block 80 mm × 60 mm × 40 mm with a central hole of 20 mm diameter.
 3. Differentiate between **visible**, **hidden**, and **center** lines and draw the correct line types used for each.
-



WEEK 4

TOPIC: SECTIONAL VIEWS

CONTENT

1. **Meaning and Purpose of Sectional Views**
 2. **Types of Sectional Views**
 3. **Conventions Used in Sectional Drawing**
 4. **Procedure for Drawing Sectional Views**
 5. **Examples of Sectional Drawings**
 6. **Applications of Sectional Views in Engineering and Design**
 7. **Evaluation and Assignment**
-

1. MEANING AND PURPOSE OF SECTIONAL VIEWS

In technical drawing, **sectional views** are used to show **internal features** of an object that are **not visible** in standard orthographic views.

When complex objects contain **holes, cavities, slots, or recesses**, these hidden details cannot be fully understood from external views. Therefore, we imagine cutting through the object along a **cutting plane** to reveal its interior.

The view obtained after the cut is called a **Sectional View**.

Definition:

A **sectional view** is an orthographic projection of an object that shows the internal features as if part of the object has been cut away by an imaginary plane.

Purpose of Sectional Views:

- To **expose internal details** of an object for clear understanding.
 - To **avoid confusion** caused by numerous hidden (dashed) lines.
 - To **enhance accuracy** in manufacturing and assembly.
 - To **reduce complexity** in interpretation.
 - To **improve neatness** and legibility of drawings.
-

2. TYPES OF SECTIONAL VIEWS

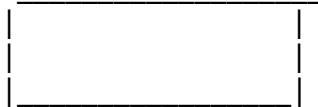
There are several types of sectional views used depending on the nature and complexity of the object.

(a) Full Section

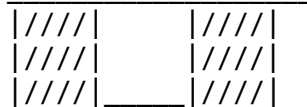
A **full section** occurs when the object is **cut completely along a single cutting plane**, and one half is removed to expose the interior. This gives a complete view of the internal shape.

Illustration (simplified):

Object before section:



After full section:



Hatched area represents the cut surface.

(b) Half Section

A **half section** is used for **symmetrical objects**, such as cylinders or brackets. The cutting plane passes **halfway through**, and **one-quarter** of the object is removed. This shows both the **inside** and **outside** details in the same view.

Illustration:

Half Section of a Cylinder



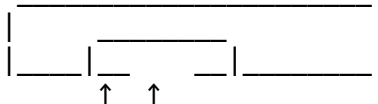
→ One half sectioned, one half external

(c) Offset Section

An **offset section** is used when the features to be shown are not in a straight line. The cutting plane is **bent or offset** to pass through important features, and the bends are not shown in the final drawing.

Illustration:

Top View Showing Offset Cutting Plane:

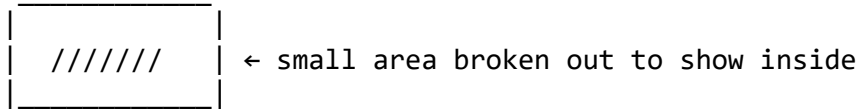


Offset cutting plane passing through key points

(d) Broken-Out Section

A **broken-out section** shows only a **small portion** of the interior, obtained by breaking away part of the view with an **irregular freehand line**. This saves time and space.

Illustration:



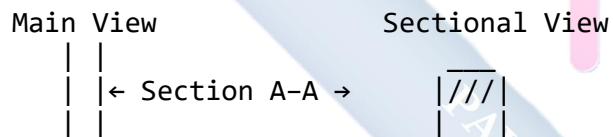
(e) Revolved Section

A **revolved section** is drawn directly on the view of the object to show the **cross-sectional shape** at a particular point. It is often used for **shafts, rods, and ribs**.

(f) Removed Section

A **removed section** is drawn **outside** the main view. The cross-section is projected and placed nearby, usually identified with a label (e.g., Section A–A).

Illustration:



3. CONVENTIONS USED IN SECTIONAL DRAWINGS

When drawing sectional views, certain **standard conventions** must be followed to ensure uniform interpretation.

(a) Cutting Plane Line

- Represented by a **chain thin line with thick ends** and arrows.
- The arrows indicate the **direction of sight**.
- It is usually labeled **A–A, B–B**, etc.

(b) Hatching (Section Lines)

- Used to represent **cut surfaces**.
- Drawn as **thin lines** at **45° angles** to the main outline.
- **Spacing:** 2–3 mm apart.
- **Adjacent parts** are hatched in **opposite directions** for clarity.

Example:

Hatching pattern:

//////// (45° thin lines)

\\\\\\\\\\\\ (Opposite 45°)

(c) Hidden Details

Hidden lines are **omitted** in the area being sectioned, since the cut reveals those internal features directly.

(d) Center Lines

Center lines are retained to show the **axes of holes, cylinders, and symmetry**.

4. PROCEDURE FOR DRAWING SECTIONAL VIEWS

To draw sectional views correctly, follow these steps:

1. **Draw the orthographic views** (front, plan, and side) of the given object.
2. **Identify the section plane** — where the object is to be cut.
3. **Represent the cutting plane line** (e.g., A–A) in the appropriate view using chain lines.
4. **Project the sectional view** in the direction of the arrows.
5. **Remove the cut portion** to reveal the interior.
6. **Apply hatching** to the surfaces cut by the plane.
7. **Label** the sectional view as *Section A–A*, *Section B–B*, etc.
8. **Dimension** the drawing as necessary.

5. EXAMPLES OF SECTIONAL DRAWINGS

Example 1: Full Section of a Hollow Cylinder

Given: A cylinder 60 mm in diameter and 80 mm high with a hole of 30 mm diameter.

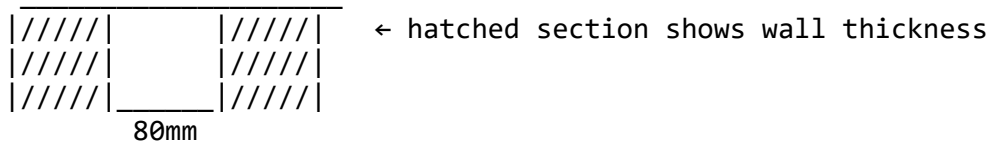
Required: Front elevation in full section and plan.

Steps:

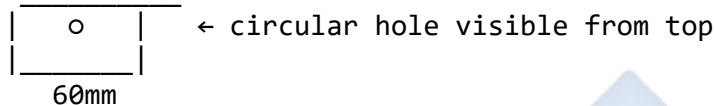
1. Draw the outer and inner circles (60 mm and 30 mm) in plan.
2. Project the front elevation as a rectangle (80 mm height).
3. Show the hole by cutting through the object.
4. Hatch the cut surfaces.
5. Label *Section A–A*.

Illustration (simplified ASCII):

Front View (Section A-A)



Top View



Example 2: Half Section of a Symmetrical Block

Given: A square block 80 mm × 80 mm with a circular hole of 20 mm at the center.

Steps:

1. Draw the front and plan.
2. Divide the object vertically.
3. Hatch one half only.
4. Leave the other half external.

Example 3: Offset Section of an L-Block

When the features are not aligned, an **offset cutting plane** is drawn through the major features, producing an accurate sectional view.

6. APPLICATIONS OF SECTIONAL VIEWS

Sectional views are used in:

- **Engineering drawing:** To show holes, slots, and cavities.
- **Machine design:** To represent interior mechanisms (e.g., pistons, valves).
- **Architectural drawing:** To show the internal layout of walls, beams, and floors.
- **Assembly drawings:** To illustrate how parts fit and interact.

They are indispensable in **production, fabrication, and inspection** processes.

7. EVALUATION

1. Define sectional view.
 2. State three purposes of sectional drawings.
 3. Mention and explain four types of sectional views.
 4. What is the purpose of a cutting plane line?
 5. State the standard hatching angle and spacing used in sectional drawings.
 6. Draw and label a full section of a cylindrical block 60 mm in diameter and 80 mm high.
 7. What type of sectional view is suitable for a symmetrical object?
 8. Why are hidden lines usually omitted in sectional views?
-

ASSIGNMENT

1. Draw the **front and plan views** of a hollow cylinder (outer $\varnothing 70$ mm, inner $\varnothing 40$ mm, height 90 mm) and show the **front elevation in full section**.
 2. Draw a **half-sectional view** of a solid cube (60 mm side) with a cylindrical hole (20 mm $\varnothing$) at its center.
 3. Sketch and label the **symbols and line conventions** used in sectional drawings (cutting plane line, hatching, and center line).
-

SUGGESTED DIAGRAMS TO INCLUDE

- Diagram 1: Symbol and representation of **cutting plane line (A–A)**
 - Diagram 2: **Full sectional view** of a hollow cylinder
 - Diagram 3: **Half section** of a symmetrical block
 - Diagram 4: **Offset section** with labeled cutting plane
 - Diagram 5: **Hatching pattern and conventions**
-

WEEK 6

TOPIC: SECTIONAL VIEWS (CONTINUED)

CONTENT OUTLINE

1. Review of Sectional Views
 2. Complex and Composite Solids
 3. Principles of Sectioning Complex Solids
 4. Types of Sectioning in Combined Solids
 5. Step-by-Step Construction of Complex Sectional Views
 6. Illustrative Examples and Diagrams
 7. Hatching Conventions
 8. Common Errors in Sectional Drawings
 9. Applications of Complex Sectional Drawings
 10. Evaluation
 11. Assignment
-

1. REVIEW OF SECTIONAL VIEWS

A **sectional view** shows the internal features of an object when it is cut by an imaginary **cutting plane**. It replaces hidden details (dotted lines) with clear, visible internal outlines.

Previously, we studied **full**, **half**, and **offset sections**. Now we'll focus on **complex or combined solids**, such as when a **cylinder** rests on a **cube** or a **cone** joins with a **prism**.

2. COMPLEX AND COMPOSITE SOLIDS

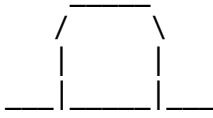
Composite solids are combinations of two or more simple solids, such as:

- Cube + Cylinder
- Cone + Prism
- Pyramid + Cylinder

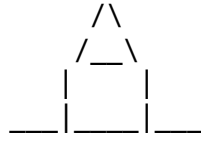
They are commonly found in machine parts, joints, and assemblies.

Illustration 1 – Examples of Composite Solids

Cylinder on Cube



Cone on Prism



3. PRINCIPLES OF SECTIONING COMPLEX SOLIDS

When a **cutting plane** passes through complex solids:

1. The plane must **reveal internal details** clearly.
2. Each solid is **considered separately** for how it intersects the cutting plane.
3. **Hatching** differentiates the material or component cut.
4. Hidden lines **within the cut area** are **not drawn**.
5. Cutting plane lines are labeled (e.g., A–A) using **chain thin lines with thick ends and arrows**.

4. TYPES OF SECTIONING IN COMBINED SOLIDS

Type	Description	Illustration
Vertical Section	Plane passes vertically through object.	
Horizontal Section	Plane passes horizontally (used for layers, bases).	Circle or rectangle cut shown from top view.
Inclined Section	Plane inclined to the base; shows slanted features.	Oblique cut across cone or prism.
Offset Section	Plane changes direction to pass through several features.	Stepped cutting plane shown as one straight view.

5. STEP-BY-STEP CONSTRUCTION OF COMPLEX SECTIONAL VIEWS

Example 1 – Cylinder on a Cube (Vertical Section)

Given: A cylinder ($\text{Ø}50 \text{ mm} \times 40 \text{ mm}$) placed centrally on a cube (60 mm sides), cut by a vertical plane through the axis.

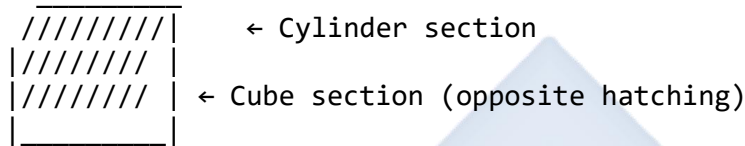
Steps:

1. Draw **plan view**: a **square** for cube base with a **circle** for cylinder base.
2. Project **elevation**: a **rectangle** (cube) and **cylinder** above it.

3. Draw **cutting plane line A-A** through the center in plan.
4. Project **sectional elevation**:
 - Cube = rectangle with inner rectangular cut area.
 - Cylinder = rectangle with semi-circular cut at top.
5. **Hatch** the cut surfaces at 45° (opposite directions for different parts).

Illustration 2 – Sectional Cylinder on Cube

SECTION A-A



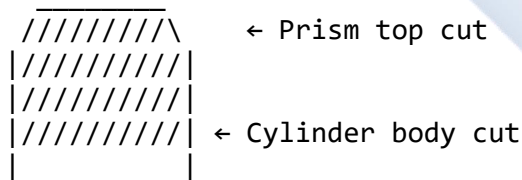
Example 2 – Prism Joined to a Cylinder (Vertical Section)

Given: A hexagonal prism (side 30 mm, height 60 mm) joined to a cylinder (Ø40 mm, height 40 mm) below it. A vertical plane passes through their common axis.

Steps:

1. Draw the **plan view** (hexagon with inscribed circle).
2. Draw **front elevation** showing both shapes.
3. Indicate **cutting plane line A-A** through center.
4. Project **sectional front elevation** — showing how the plane cuts the prism faces and the cylinder.
5. **Hatch** cut areas with opposite angles for clarity.

Illustration 3 – Prism on Cylinder (Sectional View)



Example 3 – Offset Section through an L-Shaped Block

Given: An L-block (80 × 60 × 40 mm thick) with a **vertical slot** on one arm. A **cutting plane** is offset to pass through both the slot and opposite face.

Steps:

9. EVALUATION

1. Define a **sectional view**.
 2. Distinguish between **full** and **offset** section.
 3. State **three rules** for hatching.
 4. Explain why **hidden lines** are omitted in sectional views.
 5. Draw and label the **sectional elevation** of a **cylinder on a cube** cut by a vertical plane.
 6. Describe the **steps in producing a complex sectional drawing**.
 7. Draw the **offset section** of an **L-shaped block** showing both arms.
-

10. ASSIGNMENT

1. Draw a **sectional front view and plan** of a **cone (Ø60 mm, height 70 mm)** placed on a **cube (side 60 mm)** when cut by a vertical plane through the axis.
 2. Draw the **half-sectional elevation** of a **cylinder with a central hole**.
 3. A **hexagonal prism (base 30 mm, height 50 mm)** rests on a **rectangular block (60 × 40 mm)**. Draw its **sectional elevation on A–A**.
-

SUMMARY

Complex sectional drawings require:

- Understanding of combined solids
- Clear representation of cutting planes
- Proper use of projection and hatching
- Clean line work and consistent dimensions

Through practice and neatness, students can master **accurate, professional technical sectioning**.

WEEK 8

TOPIC: AUXILIARY VIEWS

CONTENT OUTLINE

1. Definition of Auxiliary Views
2. Importance of Auxiliary Views
3. Differences Between Orthographic and Auxiliary Projections
4. Types of Auxiliary Views
5. The Principles of Auxiliary Projection
6. Construction of Auxiliary Views (Step-by-Step Procedures)
7. True Shape of Inclined Surfaces
8. Auxiliary Planes and Reference Lines
9. Practical Examples and Exercises
10. Common Mistakes and Precautions
11. Industrial and Engineering Applications
12. Evaluation
13. Assignment

1. DEFINITION OF AUXILIARY VIEWS

An **auxiliary view** is a **special orthographic projection** used to show the **true shape and size** of an **inclined surface** or an **oblique face** that cannot be shown accurately in the **principal orthographic views** (front, top, and side).

It is obtained by projecting the object onto a plane that is **parallel to the inclined surface** and **perpendicular to the line of sight**.

In simple terms, when an object has an inclined or slanted surface, its true shape appears **foreshortened** in the normal orthographic projection. To view it accurately, an auxiliary plane is introduced.

2. IMPORTANCE OF AUXILIARY VIEWS

Auxiliary views are important because they:

- Reveal the **true size and shape** of inclined or slanted surfaces.
- Aid in **accurate dimensioning** and **machining** of parts.
- Provide clarity in **complex designs** with non-perpendicular faces.

- Eliminate confusion caused by **foreshortened or distorted projections**.
- Are essential in **engineering design, architecture, and manufacturing**, where accuracy is crucial.

3. DIFFERENCES BETWEEN ORTHOGRAPHIC AND AUXILIARY PROJECTIONS

Feature	Orthographic Projection	Auxiliary Projection
Projection Plane	Perpendicular to one principal plane (HP, VP, or PP).	Parallel to the inclined surface.
View Type	Shows standard front, top, and side views.	Shows true shape of inclined or oblique surfaces.
Purpose	Represents object from main viewing directions.	Clarifies the shape and size of slanted faces.
Accuracy of Inclined Faces	Appears distorted (foreshortened).	Appears true to scale and proportion.

4. TYPES OF AUXILIARY VIEWS

1. Primary Auxiliary View:

- Drawn from one of the principal views (front, top, or side).
- Used when a surface is inclined to **one** reference plane.
- Example: A wedge with a slanted top face visible in front view.

2. Secondary Auxiliary View:

- Constructed from a **primary auxiliary view**.
- Used when a surface is inclined to **two reference planes**.
- Example: A surface slanted both to the horizontal and vertical planes.

3. Partial Auxiliary View:

- Shows only the **inclined surface** or portion of the object of interest.
- Saves time and drawing space.

5. PRINCIPLES OF AUXILIARY PROJECTION

- Projection lines (projectors) are **perpendicular to the auxiliary plane**.
- The auxiliary plane is **parallel to the inclined surface**.
- Reference lines (labeled x_1-y_1 , x_2-y_2 , etc.) represent **auxiliary reference planes**.
- Distances and dimensions are **transferred accurately** from the primary views.
- The **true shape** of the surface is obtained only when the auxiliary plane is parallel to it.

6. CONSTRUCTION OF AUXILIARY VIEWS (STEP-BY-STEP)

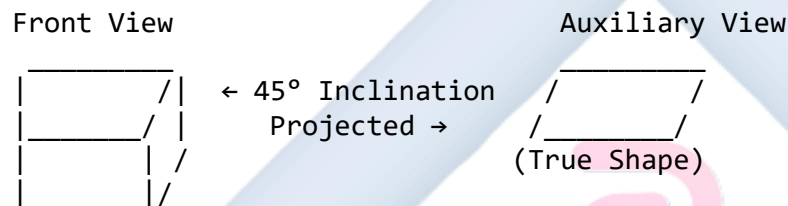
Example 1: Primary Auxiliary View of an Inclined Plane

Given: A rectangular plate ABCD (60 mm × 40 mm) is inclined at 45° to the horizontal plane (HP) and parallel to the vertical plane (VP).

Procedure:

1. Draw the **top view** of the plate as a rectangle ABCD on the HP.
2. Project its **front view**, showing it inclined at 45° to HP.
3. Draw a new **reference line** (x_1-y_1) perpendicular to the inclined surface in the front view.
4. Project perpendiculars from each corner (A, B, C, D) to the **auxiliary plane**.
5. Transfer the corresponding distances from the top view using a compass or dividers.
6. Join points A₁, B₁, C₁, and D₁ to form the **auxiliary view** — the true shape of the rectangle.

Illustration:



Example 2: Auxiliary View of a Wedge Block

Given: A wedge block (80 mm × 50 mm × 40 mm) rests on HP with its top surface inclined at 30° to HP.

Steps:

1. Draw the **top and front views** showing the inclined top face.
2. Draw a new **reference line** x_1-y_1 parallel to the inclined face.
3. Project perpendicular lines from all key points (A, B, C, D) to the reference line.
4. Transfer the vertical distances from the top view.
5. Connect the transferred points to get the **auxiliary elevation**, which shows the **true shape** of the inclined surface.

Example 3: Secondary Auxiliary View

Given: A hexagonal lamina (side 30 mm) inclined at 30° to HP and 40° to VP.

Steps:

1. Draw front and top views.
2. Construct **primary auxiliary view** to show the true shape with respect to HP.

3. Draw **secondary reference line** x_2-y_2 perpendicular to the primary auxiliary view.
4. Project perpendiculars to this line and transfer distances from the primary auxiliary view.
5. Join the points to get the **secondary auxiliary view**, which shows the **true shape** of the surface inclined to both planes.

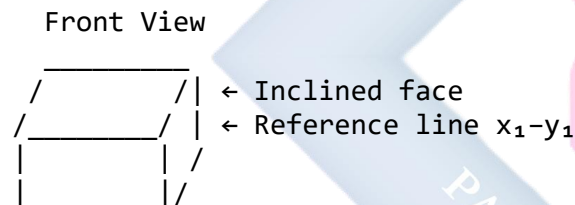
7. TRUE SHAPE OF INCLINED SURFACES

When a surface is inclined to one or more reference planes, its **true shape** can only be revealed through an **auxiliary view**. The auxiliary projection shows the actual dimensions without distortion or foreshortening.

8. AUXILIARY PLANES AND REFERENCE LINES

- Each auxiliary plane has its own **reference line** (x_1-y_1 , x_2-y_2 , etc.).
- These lines represent the intersection of the auxiliary plane with the previous projection plane.
- They serve as **datum lines** for transferring measurements accurately.

Illustration:



9. PRACTICAL EXAMPLES AND EXERCISES

1. Draw the auxiliary view of a **pentagonal lamina** resting on HP, inclined at 40° to HP and parallel to VP.
2. Construct the **primary auxiliary view** of a rectangular prism whose top face is inclined at 35° to HP.
3. From the auxiliary view of an **oblique plane**, determine its **true area and shape**.

10. COMMON MISTAKES AND PRECAUTIONS

Mistakes:

- Using wrong projection direction.
- Misplacing reference lines.
- Failing to transfer exact distances.

- Drawing the auxiliary plane at incorrect angle.

Precautions:

- Always **label reference lines** (x_1-y_1 , x_2-y_2).
- Keep **projection lines perpendicular** to the auxiliary plane.
- Ensure **neatness, accuracy, and correct scaling**.
- Use **light construction lines** for projections and **thick lines** for outlines.

11. INDUSTRIAL AND ENGINEERING APPLICATIONS

Auxiliary views are widely used in:

- **Mechanical engineering:** Machine parts with slanted faces, holes, and slots.
- **Architectural design:** Roofs, ramps, staircases, and sloped beams.
- **Aerospace and automotive design:** Wing sections, control surfaces, car body panels.
- **Civil engineering:** Bridges, road gradients, and sloping embankments.

12. EVALUATION

1. Define an auxiliary view.
2. State four importance of auxiliary projections.
3. Distinguish between primary, secondary, and partial auxiliary views.
4. Explain why auxiliary projections are used instead of standard orthographic views.
5. Draw a rectangular plane inclined at 40° to HP and obtain its auxiliary view showing the true shape.
6. List three common mistakes to avoid when constructing auxiliary views.

13. ASSIGNMENT

1. Draw the auxiliary view of a hexagonal lamina (side 30 mm) resting on HP and inclined at 45° to the horizontal plane.
2. Draw the **secondary auxiliary view** of a block with its face inclined at 30° to HP and 40° to VP.
3. Explain in your own words the **importance of auxiliary views** in real-life engineering drawing.

ILLUSTRATIVE DIAGRAMS (TEXT-BASED REPRESENTATION)

Figure 1 – Auxiliary Plane Orientation

FRONT VIEW

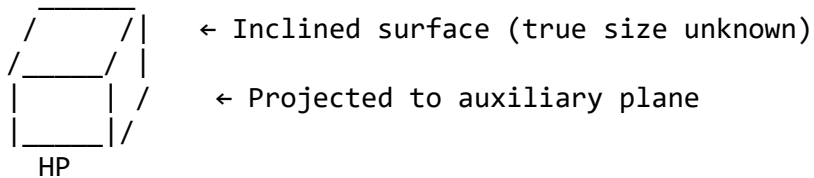


Figure 2 – Primary Auxiliary View

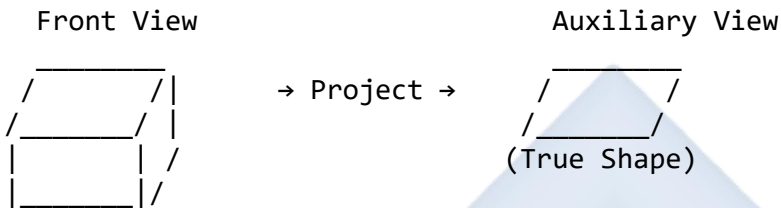


Figure 3 – Secondary Auxiliary Projection



SUMMARY

Auxiliary views are **essential techniques** in technical drawing for:

- Revealing **true shapes and dimensions** of inclined surfaces.
- Improving **clarity and precision** in engineering and architectural designs.
- Enabling **accurate measurement, manufacturing, and visualization** of complex components.

Through continuous practice in **projection accuracy**, **line discipline**, and **geometric reasoning**, students can confidently construct any auxiliary view for both simple and compound surfaces.

WEEK 9

TOPIC: ASSEMBLY DRAWING

CONTENT OUTLINE

1. Meaning and Definition of Assembly Drawing
 2. Purpose and Importance of Assembly Drawing
 3. Types of Assembly Drawings
 4. Identification of Machine Components and Fittings
 5. Difference Between Working Drawing, Detail Drawing, and Assembly Drawing
 6. Exploded and Assembled Views
 7. Steps in Producing an Assembly Drawing
 8. Sectional Assembly Drawing
 9. Conventional Symbols and Standard Practices
 10. Examples and Industrial Applications
 11. Common Errors and Precautions
 12. Evaluation
 13. Assignment
-

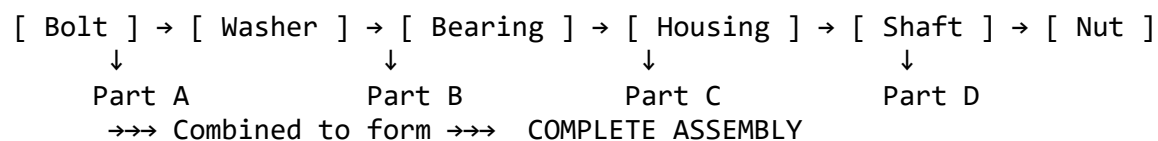
1. DEFINITION OF ASSEMBLY DRAWING

An **assembly drawing** is a **technical drawing** that shows how different parts of a machine or structure fit together to form a complete unit.

It is a **graphical representation** that illustrates the relationship between various **individual components (details)**, their **positions**, and **methods of connection** (e.g., bolts, screws, pins, nuts, shafts, or welds).

In essence, an assembly drawing gives a **complete picture** of how all the parts are joined to make a functional whole.

Illustration (Conceptual Layout)



2. PURPOSE AND IMPORTANCE OF ASSEMBLY DRAWING

Assembly drawings are essential for:

- **Visualizing the entire product** before manufacturing.
- Showing the **relationship and order of assembly** among components.
- Indicating **fit and alignment** of mating parts.
- Assisting in **inspection, maintenance, and repairs**.
- Guiding **technicians and engineers** during production.
- Serving as a reference for **exploded views** used in manuals or catalogs.
- Providing **dimensional and geometric information** for correct assembly.

3. TYPES OF ASSEMBLY DRAWINGS

Type	Description	Use
Design Assembly Drawing	Shows how components fit during the design stage.	Used by designers for visual checking.
General Assembly Drawing	Displays all parts in assembled position with overall dimensions.	Used for documentation and manufacturing.
Sub-Assembly Drawing	Represents a smaller part of a larger assembly (e.g., gear unit in a motor).	Used in complex machines.
Installation Drawing	Shows how different machines or sub-assemblies are arranged for installation.	Used in construction and factory setups.
Exploded Assembly Drawing	Displays parts slightly separated to show how they fit together.	Used in manuals, catalogs, or teaching.
Sectional Assembly Drawing	Shows internal details of an assembly by cutting through it.	Used for clarity where parts overlap.

4. IDENTIFICATION OF MACHINE COMPONENTS

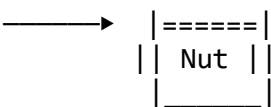
In assembly drawing, components are represented with **labels, item numbers**, and a **parts list (bill of materials)**.

Common mechanical components include:

Component	Function	Typical Symbol or Description
Bolt and Nut	Fasten two or more parts together.	Cylindrical body with threaded end.
Washer	Distributes load and prevents wear.	Thin flat ring under nut or bolt head.
Shaft	Transmits torque and rotation.	Long cylindrical member.
Bearing	Supports rotating shafts and reduces	Shown as concentric circles in

Component	Function	Typical Symbol or Description
	friction.	section.
Key and Keyway	Prevents rotation between shaft and hub.	Rectangular section fitted into slot.
Coupling	Connects two shafts together.	Two hubs joined by bolts or rubber bush.
Bush	Lining for a hole to reduce wear.	Cylindrical sleeve.
Pin	Locates or aligns parts.	Short cylindrical rod.

Diagram: Common Machine Components (Text Illustration)

Bolt → 

Shaft →  (rotating axis)

Washer → (o)

Bearing → (⊙) → [Inner + Outer race]

Key + Keyway → [] inserted between shaft and pulley hub

5. DIFFERENCE BETWEEN WORKING, DETAIL, AND ASSEMBLY DRAWINGS

Aspect	Detail Drawing	Assembly Drawing	Working Drawing
Purpose	Shows a single part.	Shows how all parts fit together.	Combines detail and assembly for manufacturing.
Content	Dimensions, material, finish.	Positions, alignment, item numbers.	Full instructions for making and assembling.
Use	Machinists or fabricators.	Assemblers or engineers.	Production and maintenance.

6. EXPLODED AND ASSEMBLED VIEWS

Exploded View

An **exploded view** shows each component **slightly separated** along the axis of assembly. It makes the **sequence of assembly** clear and helps identify every part distinctly.

Illustration (Exploded View of a Pulley Assembly):

[Nut]
 ↑
 [Washer]
 ↑
 [Pulley Hub]
 ↑
 [Key]
 ↑
 [Shaft]
 ↑
 [Bearing Housing]
 →→ Exploded View (Parts pulled apart for clarity)

Assembled View

An **assembled view** shows all the parts **joined together** in their functional position — exactly as they appear in the real object.

Illustration (Assembled View):

| Nut | Washer | Pulley Hub | Key | Shaft | Bearing |
 ↓ ↓ ↓ ↓ ↓ ↓
 →→ Combined into one continuous object

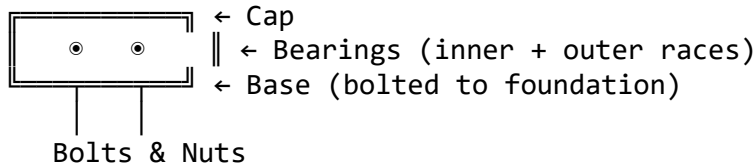
7. STEPS IN PRODUCING AN ASSEMBLY DRAWING

1. **Study the individual part drawings (details)** carefully.
2. **Understand the functional relationship** between the parts.
3. **Select appropriate views** (front, side, top, or sectional).
4. **Position each part** correctly using centerlines and reference points.
5. **Assemble in logical order**, using projection lines for alignment.
6. **Add dimensions** only where necessary (overall or critical fits).
7. **Label all parts** with balloons or leader lines.
8. **Prepare a parts list** (Bill of Materials) beside the drawing.
9. **Use sectional views** to reveal internal components.
10. **Ensure correct scale, neatness, and line types.**

8. SECTIONAL ASSEMBLY DRAWING

A **sectional assembly drawing** is used when internal features (like holes, keys, or bearings) are hidden by other parts. The assembly is cut along a **cutting plane** to show these interior details clearly.

Illustration (Sectional Assembly of a Bearing Unit):



Key points in sectional assembly:

- Alternate **hatching directions** for adjacent parts.
- **No hatching** for fasteners (bolts, nuts, shafts, washers).
- Show **centerlines** for cylindrical parts.
- Label each part clearly.

9. CONVENTIONAL SYMBOLS AND STANDARD PRACTICES

Item	Symbol/Convention	Remark
Bolt/Nut	Not sectioned	Standard fastener convention
Shaft	Continuous hatching	Axis with centerline
Bearing	Concentric circles or hatching	Inner and outer races
Hatching	45° thin lines	Adjacent parts hatched oppositely
Section Plane	Thick chain line	Arrow indicates viewing direction

10. EXAMPLES AND INDUSTRIAL APPLICATIONS

Assembly drawings are used in:

- **Mechanical engineering:** Engines, gearboxes, pumps, valves.
- **Civil/Structural engineering:** Bridges, steel joints, frame connections.
- **Electrical engineering:** Motors, generators, and switchgear.
- **Automotive industry:** Vehicle engines, clutches, and suspension systems.
- **Manufacturing:** Fabrication of equipment, tools, and machine parts.

11. COMMON ERRORS AND PRECAUTIONS

Common Errors

- Incorrect part alignment or positioning.
- Omission of fasteners or clearances.
- Lack of sectional details.
- Unlabeled or ambiguous parts.
- Inconsistent scales or projection methods.

Precautions

- Study all detail drawings carefully before assembly.
- Maintain **projection alignment** using centerlines.
- Label all parts and use consistent **ballooning**.
- Apply standard **section conventions** properly.
- Maintain **neatness, accuracy, and scale fidelity**.

12. EVALUATION

1. Define an assembly drawing.
2. State four functions of an assembly drawing.
3. Differentiate between exploded and assembled views.
4. List and describe three types of assembly drawings.
5. Explain the importance of sectional assembly drawings.
6. Identify five common components found in mechanical assemblies.
7. Describe three precautions necessary when constructing assembly drawings.

13. ASSIGNMENT

1. Draw the **exploded view** and **assembled view** of a *bolted flange coupling* (show all parts labeled).
2. Explain the **difference between a general assembly and a sub-assembly drawing**, with suitable examples.
3. Prepare a **parts list** for a *shaft and bearing assembly* including all standard components.

ILLUSTRATIVE SKETCHES (TEXT REPRESENTATIONS)

Figure 1 – Exploded vs. Assembled View

Exploded View

[Cap]
[Bolt]
[Bearing]
[Housing]

Parts pulled apart →

Assembled View

(◉) Shaft (◉)

Parts joined together →

Figure 2 – Sectional Assembly of Shaft and Bearing

FRONT SECTIONAL VIEW

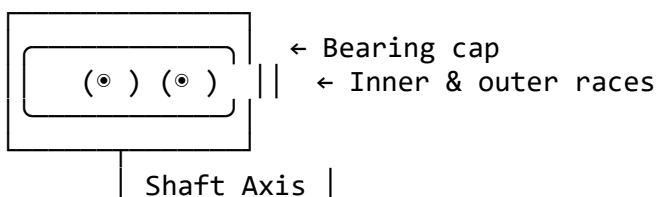


Figure 3 – Assembly Drawing Layout (With Parts List)

FRONT VIEW				
ITEM	NAME	QTY	MATERIAL	
1	Bolt	4	Steel	
2	Washer	4	Mild Steel	
3	Bearing	2	Bronze	
4	Shaft	1	Alloy	
5	Housing	1	Cast Iron	

SUMMARY

An **assembly drawing**:

- Combines all parts of a product into a unified representation.
 - Shows **relationships, positions, and functions** of components.
 - May include **sectional and exploded views** for clarity.
 - Aids in **manufacturing, repair, documentation, and education**.
 - Forms the **foundation of mechanical and industrial drafting practice**.
-

WEEK 10

TOPIC: ASSEMBLY DRAWING (CONTINUED)

CONTENT OUTLINE

1. Review of Assembly Drawing
 2. Meaning of Assembling Parts in Orthographic Projection
 3. Purpose and Importance of Assembling in Orthographic Views
 4. Orthographic Projection Principles in Assembly
 5. Steps in Assembling Parts into Complete Objects
 6. Representation of Joints and Fasteners
 7. Sectional Orthographic Assembly Drawing
 8. Parts List (Bill of Materials) and Labeling
 9. Industrial Applications
 10. Common Mistakes and Precautions
 11. Evaluation
 12. Assignment
-

1. REVIEW OF ASSEMBLY DRAWING

An **assembly drawing** shows how individual parts of a machine or structure fit together to form a complete working unit. In **Week 9**, we discussed exploded and assembled views (pictorial form). This week, we focus on how the same assembled object is **represented in orthographic projection** — i.e., in **front, top, and side views**.

2. MEANING OF ASSEMBLING PARTS IN ORTHOGRAPHIC PROJECTION

Assembling parts in orthographic projection means representing **each part of an object** accurately in **2D projection views** (front, top, and side) **as they appear when fully joined together**.

It transforms the **3D concept** of assembly into **2D technical drawings**, allowing precision in dimensioning, alignment, and detailing.

Illustration (Conceptual Overview)

3D Exploded View → Orthographic Assembly

[Cap]	Front View → □ □
[Bolt]	Top View → □ □
[Shaft]	Side View → ▨ ▨
[Base]	

This process ensures that the assembly is accurately understood from different planes — essential for manufacturing and inspection.

3. PURPOSE AND IMPORTANCE OF ORTHOGRAPHIC ASSEMBLY DRAWINGS

- To show how all parts appear when fitted together in true projection.
- To ensure **dimensional accuracy** and **alignment** of components.
- To serve as a **reference document** for engineers, machinists, and assemblers.
- To display **hidden internal details** through **sectional orthographic views**.
- To indicate the **positioning of fasteners, joints, and shafts**.
- To verify the **functionality and fit** of all parts before production.
- To create accurate **working drawings** for fabrication.

4. ORTHOGRAPHIC PROJECTION PRINCIPLES IN ASSEMBLY

In assembling parts in orthographic projection, the **First Angle** and **Third Angle Projection** principles are applied.

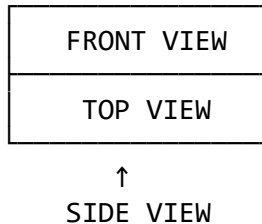
Aspect	First Angle Projection	Third Angle Projection
View Arrangement	Top view is placed below front view	Top view is placed above front view
Projection Plane	Object lies between observer and plane	Plane lies between observer and object
Usage	Common in Europe, Asia	Common in USA, Nigeria (standard in WAEC/NECO)
Symbols	ISO symbol used for identification	ISO symbol used for identification

Orthographic Views Typically Include:

- **Front View (Elevation)**
- **Top View (Plan)**
- **Side View (End Elevation)**

- Sectional View (if required)

Illustration – Basic Layout (Third Angle Projection)



5. STEPS IN ASSEMBLING PARTS INTO COMPLETE OBJECTS

To assemble given parts into complete objects in orthographic projection, follow these steps:

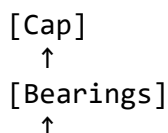
1. **Study the detail drawings** of each part — identify mating surfaces and centers.
2. **Select the appropriate views** (usually front, top, and right side).
3. **Determine the order of assembly** — from the base or foundation upward.
4. **Position the first part** (base or frame) on the drawing sheet.
5. **Align all other parts** using projection and center lines.
6. **Apply sectioning** to show internal fits (if necessary).
7. **Insert fasteners** such as bolts, nuts, keys, or pins correctly.
8. **Add dimensions** only for overall and functional purposes (avoid crowding).
9. **Label all parts** using balloons or numbered circles.
10. **Prepare a parts list (Bill of Materials)** beside or below the drawing.

Example: Assembly of a Simple Bearing Unit

Given Parts:

1. Bearing base
2. Shaft
3. Bearings
4. Bearing cap
5. Bolts and nuts

(a) Exploded View (Conceptual Sketch)



[Shaft]
↑
[Base + Bolts]

(b) Assembled Orthographic Projection (Sectional Front View)

FRONT VIEW (SECTION)

(⊙)	Shaft	(⊙)	← Bearing Cap
			← Bearings
Bolt	Bolt	Bolt	← Base
			← Foundation bolts

TOP VIEW

Bolt	Bolt	Bolt
Shaft		

SIDE VIEW

Cap
(⊙) Shaft

6. REPRESENTATION OF JOINTS AND FASTENERS IN ORTHOGRAPHIC ASSEMBLY

In orthographic projection, **fasteners** (bolts, nuts, rivets, pins) are shown according to standard conventions.

Fastener	Representation in Orthographic View	Remarks
Bolt and Nut	Hexagonal heads shown in top view, thread shown in side view	Not hatched in section
Washer	Thin ring between nut and surface	Prevents damage to material
Rivet	Head shown as circle in plan, profile in elevation	Used for permanent joints
Pin	Solid circular end view	Used for alignment
Screw	Helix indicated with thin lines	For detachable joints

Illustrative Symbols (Textual Diagrams)

Bolt + Nut: $\bigcirc === | === \bigcirc$
Washer: $(\bigcirc)$
Rivet: $\bullet$
Screw: $(\neg)$

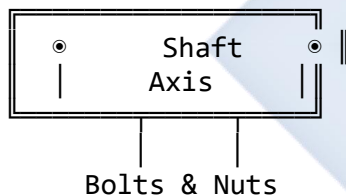
7. SECTIONAL ORTHOGRAPHIC ASSEMBLY DRAWING

Sectioning is essential to show **hidden internal parts** like holes, bearings, or keys.

Rules for Sectional Assembly Drawings:

- Section the object along a **cutting plane** passing through the axis.
- Hatch all solid parts at **45°**.
- Do not hatch bolts, nuts, shafts, or screws.
- Alternate hatching directions for adjacent parts.
- Show **centerlines** clearly for cylindrical parts.

Illustration – Sectional Front View (Bearing Assembly)



8. PARTS LIST (BILL OF MATERIALS)

Every assembly drawing includes a **parts list**, showing all components used.

ITEM NO.	NAME	QTY.	MATERIAL
1	Shaft	1	Mild Steel
2	Bearing	2	Bronze
3	Bearing Cap	1	Cast Iron
4	Base Plate	1	Cast Iron
5	Bolts and Nuts	4	Steel
6	Washers	4	Mild Steel

9. INDUSTRIAL APPLICATIONS

Assembly in orthographic projection is applied in:

- Machine design (motors, pumps, valves).
- Automobile engineering (engines, brakes, clutches).
- Structural drawing (frame joints, supports).
- Electrical and electronic assembly (switches, terminals).
- Manufacturing industries (fabrication, inspection).

10. COMMON MISTAKES AND PRECAUTIONS

Mistakes

- Misalignment of parts in projection views.
- Incorrect sectioning of fasteners.
- Missing centerlines or projection links.
- Over-dimensioning or crowding the drawing.
- Inconsistent projection methods (mixing first and third angle).

Precautions

- Maintain strict orthographic projection alignment.
- Apply correct sectioning conventions.
- Clearly label parts using leader lines.
- Use neat, sharp lines and consistent lettering.
- Review the sequence of assembly before final drawing.

11. EVALUATION

1. Define “assembling parts in orthographic projection.”
2. State three differences between pictorial and orthographic assembly drawings.
3. List five steps in constructing an orthographic assembly drawing.
4. Explain why bolts and nuts are not hatched in sectional views.
5. Draw and label an orthographic assembly of a simple bearing unit (front, top, side).
6. Describe the role of the parts list in an assembly drawing.

12. ASSIGNMENT

1. Using orthographic projection, draw the **assembled front, top, and side views** of a *shaft, bearing, and cap* unit.
2. Identify **five standard conventions** used in sectional assembly drawing.
3. Prepare a **parts list** for the above drawing, indicating materials and quantity.

SUMMARY

- Orthographic assembly drawing converts 3D assemblies into accurate 2D technical projections.
 - It helps engineers and machinists understand the **fit and function** of each part.
 - Proper sectioning and labeling reveal both **external and internal details**.
 - Every assembly must include a **parts list** and follow **standard conventions**.
 - Accuracy, neatness, and projection discipline are vital.
-

TEXTUAL DIAGRAM SUMMARY

Figure 1 – Orthographic Arrangement

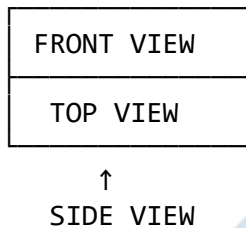


Figure 2 – Sectional Bearing Assembly (Front View)

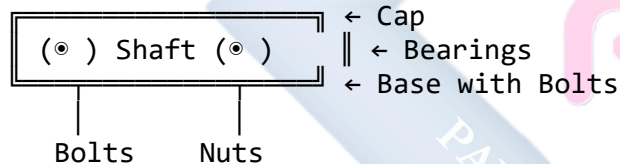


Figure 3 – Bolt Representation

Top View: ○
Front View: |====○
Side View: |====○

WEEK 11

TOPIC:

FASTENERS

CONTENT:

- Meaning and function of fasteners
 - Classification of fasteners
 - Identification and description of types (nuts, bolts, screws, rivets, etc.)
 - Conventional representations of fasteners in technical drawing
 - Standard thread conventions (metric and British)
 - Illustrative examples and labelled diagrams
-

DETAILED LESSON CONTENT

1. MEANING OF FASTENERS

Fasteners are **mechanical devices** used to **join two or more parts** together, either **temporarily** or **permanently**. They are essential in engineering and construction, where components must be assembled, maintained, and sometimes disassembled.

Examples include **nuts, bolts, screws, rivets, studs, and washers**.

Classification of Fasteners:

- **Permanent Fasteners:** Cannot be removed without damaging the parts (e.g., rivets, welding).
 - **Temporary Fasteners:** Can be removed and reused (e.g., bolts, screws, nuts, pins).
-

2. FUNCTIONS OF FASTENERS

- To hold or join machine parts together.
 - To enable easy assembly and disassembly.
 - To transmit forces and motion between joined parts.
 - To provide alignment and structural rigidity.
 - To ensure safety and stability of mechanical systems.
-

3. TYPES OF FASTENERS AND THEIR DESCRIPTIONS

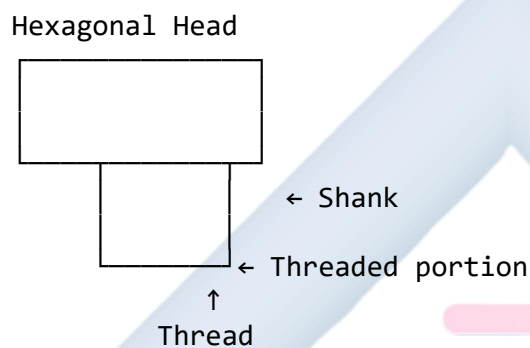
A. BOLTS

A **bolt** is a cylindrical metal rod with an external thread and a hexagonal or square head at one end. It is used with a **nut** to fasten parts together.

Parts of a Bolt:

- **Head:** Used to apply torque (usually hexagonal).
- **Shank:** The smooth part between head and threads.
- **Thread:** The helical ridge that engages the nut.

Illustration:



Common Types of Bolts:

- Hexagonal head bolt
- Square head bolt
- Eye bolt
- Hook bolt
- Countersunk head bolt

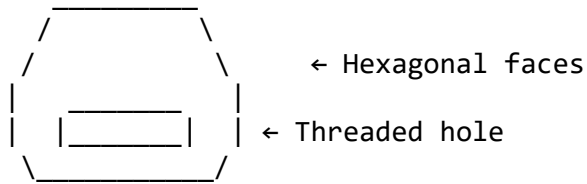
B. NUTS

A **nut** is an internally threaded component used with a bolt to hold two or more parts together.

Common Types:

1. **Hexagonal Nut** – Six-sided, standard type.
2. **Square Nut** – Four-sided, used for rough or heavy work.
3. **Cap Nut** – Has a domed top to protect threads.
4. **Wing Nut** – Has two wings for hand tightening.
5. **Lock Nut** – Used with another nut to prevent loosening.

Illustration:



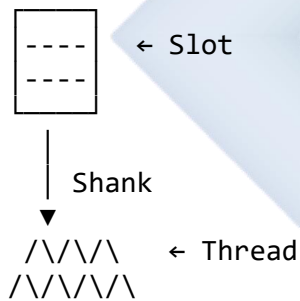
C. SCREWS

A **screw** is similar to a bolt but designed to be screwed directly into a tapped hole or material (like wood, plastic, or metal).

Common Types:

- **Slotted round-head screw**
- **Countersunk-head screw**
- **Cheese-head screw**
- **Machine screw**
- **Wood screw**

Illustration:



D. RIVETS

A **rivet** is a **permanent fastener** used to join metal sheets or plates by deforming one end after insertion.

Parts of a Rivet:

- **Head:** Factory-formed portion.
- **Shank:** Smooth cylindrical part.
- **Tail:** Deformed end after riveting.

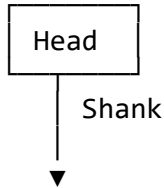
Types of Rivets:

1. Snap head (or round head) rivet
2. Countersunk head rivet

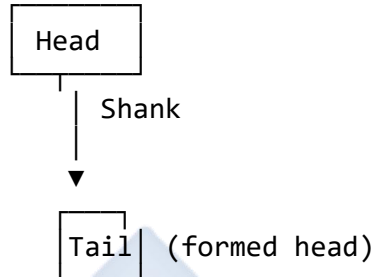
3. Pan head rivet
4. Flat head rivet

Illustration:

Before Riveting:



After Riveting:



4. CONVENTIONAL REPRESENTATIONS IN TECHNICAL DRAWING

In **engineering drawings**, fasteners are not always drawn in full detail — **symbols and conventions** are used to represent them clearly and uniformly.

Fastener	Symbolic Representation	Description
Bolt and Nut		Parallel lines with hexagonal heads represent bolts and nuts.
Screw		Line with short strokes showing thread direction.
Rivet		Circles at ends of a line showing heads on both sides.

Engineering conventions:

- **Threads** are drawn using **thin lines** for the core and **thick lines** for the crest.
- **Hidden threads** are represented with **dashed lines**.
- **Screw pitch** and direction are not shown unless required.

5. THREAD CONVENTIONS

A. Metric Thread System

- Designated as $M8 \times 1.25$, meaning 8 mm diameter and 1.25 mm pitch.
- Commonly used in modern engineering.

B. British Standard Whitworth (BSW)

- Based on inch system, e.g., $\frac{1}{2}$ " BSW
- Thread angle: 55° , rounded crests and roots.

C. Conventional Thread Representation in Drawing

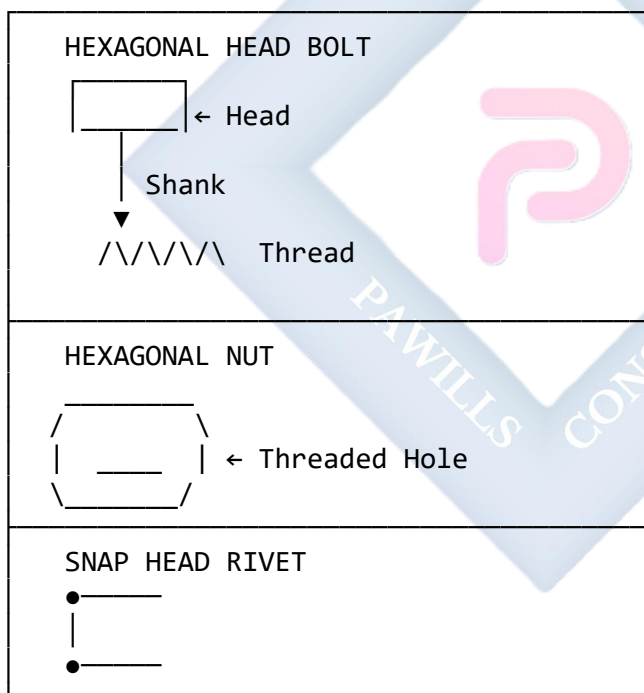
Thick Line (Crest)
Thin Line (Root)
Pitch shown by spacing

6. APPLICATIONS OF FASTENERS

- Used in mechanical assemblies like **engine blocks**, **frames**, **machine beds**, **vehicles**, **aerospace parts**, and **structural steel**.
- Rivets are used in **aircraft fuselage**, **bridges**, and **metal tanks**.
- Screws and bolts are used in **machines**, **furniture**, and **electronic housings**.

7. ILLUSTRATIVE DIAGRAMS

A. Common Fasteners (Orthographic Sketch)



8. SAFETY AND GOOD PRACTICES

- Always select the **right size and material** for the job.
- Apply correct **torque** when tightening bolts and screws.
- Use **washers** to distribute pressure evenly.
- Avoid mixing **metric and imperial threads**.

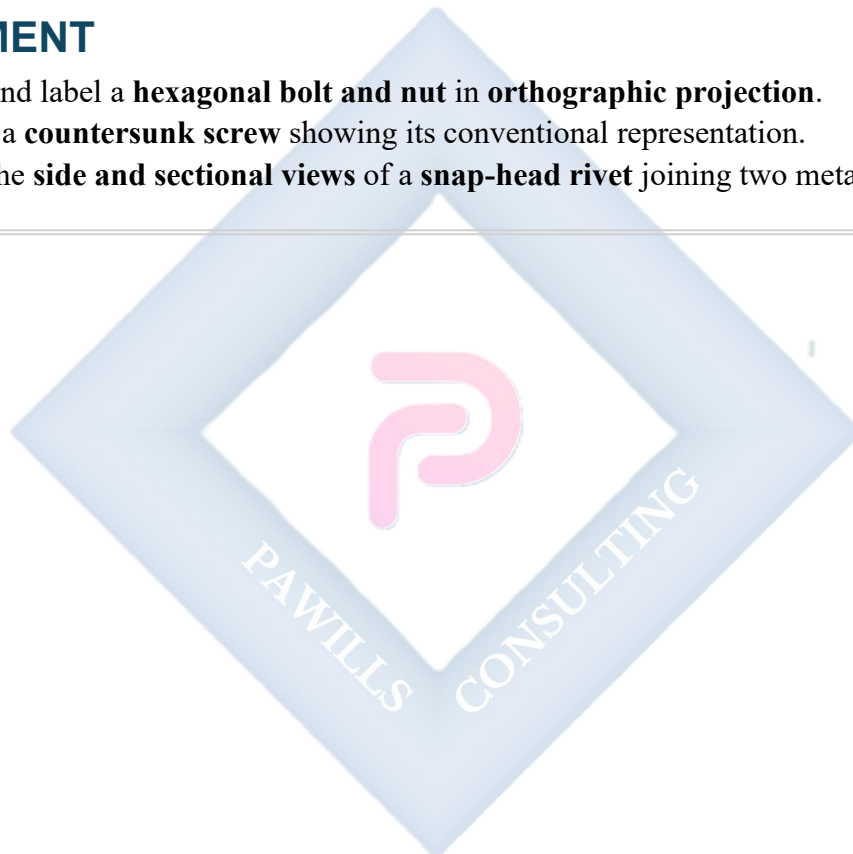
- Check for **corrosion and wear** during maintenance.
-

EVALUATION

1. Define *fastener* and give two classes of fasteners.
 2. Mention five examples of fasteners and state their uses.
 3. Differentiate between a bolt and a screw.
 4. Describe the conventional representation of a bolt and nut in drawing.
 5. Explain the difference between **metric** and **Whitworth** threads.
-

ASSIGNMENT

1. Draw and label a **hexagonal bolt and nut** in **orthographic projection**.
 2. Sketch a **countersunk screw** showing its conventional representation.
 3. Draw the **side and sectional views** of a **snap-head rivet** joining two metal plates.
-



SS2 TECHNICAL DRAWING LESSON NOTES

THIRD TERM

WEEK 1

TOPIC:

INTERPENETRATION AND DEVELOPMENT OF SOLIDS

CONTENT:

- Definition of *Interpenetration of Solids*
 - Explanation of *Development of Solids*
 - Practical applications of interpenetration and development
 - Construction of interpenetration of two solids
 - Techniques of finding lines of intersection
 - Illustrative diagrams and examples
-

1. INTRODUCTION

In technical drawing, solid objects are often combined or intersected to form complex structures. To understand how these objects interact, we must study **interpenetration** and **development**.

These concepts are crucial in engineering, architecture, and fabrication — where parts such as **ducts, pipes, joints, elbows, and cones** meet or connect.

2. DEFINITION OF TERMS

A. INTERPENETRATION OF SOLIDS

Interpenetration refers to the *intersection of two or more solids* that occupy the same space or cut through each other. It shows the **curve or line of intersection** formed where the surfaces of the solids meet.

For example:

- A **cylinder penetrating a prism**.
- A **pipe intersecting another pipe at 90°** (as in plumbing or ventilation).
- A **cone penetrating a cylinder**.

Illustration (simplified text diagram):

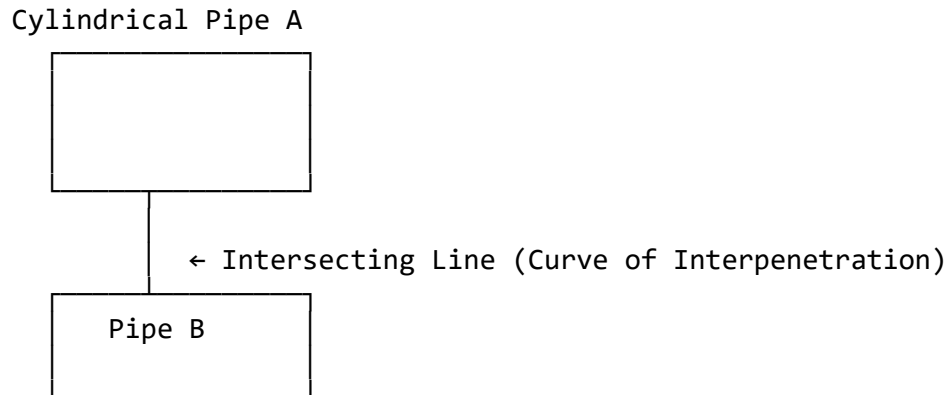


Figure 1: Two cylinders intersecting to form a curve of interpenetration.

B. DEVELOPMENT OF SOLIDS

Development is the process of *unfolding or laying out* the surfaces of a 3D solid on a flat plane to show its true shape and size. It allows sheet metal workers, welders, and designers to cut materials accurately before bending or joining.

Illustration:



Figure 2: Development of a cylinder showing its lateral surface as a rectangle.

3. PRACTICAL APPLICATIONS

Interpenetration and development are used in various engineering fields:

Application	Example
Pipe Fittings	Junction of two pipes or ducts (T-joint or Y-joint)
Sheet Metal Work	Development of cones, hoppers, or tanks
Plumbing and HVAC	Designing duct elbows and intersections
Aerospace/Automobile Design	Fuselage or body panel development
Structural Engineering	Intersection of beams or trusses

4. METHODS OF CONSTRUCTING INTERPENETRATION OF SOLIDS

The *curve of interpenetration* can be found using the **generator or projection method** depending on the solids involved.

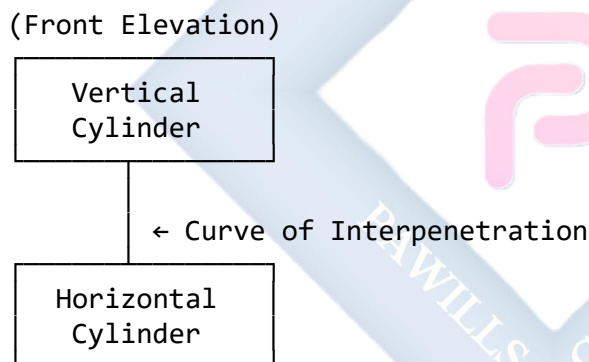
A. INTERPENETRATION OF TWO CYLINDERS AT RIGHT ANGLES

Example Problem: A horizontal cylinder penetrates a vertical cylinder of equal diameter at right angles. Find and draw the curve of intersection.

Steps:

1. **Draw the elevation** of the two cylinders as intersecting rectangles.
2. **Project the base circles** in plan to show their intersection.
3. **Divide one base circle** (of the penetrating cylinder) into equal parts (say 12).
4. **Project generators (lines)** from these points to intersect the surface of the other cylinder.
5. **Mark the points of intersection** in both views (plan and elevation).
6. **Join the points smoothly** to form the curve of interpenetration.

Illustration (simplified text):



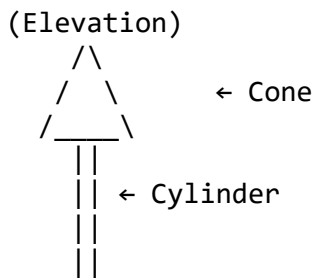
B. INTERPENETRATION OF A CONE AND A CYLINDER

Example Problem: A cone penetrates a cylinder coaxially (having the same axis). Determine the curve of intersection.

Steps:

1. Draw both solids in **elevation and plan**.
2. Divide the cone's base circle into equal parts (say 12).
3. Project **generators** from the cone's surface to intersect the cylinder.
4. Transfer intersection points between views.
5. Join these points smoothly to obtain the intersection curve.

Illustration:



The curve where the cone and cylinder meet represents the interpenetration.

5. DEVELOPMENT OF SURFACES OF INTERPENETRATING SOLIDS

When solids interpenetrate, their **developments** must include the **curve of intersection** projected onto the flattened surface.

Steps to Develop the Lateral Surface of a Cylinder Penetrated by Another Cylinder

1. Draw the **true shape** of the base circle of the cylinder.
2. Divide it into equal parts.
3. Use projection lines to mark intersection points on the curved surface.
4. Unfold (develop) the surface into a **rectangle** whose width equals the **circumference** (πD).
5. Transfer the intersection points onto the development.
6. Join them smoothly — this gives the **true shape of the penetration line**.

Illustration:

(Development of Cylinder)

Curve of
Interpenetration (true)

6. CONVENTIONAL SYMBOLS AND DRAWING TIPS

- Use **center lines** to represent the axes of solids.
- Show **hidden lines** (dashed) for invisible intersections.
- Label views properly: *Front Elevation, Plan, Side Elevation, Development.*

- Use **light construction lines** for projections.
- Ensure all drawings are **to scale**.

7. PRACTICAL EXERCISE

Task: Construct the interpenetration of two equal cylinders intersecting at right angles. Then, develop the lateral surface of one cylinder showing the curve of penetration.

Materials Needed:

- Drawing board
- Set squares
- Compasses
- T-square
- Drawing paper
- HB and 2H pencils
- Eraser and French curves

8. ILLUSTRATIVE DIAGRAMS (TEXT REPRESENTATION)

Figure 3: Cylinders Intersecting at 90°

ELEVATION:

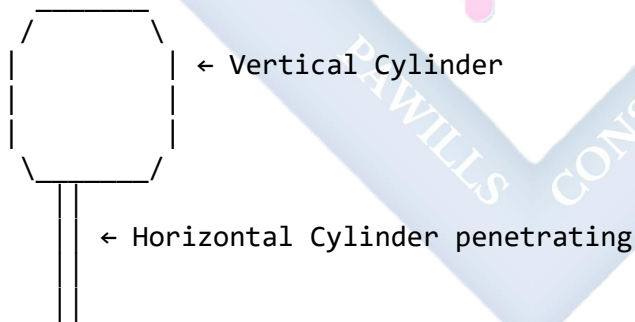


Figure 4: Development of Vertical Cylinder Surface

Curve of Intersection
(Transferred from plan)

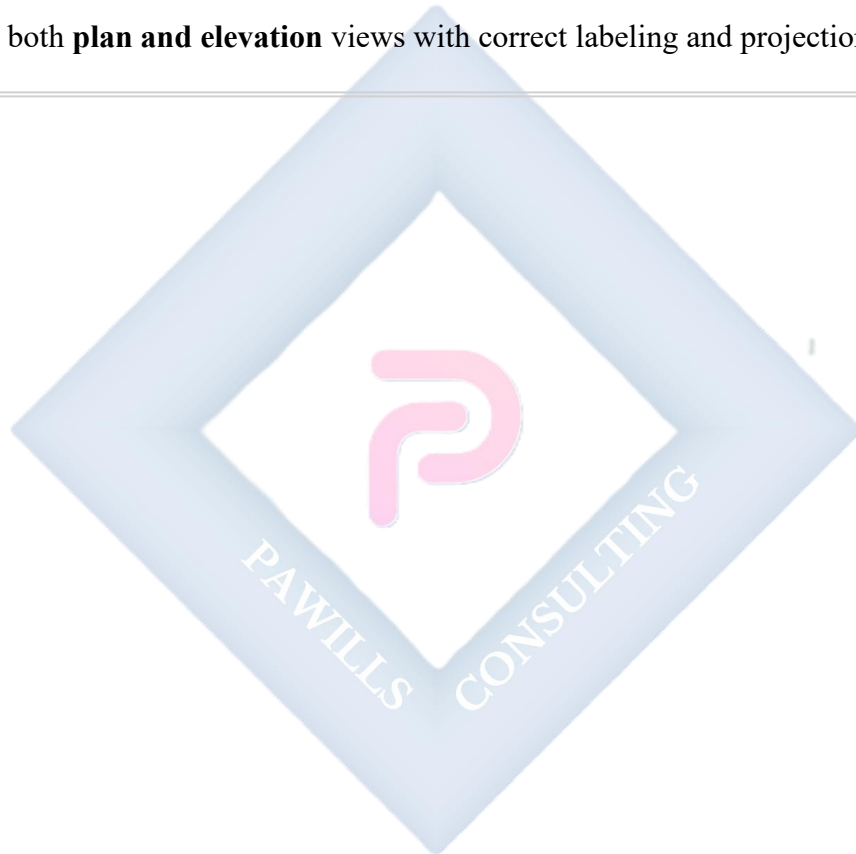
EVALUATION QUESTIONS

1. Define **interpenetration of solids**.

2. Explain the **difference between interpenetration and development**.
3. List **five applications** of interpenetration in engineering.
4. Describe **two methods** used to determine the curve of intersection between solids.
5. Sketch the **curve of interpenetration** of two cylinders meeting at right angles.

ASSIGNMENT

1. Using standard drawing instruments, **construct the interpenetration of a cone and a cylinder** (axes intersecting at 90°).
 2. **Develop the lateral surface** of the cylinder, showing the **true shape of the intersection curve**.
 3. Submit both **plan and elevation** views with correct labeling and projection lines.
-



WEEK 2

TOPIC:

INTERPENETRATION AND DEVELOPMENT OF SOLIDS (CONTINUED)

CONTENT

1. Meaning of Development of Surfaces

Development of surfaces refers to the process of unfolding or laying out the surfaces of a solid object on a flat plane. It is a two-dimensional representation of a three-dimensional object, showing all its surfaces in true shape and size.

Development helps in:

- Fabricating objects from sheet materials (like tins, ducts, and pipes).
- Understanding how complex solids fit together.
- Determining material quantity and cutting patterns.
- Understanding the geometry of intersection between solids.

2. Practical Applications of Development

Development of solids is used in:

- **Sheet metal work** (e.g., ducts, funnels, pipes).
- **Packaging design** (e.g., boxes, containers).
- **Architecture** (e.g., roofs, ventilation ducts).
- **Engineering fabrication** (e.g., boilers, tanks).
- **Pipework intersections** (e.g., where two pipes meet at right angles).

3. Interpenetration of Solids

Interpenetration occurs when **two or more solids intersect** each other. The **line of interpenetration** is the curve formed by their intersection.

Common examples include:

- Cylinder penetrating a prism.
 - Cylinder intersecting another cylinder.
 - Prism penetrating a cone.
-

4. Development of Interpenetrating Solids

The development of interpenetrating solids shows how the surfaces of the two solids unfold after intersection. This requires finding the **line of intersection** and projecting it onto the developed surface.

5. Step-by-Step Construction: Cylinder Penetrating Another Cylinder at Right Angles

Example:

A cylinder of 60 mm diameter penetrates another cylinder of 90 mm diameter at right angles.
Draw:

1. The projections of the intersection.
2. The development of the lateral surface of the larger cylinder.

Step 1: Draw the Elevation and Plan

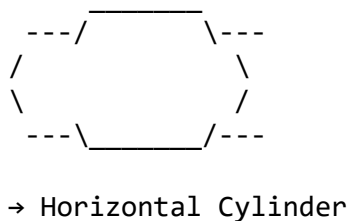
- Draw the **vertical cylinder** in elevation as a rectangle (height = given, width = 90 mm).
- Draw the **penetrating horizontal cylinder** in plan as a circle (diameter = 60 mm).
- The point of penetration should intersect the center of the vertical cylinder.

Illustration:

ELEVATION VIEW



PLAN VIEW



Step 2: Locate the Line of Intersection

- Divide the circle (in plan) of the horizontal cylinder into **12 equal parts**.

- From each division, project vertical lines upward to intersect the surface of the vertical cylinder in elevation.
- The points of intersection form a **curve**, known as the **curve of interpenetration**.

Illustration:

Elevation showing curve of interpenetration

```

(a1)---*-----*---(a7)
      *         *
      *         *
      *         *
      *-----*
  
```

Step 3: Development of the Vertical Cylinder

1. Draw a **rectangle** representing the lateral surface of the vertical cylinder.
 - Length = $\pi \times \text{Diameter} = 3.142 \times 90 = 283 \text{ mm}$.
 - Height = same as the height of the cylinder.
2. Divide the base into **12 equal parts** to correspond with the divisions used in the plan.
3. Transfer the intersection points from the elevation to the developed surface.
4. Join these points with a smooth curve — this represents the **development of the interpenetration**.

Development Sketch:

DEVELOPMENT OF VERTICAL CYLINDER

```

0---1---2---3---4---5---6---7---8---9---10---11---12
 *       *       *       *       *       *       *
 \_____\ \_____\ \_____\ \_____\ \_____\ \_____\ \_____ Curve of
Interpenetration
  
```

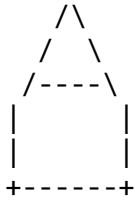
6. Development of a Prism Penetrating a Cylinder

For a **square prism penetrating a cylinder**, the procedure is similar:

1. Draw the projections (plan and elevation).
2. Determine the line of intersection by projecting the edges of the prism onto the cylindrical surface.
3. Transfer these intersection points onto the developed surface of the cylinder.
4. Draw the development with the curve (or line) of interpenetration.

Illustration (simplified)

Prism edges intersecting cylinder
in elevation:



→ When developed, the curve of intersection becomes an irregular shape on the cylinder's surface.

7. Rules for Accurate Development

- All edges must be projected **in true length**.
- Divisions on circular bases must be **equal**.
- Lines of intersection should be transferred **accurately** using projection lines.
- Smooth curves should be drawn **with French curves or flexible curves**.

ILLUSTRATIONS (LABELLED)

Figure 1: Line of Interpenetration between Two Cylinders

(Label: A–Base of vertical cylinder, B–Horizontal cylinder, C–Curve of interpenetration)
 Vertical Cylinder Penetrating Cylinder ● Line of Intersection

Figure 2: Development of Interpenetrating Cylinders

(Label: D–Division lines, E–Curve transferred, F–True shape of development)

Figure 3: Development of a Prism Penetrating a Cylinder

(Label: G–Edges of prism, H–Cylinder surface, I–Projected intersection line)

EVALUATION

1. Define **development of surfaces** and explain its importance.
2. Differentiate between **interpenetration** and **development**.
3. Describe the **procedure** for constructing the development of a cylinder penetrated by another cylinder.
4. Draw and label the **development of a square prism penetrating a cylinder**.
5. Mention **three rules** for accurate development of solids.

ASSIGNMENT

1. Explain **three practical applications** of the development of interpenetrating solids.
 2. Draw a **neat development** of two cylinders intersecting at right angles (diameters 80 mm and 50 mm).
 3. Describe **step by step** how to locate the curve of interpenetration in such solids.
-



WEEK 3

TOPIC:

BUILDING DRAWINGS

CONTENT

1. **Meaning and Importance of Building Drawings**
 2. **Types of Building Drawings**
 3. **Symbols and Conventions Used in Building Drawings**
 4. **Parts of a Building Plan**
 5. **Reading and Interpreting Simple Floor Plans**
 6. **Practical Example and Illustrations**
-

1. MEANING AND IMPORTANCE OF BUILDING DRAWINGS

Building Drawing is a specialized type of **technical** drawing used in the design and construction of buildings. It represents the layout, dimensions, structure, and other specifications of a building in graphical form.

Building drawings are essential communication tools between:

- **Architects** (who design),
- **Engineers** (who calculate and construct),
- **Builders/Contractors** (who execute), and
- **Clients** (who approve or fund the project).

Importance:

- They serve as **blueprints** for construction.
 - Ensure **accuracy** and **uniformity** during building.
 - Provide **cost estimation** data.
 - Serve as **records** for future maintenance or renovations.
 - Allow **regulatory authorities** to approve or reject building plans.
-

2. TYPES OF BUILDING DRAWINGS

Building drawings can be classified according to their purpose and view.

Type	Description / Purpose
Site Plan	Shows the entire building plot, including boundaries, access roads,

Type	Description / Purpose
	drainage, and the position of the building on the site.
Floor Plan	A horizontal section through the building (usually 1 meter above ground) showing room arrangement, walls, doors, and windows.
Elevation	The external view of the building seen from different directions (front, rear, left, right).
Section	A vertical cut through the building showing internal details like beams, floors, stairs, etc.
Foundation Plan	Shows the layout and details of the foundation structure beneath the ground.
Roof Plan	Represents the shape, slope, and covering of the roof.
Detail Drawings	Enlarged views showing specific features such as doors, windows, or joints.
Electrical and Plumbing Layouts	Show the arrangement of wiring, sockets, pipes, and fixtures.

3. SYMBOLS AND CONVENTIONS IN BUILDING DRAWINGS

Building drawings use **standardized graphical symbols** and **line types** to represent different components. These are governed by **ISO** and **Nigerian Standard Organization (SON)** drawing conventions.

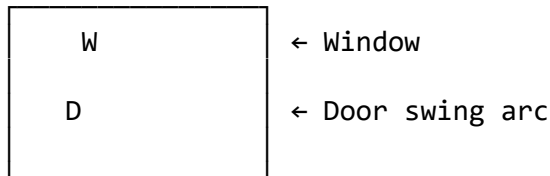
A. Line Types

Type of Line	Purpose
Thick continuous line	Outline of walls and main structural parts
Thin continuous line	Dimension and projection lines
Dashed line	Hidden features (e.g., overhead beams, roof edges)
Chain line	Center lines and axes of symmetry

B. Symbols

Component	Symbol / Representation
Door	
Window	
Wall	Two parallel thick lines
Staircase	A series of diagonal lines with arrow showing direction of ascent
Sink/Basin	Small rectangle with inner circle
Water closet (WC)	Oval or circle labeled "WC"
Switch	Small "S" or circle with line
Socket outlet	Circle with "SO"

Illustration: Typical Floor Plan Symbols (schematic)



4. PARTS OF A BUILDING PLAN

A **floor plan** is the most common and fundamental drawing in architecture. It shows the layout of rooms and internal spaces as seen from above.

Key components include:

1. **Walls** – Indicate room boundaries; external walls are thicker.
2. **Doors** – Shown by a thin line and an arc indicating the swing direction.
3. **Windows** – Shown as gaps in the wall with double lines.
4. **Rooms and Labels** – Each room (e.g., LIVING ROOM, KITCHEN) is labeled.
5. **Dimensions** – Show measurements between walls and overall sizes.
6. **Orientation (North Point)** – Indicates direction on the plan.
7. **Stairs, Columns, and Beams** – Shown for multi-storey structures.
8. **Scale** – Usually **1:50** or **1:100** for floor plans.
9. **Title Block** – Contains project title, name, scale, date, and drafter's details.

5. READING AND INTERPRETING SIMPLE FLOOR PLANS

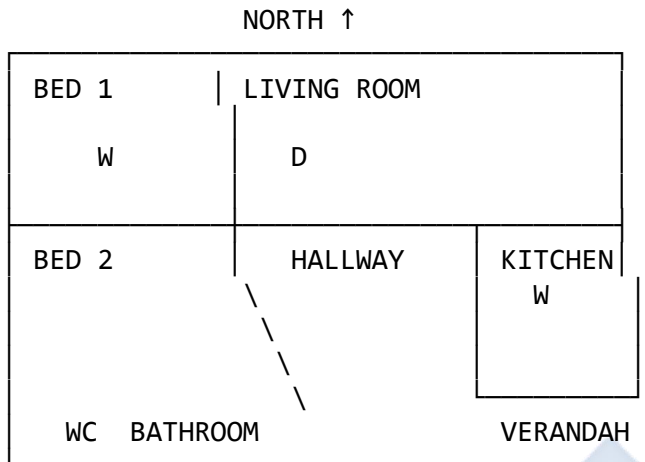
Reading a floor plan involves understanding the layout, room functions, and circulation flow in a building.

A. Steps to Read a Floor Plan

1. **Identify the orientation** – Locate the *north arrow*.
2. **Locate external walls** – Thicker lines represent external boundaries.
3. **Trace the entrances and exits** – Identify main and back doors.
4. **Identify rooms and their labels** – Each labeled according to use.
5. **Observe windows and doors** – Arcs show door swing; small breaks show window positions.
6. **Interpret dimensions** – Read from the extension and dimension lines.
7. **Note internal fittings** – Such as sinks, WCs, wardrobes, etc.
8. **Observe flow and circulation** – How people move through spaces.

B. Example: Simple Two-Bedroom Floor Plan

(Scale: 1:100)



Legend: W = Window, D = Door, WC = Toilet/Bathroom

This plan shows two bedrooms, a living room, kitchen, and bathroom. The doors swing in the direction indicated by arcs. Windows are placed along walls for ventilation and lighting.

6. SCALE AND DIMENSIONING IN BUILDING DRAWINGS

Building drawings are drawn to **scale**, which is the ratio between actual dimensions and drawing size.

Common architectural scales include:

- **1:50** → 1 unit on drawing = 50 units in real life
- **1:100** → 1 unit on drawing = 100 units in real life
- **1:200** → Used for site plans

Dimensioning Rules:

- Always dimension **overall lengths** and **room sizes**.
- Use **thin lines** for dimension lines and arrows.
- Write dimensions **above or beside** the line.
- Indicate all **openings (doors/windows)** on wall lines.

7. PRACTICAL EXERCISE: READING A SIMPLE RESIDENTIAL FLOOR PLAN

Objective: Interpret and draw a 2-bedroom flat using given data.

Given Data:

- Living room: 4200 × 3600 mm
- Bedrooms: 3600 × 3000 mm each

- Kitchen: 3000 × 2700 mm
- Toilet/Bath: 2100 × 1500 mm
- Wall thickness: 230 mm
- Scale: 1:100

Tasks:

1. Draw the external walls.
2. Mark out the rooms according to given dimensions.
3. Add doors and windows at appropriate positions.
4. Indicate the labels of each room.
5. Add dimension lines and a title block.

Illustration: Scaled Drawing (Simplified Floor Layout)

LIVING ROOM D	
BEDROOM 1 W	BEDROOM 2 W
KITCHEN W	TOILET/BATH

EVALUATION QUESTIONS

1. Define **building drawing**.
2. Mention **four types** of building drawings and state their uses.
3. List **five symbols** commonly used in building drawings and their meanings.
4. Explain the difference between **floor plan** and **elevation**.
5. Interpret the following symbols on a building drawing: (a) Arc with D inside (b) Parallel lines with a gap (c) Circle with “WC”

ASSIGNMENT

1. Using a scale of 1:100, draw a **simple floor plan** for a one-bedroom flat with a living room, kitchen, bathroom, and veranda.
2. Identify and label all **doors, windows, and sanitary fittings**.
3. List **five reasons** why scale and symbols are important in building drawings.

WEEK 4

TOPIC:

BUILDING DRAWINGS (CONTINUED)

CONTENT OUTLINE

1. Meaning and Purpose of Floor Plans and Elevations
 2. Steps in Drawing Floor Plans
 3. Steps in Drawing Elevations
 4. Line Conventions and Symbols Used in Building Drawings
 5. Orthographic Relationship between Floor Plans and Elevations
 6. Worked Example: Floor Plan and Elevation of a Simple Two-Bedroom Flat
 7. Illustrations (Vector Diagrams Descriptions)
 8. Evaluation
 9. Assignment
-

1. MEANING AND PURPOSE OF FLOOR PLANS AND ELEVATIONS

A. Floor Plan

A **floor plan** is a **horizontal section** through a building, usually taken at a height of **about 1 meter above the floor level**, showing the **arrangement of spaces (rooms)**, **walls**, **doors**, **windows**, and **fittings** as viewed from above.

It shows:

- The **layout** of the building.
- **Circulation pattern** (movement within rooms).
- **Location of openings** (doors and windows).
- **Relationship** between different rooms.
- **Dimensions and proportions** of spaces.

B. Elevation

An **elevation** is a **vertical orthographic projection** of the exterior face of a building. It shows the **appearance**, **height**, **door and window placements**, **roof design**, and **materials** used.

Types of elevations include:

- **Front Elevation** (View from main entrance)

- **Rear Elevation** (Opposite the front)
- **Left/Right Side Elevations** (Lateral views)

Purpose of Elevations:

- To visualize the **appearance and style** of the building.
- To determine **heights and levels** of different components.
- To specify **external finishes** and openings.
- To serve as **guidelines for masons and builders** during construction.

2. STEPS IN DRAWING FLOOR PLANS

To draw a floor plan correctly, the following systematic steps are followed:

Step 1: Choose a Scale

Select an appropriate scale (usually **1:50** or **1:100**). E.g., at a scale of 1:100, 1 cm on the drawing represents 1 meter on site.

Step 2: Draw the External Walls

- Start by drawing a **rectangular or L-shaped outline** of the building.
- Indicate **wall thickness** (e.g., 230 mm for external, 115 mm for internal).
- Use **thick continuous lines** for walls.

Step 3: Mark Internal Partitions

- Divide the internal spaces into rooms using thinner lines for **partition walls**.
- Label each room (e.g., LIVING ROOM, BEDROOM, KITCHEN).

Step 4: Add Doors and Windows

- Represent doors by thin lines and arcs indicating the **direction of swing**.
- Windows are shown as **thin double lines** across walls.
- Use standard door/window symbols according to conventions.

Step 5: Add Sanitary Fittings and Fixtures

- Include WCs, basins, sinks, wardrobes, and staircases using correct symbols.

Step 6: Dimensioning

- Dimension external and internal walls.
- Indicate room sizes and overall building dimensions using **thin continuous lines** and **arrowheads**.

Step 7: Add the Title Block and North Arrow

- Include the **project title, scale, drafter's name, and date**.
- Add the **north arrow** for orientation.

3. STEPS IN DRAWING ELEVATIONS

To project an elevation from a floor plan:

Step 1: Select Viewpoint

Decide which side of the building to view — typically **Front Elevation** from the main entrance.

Step 2: Project Vertical Lines Upward

- From the **corners of the floor plan**, project **vertical projection lines** upwards.
- These form the outline of the elevation.

Step 3: Add Doors and Windows

- Transfer door and window positions from the floor plan using projection lines.
- Show **heights** and **lintel levels** (standard door height = 2100 mm).

Step 4: Indicate Roof Type

- Choose appropriate roof design (flat, gable, or hipped).
- Show roof slopes with **ridge lines** and **eaves projections**.

Step 5: Add Finishing Details

- Include **plinth level**, **floor level**, and **roof level lines**.
 - Represent materials (brick, glass, block, etc.) with standard symbols.
-

4. LINE CONVENTIONS AND SYMBOLS USED

Type of Line	Purpose
Thick continuous line	Visible outline of walls and building parts
Thin continuous line	Dimension lines and hatching
Dashed line	Hidden features (e.g., roof edges, beams)
Chain thin line	Center lines
Zigzag line	Cutting plane in section views

Common Building Symbols

Component	Symbol / Representation
Door	Thin line with arc swing
Window	Double thin parallel line
Stairs	Series of diagonal lines with an arrow indicating ascent
WC	Oval or circle labeled “WC”
Wash Hand Basin	Small rectangle with a circle inside

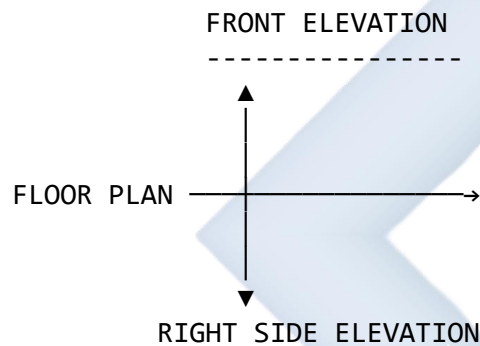
Component	Symbol / Representation
Wall	Two parallel thick lines

5. ORTHOGRAPHIC RELATIONSHIP BETWEEN FLOOR PLAN AND ELEVATION

In architectural drawing, **floor plan** and **elevation** are related as **orthographic projections** of the same structure:

- The **floor plan** is a **horizontal cut**.
- The **elevation** is a **vertical projection**.
- Both share **projection lines** — vertical from plan to elevation. This helps visualize the **complete 3D form** of the building.

Illustrative Diagram: Orthographic Relationship



6. WORKED EXAMPLE: FLOOR PLAN AND FRONT ELEVATION OF A SIMPLE TWO-BEDROOM BUNGALOW

Given Data:

- Living Room: 4200 × 3600 mm
- Bedrooms: 3600 × 3000 mm each
- Kitchen: 3000 × 2700 mm
- Bath/WC: 2100 × 1500 mm
- Wall thickness: 230 mm (external), 115 mm (internal)
- Scale: 1:100

Step-by-Step Floor Plan Construction

1. Draw a **rectangular outline** 9 m × 7 m (scaled to 90 mm × 70 mm).
2. Divide into rooms according to given dimensions.

3. Insert **doors** (width 900 mm) and **windows** (width 1200 mm).
4. Label each room and add **door swing arcs**.
5. Add **dimension lines** and **north arrow**.

Illustration (Simplified ASCII Plan):

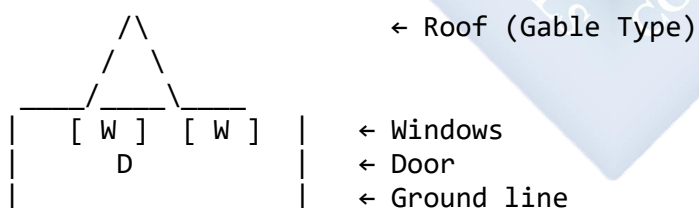
LIVING ROOM D	
BEDROOM 1 W	BEDROOM 2 W
KITCHEN W	TOILET / BATH

Legend: W = Window, D = Door, 230mm = External wall, 115mm = Internal partition.

Step-by-Step Front Elevation Construction

1. Project **vertical lines** upward from the floor plan corners.
2. Draw the **base (ground) line** and **roof line**.
3. Insert **doors and windows** in correct positions.
4. Add **lintel** and **roof slope** (gable type shown below).
5. Label height dimensions (e.g., door height 2100 mm, window sill 900 mm, wall height 3000 mm).

Simplified Illustration of Front Elevation (ASCII):



7. ILLUSTRATIVE EXAMPLES (VECTOR DIAGRAMS – DESCRIPTIONS)

Figure 1: Typical Floor Plan Symbols

Shows graphical symbols for door swings, window lines, wall thicknesses, stairs, WC, and kitchen fittings.

Figure 2: Orthographic Projection Relationship

A 3-view projection linking the plan to the front and side elevations.

Figure 3: Front Elevation Rendering

Depicts architectural elements — plinth, windows, doors, roof slope, fascia board, and floor level.

Figure 4: Dimensioning of a Floor Plan

Displays properly spaced dimension lines, arrowheads, and annotated room sizes.

8. EVALUATION

1. Define a **floor plan** and **elevation**.
2. Differentiate between a **front elevation** and a **side elevation**.
3. State the **purpose** of drawing floor plans in building design.
4. List the **steps** involved in producing an elevation from a plan.
5. Explain the **orthographic relationship** between floor plan and elevation.
6. Draw and label the **front elevation** of a simple one-room apartment using a given floor plan.

9. ASSIGNMENT

1. Using a scale of 1:100, draw the **floor plan** and **front elevation** of a 2-bedroom bungalow with a veranda, kitchen, and bath/WC.
2. Include doors, windows, and appropriate architectural symbols.
3. Label all rooms, add dimension lines, title block, and north arrow.

✓ TEACHER'S NOTE:

For demonstration:

- Use **drawing instruments** (T-square, set squares, compasses).
 - Begin with a **constructional layout**, then trace in **ink or fine liner** for clarity.
 - Discuss **architectural symbols** and **room labeling conventions** in class.
 - Display a **printed vector illustration** of a simple bungalow for visual aid.
-

WEEK 6

TOPIC: PICTORIAL DRAWING OF BUILDINGS

CONTENT

1. Meaning of Pictorial Drawing of Buildings

Pictorial drawing is a **three-dimensional (3D)** representation of an object or building that shows height, width, and depth in a single view. In building drawing, pictorial representation helps architects, builders, and clients **visualize** what the final structure will look like before construction begins.

There are **three main types of pictorial drawings** used in architecture and technical drawing:

1. **Isometric Projection** – The building is drawn with its axes inclined at **120° to each other**, allowing equal scaling along all three dimensions.
2. **Oblique Projection** – The front view is drawn in true shape while the sides are projected at **45°**, showing depth.
3. **Perspective Projection** – The most realistic type, where lines converge at one or more **vanishing points** to simulate the way the human eye sees.

2. Uses of Pictorial Drawings in Building Design

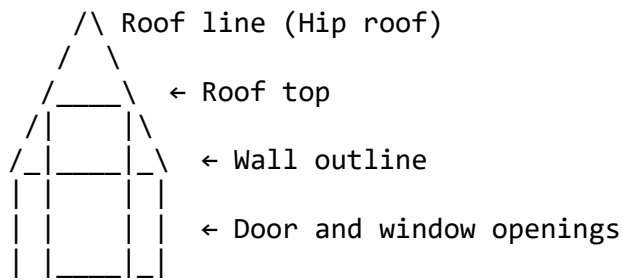
- To give a **realistic appearance** of the structure.
- To help in **client presentation** and architectural visualization.
- To serve as a **reference for 3D modeling** and construction planning.
- To assist in **estimating materials** and spatial arrangement.

3. Basic Steps in Constructing Pictorial Drawings of Buildings

A. Isometric Projection Method

1. **Draw the Isometric Axes:** Three lines at **120°** to each other — one vertical, two inclined at 30° to the horizontal.
2. **Mark Out the Plan:** Reproduce the building plan on the isometric base using the **isometric scale**.
3. **Project the Walls Vertically:** From each corner of the base, project vertical lines equal to the wall height.
4. **Draw the Roof:** Connect roof edges according to the building's roof design (gable, flat, hipped, etc.).
5. **Add Openings:** Show **doors, windows, and ventilations** in correct proportion and placement.
6. **Finalize with Details:** Include the staircase, veranda, columns, and boundary lines.

Illustration (Vector Description):



(Isometric pictorial of a simple rectangular building)

B. Oblique Projection Method

1. Draw the **front elevation** in true shape.
2. Project the sides backward at **45°** to represent depth.
3. Complete the roof, windows, and door outlines using proper projection distances.
4. Use **cavalier** (full depth) or **cabinet** (half depth) projection depending on scale accuracy.

Illustration (Vector Description):

Front view (true shape)



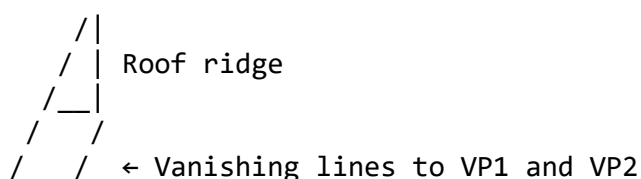
C. Perspective Projection Method

Used mainly for **architectural renderings** and visual presentations.

Steps:

1. Establish the **ground line (GL)** and **horizon line (HL)**.
2. Mark the **station point (SP)** and **vanishing points (VP1, VP2)**.
3. Draw the plan of the building and project its corners toward the vanishing points.
4. Raise verticals to show height and connect tops to corresponding vanishing lines.
5. Add doors, windows, and roof in perspective view.

Illustration (Vector Description):



(Two-point perspective of a simple rectangular building)

4. Proportion and Scale

- Ensure **correct proportion** between walls, roof height, and openings.
 - Use **architectural scales** (e.g., 1:100, 1:50) for accurate dimensions.
 - Apply **isometric scale correction** (multiply true length by 0.816) when necessary.
-

5. Conventions in Building Pictorials

- **Walls:** Drawn thick and dark to indicate solid masonry.
 - **Openings:** Shown with thinner lines.
 - **Hidden parts:** Dashed lines (if necessary).
 - **Ground line:** A bold base line representing floor level.
 - **Roof edges:** Shown lighter unless sectioned.
-

6. Applications

- Residential building representation
 - Classroom or office layout sketches
 - 3D concept visualization for architectural projects
 - Technical project submission and modeling
-

EVALUATION

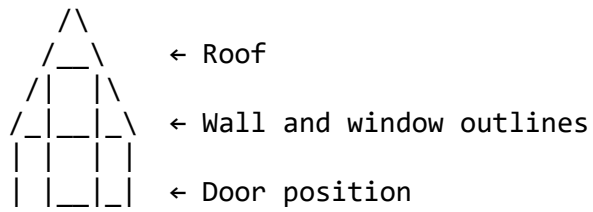
1. Define pictorial drawing of buildings.
 2. List and explain the three main types of pictorial drawings.
 3. Outline the steps involved in constructing an isometric view of a simple building.
 4. Differentiate between isometric, oblique, and perspective projections.
 5. Why are pictorial drawings important in building design and construction?
-

ASSIGNMENT

1. Draw the isometric pictorial of a **two-room building** with a gable roof using given plan dimensions (6 m × 4 m).
 2. Using oblique projection, draw the front and side of a **simple store building** with doors and windows.
 3. Produce a **two-point perspective view** of a classroom block based on a provided plan.
-

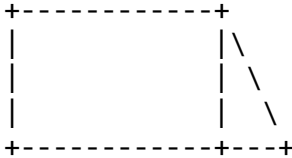
ADDITIONAL ILLUSTRATIONS

1. Isometric Building Sketch

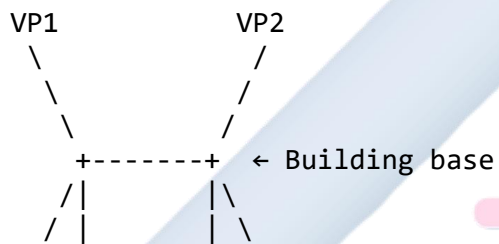


2. Oblique Projection of Building

Front elevation (true shape)



3. Two-Point Perspective



WEEK 8

TOPIC: PICTORIAL DRAWING OF BUILDINGS (CONTINUED)

CONTENT

1. Meaning of Complex Pictorial Drawings of Buildings

A **complex pictorial drawing** is a **three-dimensional (3D)** graphical representation of a building that includes **multiple components** such as verandas, staircases, roofs with varying slopes, balconies, extensions, or annexes.

These drawings go beyond simple rectangular structures to illustrate the **interconnection of several parts** of a building in a single, realistic view.

They provide a complete spatial visualization that aids **architects, builders, and clients** in understanding the building's **form, design, and function** before construction.

2. Types of Complex Pictorial Projections Used in Building Drawings

1. Isometric Projection (Advanced Application)

- Uses 120° angles between the axes.
- Shows complex details like roofs with dormers, porches, columns, stairs, and multiple levels.
- All dimensions along the axes are drawn to scale (isometric scale = $0.816 \times \text{true length}$).

2. Oblique Projection (Advanced Application)

- The front elevation is drawn true, while the sides are projected backward at 45°.
- Best used when the front face of the building must remain undistorted.
- Commonly used for buildings with wide frontages or visible depth in extensions.

3. Perspective Projection (Advanced Application)

- Produces the most **realistic view** of the building as seen by the eye.
 - Incorporates **one-point** or **two-point vanishing** systems.
 - Used for architectural presentations and renderings.
-

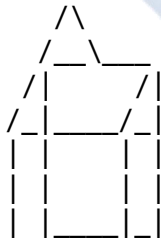
3. Steps in Drawing Complex Pictorial Views of Buildings

A. Isometric Projection of a Complex Building

Procedure:

1. **Draw the isometric axes:**
 - A vertical axis and two inclined axes at **30°** to the horizontal.
2. **Reproduce the building plan:**
 - Draw the floor plan using **isometric scale**, marking room partitions and walls.
3. **Project wall heights:**
 - Project verticals from each plan corner equal to wall height (e.g., 3000 mm).
4. **Outline the roof structure:**
 - For a **hipped or gable roof**, locate ridge points, slopes, and junctions accurately.
5. **Add architectural features:**
 - Doors, windows, veranda posts, columns, railings, and step projections.
6. **Indicate hidden or recessed features:**
 - Use light dashed lines to indicate hidden walls, beams, or internal openings.
7. **Finalize:**
 - Add textures for roofing, wall bricks, and floor tiling.
 - Darken visible outlines and erase construction lines.

Illustrative Sketch (Isometric Building View):



← Roof ridge

← Roof slopes and wall outlines

← Door and window openings

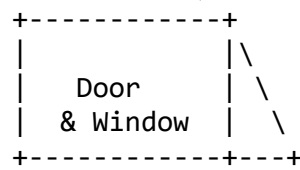
B. Oblique Projection of a Complex Building

Steps:

1. Draw the **front elevation** in true shape.
2. Project the sides backward at **45°**.
3. Transfer dimensions of side walls and roof slopes along the receding axes.
4. Construct additional structures like **porches**, **steps**, and **balconies** in their correct positions.
5. Add openings (doors, windows) to both front and side elevations.
6. Darken visible edges and apply tonal shading to emphasize depth.

Illustrative Sketch (Oblique Projection):

Front View (true shape)



←45° lines represent receding sides

C. Perspective Projection of a Complex Building

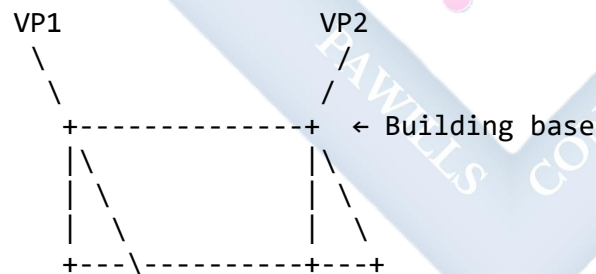
Types of Perspective Used:

- **One-point perspective:** For frontal buildings such as halls or offices.
- **Two-point perspective:** For corner views or buildings with visible depth.

Steps:

1. Draw the **ground line (GL)** and **horizon line (HL)**.
2. Place the **station point (SP)** and **vanishing points (VP1, VP2)**.
3. Draw the plan of the building on the GL.
4. Project the plan points toward VP1 and VP2.
5. Raise vertical lines for the walls, and connect top corners to VPs to define the 3D form.
6. Add the roof, windows, doors, and other architectural details.

Illustrative Sketch (Two-Point Perspective):



4. Proportion and Detailing

- Maintain consistent **scaling and proportion** among building components.
- Ensure **roof angles**, **window positions**, and **door heights** are realistic.
- Include **balustrades**, **steps**, **columns**, **fence walls**, and **canopies**.
- Label parts properly, e.g.:

Example Labels (Isometric House):

- A – Main building wall
- B – Veranda

- C – Roof ridge line
 - D – Door opening
 - E – Window opening
-

5. Line Work and Shading Techniques

- **Thick continuous lines:** Visible outlines
 - **Thin continuous lines:** Construction lines
 - **Dashed lines:** Hidden or internal edges
 - **Shading:** Apply lighter tones on the right side and darker tones below for depth
 - **Roof texture:** Slanted hatch lines at 45°
-

6. Applications of Complex Pictorial Building Drawings

- **Architectural visualization** and 3D modeling
 - **Client presentation** for housing projects
 - **Town planning documentation**
 - **Design evaluation** in technical colleges
 - **CAD modeling** and rendering reference
-

EVALUATION

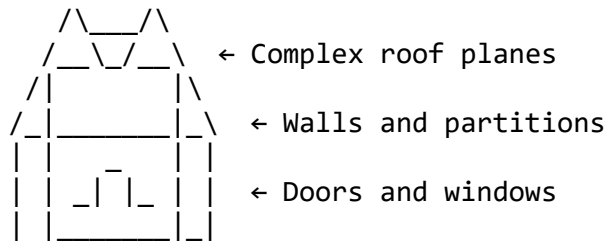
1. What is a complex pictorial drawing?
 2. Mention and explain the three types of complex pictorial projections used in building drawing.
 3. List five important steps in constructing an isometric view of a complex building.
 4. Differentiate between isometric, oblique, and perspective projections.
 5. Explain three advantages of using pictorial drawings in building design.
-

ASSIGNMENT

1. Draw the **isometric projection** of a **three-room building** with a central veranda and a gable roof (use 1:100 scale).
 2. Construct the **oblique projection** of a **two-storey office block**, showing the main staircase and balcony.
 3. Draw the **two-point perspective projection** of a **bungalow** with hip roof and front porch.
-

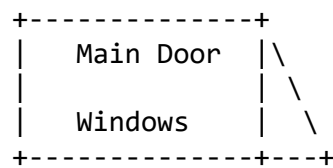
SUPPLEMENTARY ILLUSTRATIONS

1. *Isometric Projection of a Multi-Room House*

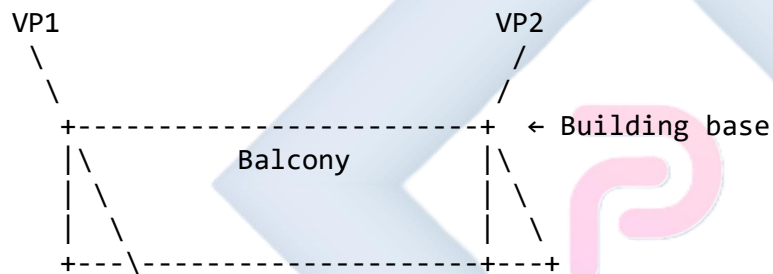


2. *Oblique Projection with Depth*

Front Elevation



3. *Perspective View of a Complex Structure*



WEEK 9

TOPIC: FREEHAND SKETCHING

SUBTOPIC: SKETCHING BUILDING COMPONENTS AND FITTINGS ACCURATELY

CONTENT OUTLINE

1. Meaning and importance of freehand sketching
 2. Materials and techniques used in freehand sketching
 3. Common building components and fittings
 4. Procedures for sketching building components accurately
 5. Shading, proportion, and line application
 6. Evaluation and practice tasks
-

1. MEANING OF FREEHAND SKETCHING

Freehand sketching is the process of **drawing without the use of drawing instruments**, using only the hand and pencil to represent an object quickly and accurately. In technical drawing, especially in building design, it is used to **communicate ideas, record observations, and design components** before producing instrument drawings.

In architecture and building construction, freehand sketching is essential for:

- Making quick design concepts.
 - Recording details of building fittings (windows, doors, hinges, staircases).
 - Visualizing shapes, forms, and proportions of structures.
 - Communicating ideas between architects, builders, and clients.
-

2. IMPORTANCE OF FREEHAND SKETCHING IN BUILDING DRAWINGS

- Helps to **visualize design concepts** before detailed working drawings.
 - Allows **quick expression** of ideas on-site or during discussions.
 - Serves as a **foundation** for later instrument or computer-aided drawings.
 - Enhances **creative ability** and line confidence.
 - Useful in **field sketches** for recording construction details.
-

3. MATERIALS AND TECHNIQUES USED IN FREEHAND SKETCHING

Materials:

- Soft pencil grades (HB, B, 2B, 4B)
- Sketch pad or plain drawing paper
- Eraser and sharpener
- Ruler (for guidance only, not for drawing lines)

Techniques:

- Hold the pencil loosely and comfortably.
- Use **short, light strokes** for initial outlines.
- Maintain correct **proportions** of height, width, and depth.
- Add **details gradually**—doors, handles, hinges, and glazing.
- Shade or tone to give a sense of depth and realism.

Illustrative Sketch — Pencil Grip and Line Practice:

/ Pencil held at 45°
() → free movement of wrist

LINE PRACTICE

----- (continuous)
- - - - (broken)
_____ (thick)

4. COMMON BUILDING COMPONENTS AND FITTINGS TO BE SKETCHED

1. **Doors** – panel door, flush door, glazed door, double door.
2. **Windows** – casement window, louver window, sliding window.
3. **Roofs** – gable, hip, flat, and combination roof types.
4. **Staircases** – straight flight, dog-leg, open-well.
5. **Columns and Beams** – structural members of a building.
6. **Wall Sections** – block walls, partition walls, cavity walls.
7. **Sanitary fittings** – wash-hand basins, water closets (WC), sinks, bathtubs.
8. **Electrical fittings** – light switches, sockets, ceiling fittings.

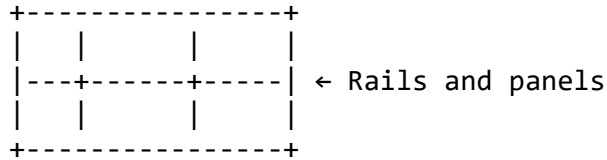
5. PROCEDURE FOR SKETCHING BUILDING COMPONENTS

(A) *Sketching a Panel Door*

1. Draw a **rectangle** to represent the frame.
2. Divide it vertically and horizontally to form panels.

3. Add the **rails and stiles** (top, middle, and bottom rails).
4. Include the **handle** and hinges.

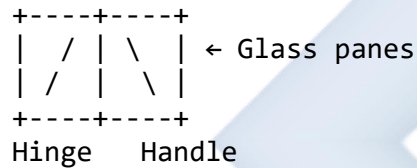
Illustrative Sketch (Panel Door):



(B) Sketching a Casement Window

1. Draw a rectangle representing the frame.
2. Divide it into two equal vertical parts.
3. Sketch the **hinges** on one side and **handles** on the opposite side.
4. Indicate the **glass panes** using light diagonal lines.

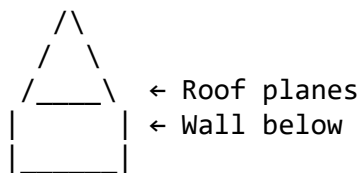
Illustrative Sketch (Casement Window):



(C) Sketching a Gable Roof

1. Draw the base rectangle of the building.
2. Project the ridge line at the centre.
3. Join roof slopes from ridge to eaves.
4. Add fascia and roof edge details.

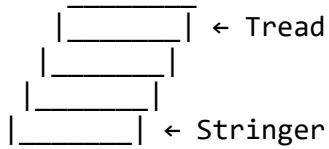
Illustrative Sketch (Gable Roof):



(D) Sketching a Staircase (Straight Flight)

1. Draw two parallel inclined lines for the **stringers**.
2. Divide into equal steps (treads and risers).
3. Add **handrails** and **balusters** if required.

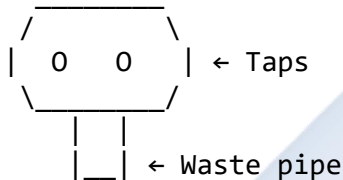
Illustrative Sketch (Staircase):



(E) Sketching a Wash-Hand Basin

1. Draw a curved bowl shape.
2. Add a tap on the top and waste pipe below.
3. Indicate wall bracket or pedestal support.

Illustrative Sketch (Wash-Hand Basin):



6. RULES FOR ACCURATE FREEHAND SKETCHING

- Always start with light construction lines before darkening outlines.
- Keep proportions correct; compare heights and widths visually.
- Add labels where necessary (e.g., "door frame", "handle", "hinge").
- Practice different **line types**: outline, hidden, and sectional.
- Maintain neatness and clarity—technical sketches must be legible.

7. SHADING AND TONING

- **Shading** gives depth and a sense of light direction.
- Use parallel strokes (hatching) for darker areas.
- Apply lighter tones where light falls directly.
- Cross-hatching may be used for texture representation (e.g., wood grain, masonry).

Illustration (Shading Technique):

//////// ← Light hatch
\\\\\\\\\\\\ ← Cross hatch

8. APPLICATION OF FREEHAND SKETCHING IN BUILDING DRAWING

Freehand sketching is applied to:

- **Architectural concept design** – initial sketches of buildings.
 - **Site analysis** – quick representation of land features.
 - **Interior design** – furniture, fittings, and decor sketches.
 - **Building detailing** – on-site explanations and modifications.
-

9. PRACTICAL EXERCISES

Students should sketch the following:

1. A **panel door** with frame and handle.
 2. A **casement window** showing glazing bars and hinges.
 3. A **staircase** with four risers and two treads.
 4. A **wash-hand basin** in elevation view.
 5. A **gable roof** over a rectangular building.
-

EVALUATION

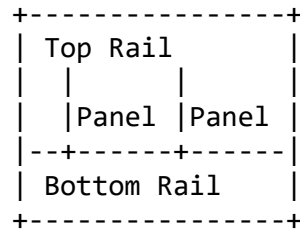
1. Define freehand sketching.
 2. State five importance of freehand sketching in technical drawing.
 3. Mention five building components that can be represented through freehand sketching.
 4. List three materials required for freehand sketching.
 5. Explain the steps involved in sketching a casement window.
-

ASSIGNMENT

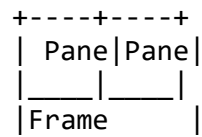
1. Sketch in freehand a **panel door** showing rails, stiles, and panels.
 2. Draw a **staircase** in elevation and plan views using freehand technique.
 3. Sketch and label a **louver window** showing its components.
-

ADDITIONAL ILLUSTRATIONS

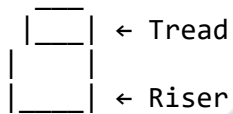
1. Door Components



2. Window Components



3. Staircase Elements



✓ **TEACHER'S NOTE:** Encourage students to develop line confidence through repeated sketching. Begin with large, bold outlines, then refine to include details. Incorporate both **orthographic** (flat) and **pictorial** (3D) sketches for each component.

WEEK 10

TOPIC: FREEHAND SKETCHING (CONTINUED)

CONTENT: SKETCHING MORE COMPLEX BUILDING COMPONENTS

CONTENT

1. INTRODUCTION

Freehand sketching is an essential skill in technical drawing and architecture. It allows designers, builders, and students to quickly represent ideas, construction details, and architectural components **without using drawing instruments**.

In this lesson, we focus on **complex building components** — those that contain multiple elements, curves, or structural details — and learn how to represent them accurately and proportionally by hand.

2. MEANING OF COMPLEX BUILDING COMPONENTS

Complex building components are architectural parts that involve **more than one geometric shape, compound curves, or interacting structural members**. They are often parts of the building that require precision, visual communication, and craftsmanship.

Examples of complex building components include:

- Archways and vaults
- Staircases and balustrades
- Roof trusses and gables
- Window assemblies (with frames and glazing)
- Door frames and moldings
- Columns and capitals
- Ceiling cornices and beams
- Cupolas and domes

These components form critical structural and aesthetic parts of buildings.

3. PRINCIPLES OF FREEHAND SKETCHING COMPLEX BUILDING COMPONENTS

To ensure accuracy and clarity in freehand sketching:

1. **Observation:** Study the object carefully to understand its form, proportions, and symmetry.
2. **Simplification:** Break down the complex object into simple geometric forms (cubes, cylinders, cones, spheres).
3. **Proportioning:** Ensure the relative sizes of parts are correct.
4. **Guidelines:** Lightly draw reference lines to maintain symmetry and alignment.
5. **Contours and Details:** Gradually add secondary features such as edges, frames, and joints.
6. **Shading and Texturing:** Add tonal values to indicate depth and materials.
7. **Annotations:** Label parts (e.g., “beam,” “arch keystone,” “handrail”).

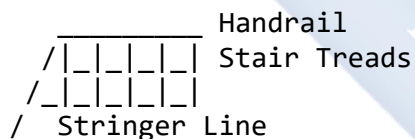
4. PROCEDURE FOR SKETCHING COMPLEX BUILDING COMPONENTS

Let's consider some examples:

A. Sketching a Staircase (with Balustrade)

1. Start with the **outline of the flight** using light lines to establish the angle of ascent.
2. Divide the flight into equal treads and risers.
3. Add the **stringer line** along the underside of the stairs.
4. Sketch the **handrail** parallel to the flight.
5. Add **balusters** evenly spaced between the treads and handrail.
6. Add light shading to show form and direction.

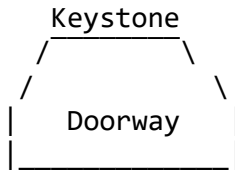
Illustration:



B. Sketching an Arched Doorway

1. Draw the vertical jambs (door sides) first.
2. Locate the **springing line** — the point where the curve begins.
3. Lightly sketch the **semi-circular or segmental arch** connecting the jambs.
4. Add **voussoirs** (wedge-shaped stones or bricks forming the arch).
5. Indicate the **keystone** at the top center.
6. Add the door frame and leaf below the arch.

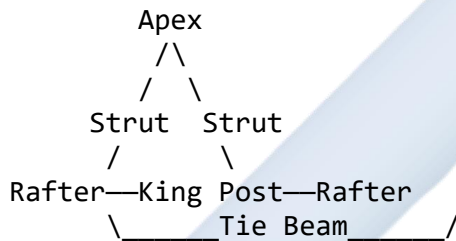
Illustration:



C. Sketching a Roof Truss

1. Draw the base line (span).
2. Sketch two inclined rafters meeting at the apex.
3. Add the tie beam, struts, and king post (or queen posts) as internal supports.
4. Label all members clearly.

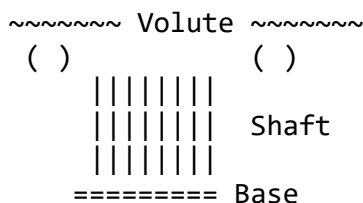
Illustration (King Post Truss):



D. Sketching a Classical Column (with Capital and Base)

1. Draw the vertical axis line to maintain alignment.
2. Sketch the cylindrical shaft using light guidelines.
3. Add the base (square or circular plinth).
4. Sketch the **capital** — Doric, Ionic, or Corinthian style.
5. Indicate fluting (vertical grooves) along the shaft for realism.

Illustration (Ionic Column):



5. IMPORTANCE OF FREEHAND SKETCHING COMPLEX BUILDING COMPONENTS

- Facilitates communication of design ideas on-site or in meetings.

- Helps architects and engineers visualize and modify details quickly.
- Useful for recording construction details and measurements.
- Essential in preliminary design stages before formal drafting.
- Aids craftsmen in interpreting architectural details.

6. MATERIALS AND TOOLS FOR FREEHAND SKETCHING

Even though no instruments are used, the following materials enhance presentation:

- Drawing paper or sketch pad
- Pencil grades (HB, 2B, 4B)
- Eraser and sharpener
- Ruler (for rough guiding lines)
- Shading tools (stump or tissue)

7. TECHNIQUES FOR PRESENTATION

- Use **varying line weights** (thicker outlines, lighter inner lines).
- Apply **consistent proportioning** across all sketches.
- Add **titles and part labels** neatly.
- Include **shadows** and **hatching** for a 3D effect.

EVALUATION

1. Define complex building components.
2. List five examples of complex building components.
3. Sketch neatly, in freehand, any two of the following:
 - A staircase with handrail
 - A king post truss
 - An arched doorway
 - A classical column with base and capital
4. Explain the steps involved in sketching a roof truss.
5. State three reasons why freehand sketching is important in building design.

ASSIGNMENT

1. Sketch a **semi-circular arched doorway** showing voussoirs and keystone.
2. Draw a **king post truss** in freehand and label all the parts.

3. Sketch an **Ionic column** with fluted shaft and volute.
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ADDITIONAL ILLUSTRATIONS (for inclusion in the lesson note)

- 3D perspective of a staircase
 - Roof truss assembly (king post and queen post types)
 - Doorway elevation with arch detail
 - Freehand sketch of window frame and sill
 - Column and entablature details
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WEEK 11

TOPIC: SPECIFICATION WRITING

CONTENT: DEFINITION OF SPECIFICATION WRITING; WRITING SIMPLE SPECIFICATIONS FOR CONSTRUCTION WORKS

CONTENT

1. INTRODUCTION

In building and construction, **technical drawings alone are not sufficient** to describe the full intent of a design. Drawings provide graphical representation, while **specification writing** provides written details about **materials, workmanship, quality, finishes, dimensions, and methods** to be used in construction.

Together, drawings and specifications form a **complete communication tool** between the designer, builder, and client.

Specification writing is therefore an essential skill in technical drawing, architecture, and engineering.

2. DEFINITION OF SPECIFICATION WRITING

Specification writing is the **detailed written description of the materials, workmanship, and methods** required for a construction project.

It provides information that **cannot be conveniently or clearly shown on drawings**.

In simple terms:

Specification writing defines “**what**” and “**how**” a construction task should be executed to achieve the desired quality and standard.

3. PURPOSE AND IMPORTANCE OF SPECIFICATION WRITING

Specification writing plays a critical role in the construction process. It ensures that everyone involved understands exactly what is required.

The main purposes include:

1. **To complement drawings** — by describing aspects not clearly represented graphically.
2. **To guide construction** — ensuring materials and workmanship meet required standards.
3. **To ensure uniformity and quality control.**

4. **To aid cost estimation and budgeting** — since specifications define material types and quantities.
 5. **To serve as a contractual document** — forming part of the agreement between client and contractor.
 6. **To reduce ambiguity and disputes** — by providing clear, written instructions.
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4. CHARACTERISTICS OF A GOOD SPECIFICATION

A well-written specification should possess the following qualities:

1. **Clarity:** The description should be simple, direct, and unambiguous.
 2. **Completeness:** It should cover all necessary details about materials, dimensions, finishes, and workmanship.
 3. **Accuracy:** Technical terms, measurements, and material grades must be correct.
 4. **Consistency:** It must not contradict the accompanying drawings.
 5. **Standardization:** Should conform to recognized building codes and standards (e.g., SON, ISO, BSI).
 6. **Brevity:** Should be concise but complete—no unnecessary words or repetition.
 7. **Legibility and Proper Layout:** Should be neatly organized under headings and subheadings.
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5. TYPES OF SPECIFICATIONS

Specifications may be classified based on purpose and detail level. The main types include:

A. *General Specifications*

- Describe the general quality and nature of materials and workmanship.
- Do not include precise quantities or dimensions.
- Used in **preliminary stages** or **small-scale projects**.

Example:

“The walls shall be constructed with sandcrete blocks of good quality, properly bonded and plastered with cement-sand mortar.”

B. *Particular Specifications*

- Give detailed, precise, and measurable information about materials and methods.
- Used for **large or complex projects**.
- Include brand names, sizes, proportions, finishes, and installation techniques.

Example:

“External walls shall be constructed of 225 mm sandcrete blocks (1:8 cement-sand mix), plastered with cement mortar (1:4 mix), and finished with two coats of oil-based paint over one coat of primer.”

C. Detailed Specifications

- Include **all technical and quantitative details**.
- State **tests, standards, and tolerances** for quality control.
- Often used in **tendering and contract documents**.

Example:

“Cement shall conform to NIS 444:2003 (Portland cement standard). Sand shall be clean, well-graded, and free from clay. Reinforcement shall be mild steel rods of 12 mm diameter with yield strength of 250 N/mm².”

6. COMPONENTS OF A BUILDING SPECIFICATION

A complete specification document should include:

1. **Title Page:** Project name, client, date, and preparer’s name.
 2. **Table of Contents:** Lists sections and page numbers.
 3. **General Conditions:** Definitions, scope, and legal framework.
 4. **Materials:** Description, grade, and source of materials.
 5. **Workmanship:** How materials are to be handled, mixed, and installed.
 6. **Tests and Standards:** Codes or tests for verifying material quality.
 7. **Measurements:** Units, dimensions, and tolerances.
 8. **Finishes:** Type, colour, and quality of surface finishes.
 9. **Approval:** Space for signatures of the architect, engineer, and client.
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7. EXAMPLES OF SIMPLE SPECIFICATIONS FOR CONSTRUCTION WORKS

A. Brickwork

Brickwork shall be executed with well-burnt red clay bricks of uniform size and colour. Mortar shall consist of 1 part cement to 6 parts clean sand by volume. The brick joints shall not exceed 12 mm in thickness and shall be neatly struck flush.

B. Concrete Work

Concrete shall be made using one part of cement, two parts of clean sharp sand, and four parts of coarse aggregate by volume. Water shall be clean and free from

impurities. Reinforcement shall be placed as shown in the drawings and covered with at least 25 mm of concrete.

C. Plastering and Finishing

All plastering shall be carried out in cement-sand mortar (1:4). Surfaces shall be finished smooth, even, and free from cracks. Painting shall include one coat of primer and two coats of emulsion paint.

D. Carpentry Work

All timber shall be well-seasoned, free from defects, and conform to NIS standards. Joints shall be tight and properly secured. Exposed wood surfaces shall be sanded smooth and polished.

E. Roofing

Roofing sheets shall be aluminium, 0.45 mm thick, fixed with aluminium nails and washers. Trusses shall be made of seasoned hardwood or mild steel sections according to drawings.

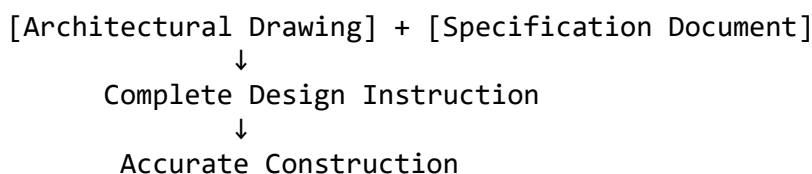
8. LAYOUT AND PRESENTATION OF A SPECIFICATION DOCUMENT

Section	Details to Include
Title Page	Project name, location, designer, client, date
General Description	Type of building, overall purpose
Materials	Types, grades, and proportions
Workmanship	Methods of application and finishing
Tests and Standards	Required codes (NIS, BS, ISO, etc.)
Finishes	Paint types, coatings, or polish
Notes	Safety, environment, or maintenance instructions

9. ILLUSTRATIVE DIAGRAMS

Below are diagrams showing how **specification writing supports construction drawings**:

A. Relationship Between Drawing and Specification



B. Example of Specification Sheet Layout

PROJECT TITLE: BUNGALOW HOUSE PLAN
CLIENT: MR. JOHN DOE
DATE: 10/11/2025

SECTION 1: MATERIALS

Cement: Portland (NIS 444:2003)
Sand: Clean, sharp, well-graded
Aggregate: Granite 20 mm max

SECTION 2: WORKMANSHIP

Concrete mix ratio 1:2:4 by volume
Reinforcement: Mild steel rods 12 mm dia
Cover: Minimum 25 mm

SECTION 3: FINISHING

Paint: Emulsion, two coats, white colour

10. COMMON MISTAKES IN SPECIFICATION WRITING

- Using vague terms like “good quality” without definition.
- Contradicting the drawings.
- Failing to specify standards or test requirements.
- Over-specifying materials (unrealistic for budget).
- Poor grammar or unclear sentence structure.

11. RELATIONSHIP BETWEEN SPECIFICATION AND DRAWINGS

Aspect	Drawings	Specifications
Representation	Graphical	Written
Focus	Size, shape, layout	Materials, quality, workmanship
Nature	Visual	Descriptive
Use	To visualize	To explain
Combined Role	Ensure full understanding of design intent	Ensure correct execution

EVALUATION

1. Define specification writing.
2. State five characteristics of a good specification.
3. Differentiate between general and detailed specifications.

4. Write a simple specification for: a. Brickwork b. Concrete work
 5. Explain three purposes of specification writing in construction.
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ASSIGNMENT

1. Write detailed specifications for **a one-bedroom bungalow**, covering:
 - Foundation
 - Walling
 - Roofing
 - Finishing
 2. Prepare a simple specification sheet layout for **a school classroom block**.
 3. Explain how **drawings and specifications** complement each other in a building project.
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ADDITIONAL VISUAL AIDS (recommended for inclusion)

- Diagram showing **sample specification sheet format**
 - Flowchart showing **relationship between design, specification, and construction**
 - Sample **extract from a construction contract specification**
 - Comparison chart of **general vs. particular specification**
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